CONTRIBUTED PAPER

# Reactive Power Implications of Penetrating Inverter-Based Renewable and Storage Resources in Future Grids Toward Energy Transition—A Review

*This article addresses the gap in understanding the reactive power transition in the broader context of energy transformation and examines the reactive capability of major inverter-based resources for utilization in future decarbonized grids.*

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**ABSTRACT** | Transitioning to net-zero emission energy systems is currently on the agenda in various countries to tackle climate change, a global challenge that threatens the lives of future generations. To fully decarbonize energy systems, a radical paradigm shift through deep integration of renewable resources supported by storage technologies is envisaged in multisector energy systems, especially in the electric power sector. As a result, inverter-based resources (IBRs), mainly wind, photovoltaics (PVs), and batteries, will dominate the electric power grids. This transition involves phasing out conventional fossil fuel-based plants and decommissioning associated synchronous machines, the grid's primary reactive power sources. The ongoing removal of these primary reactive power sources introduces critical operational challenges that could compromise the reliability and stability of the grid. The inverters used for integrating IBRs can deliver diverse crucial ancillary services, particularly reactive power support. However, the potential of IBRs to address reactive power requirements in future decarbonized grids still needs to be fully addressed. The existing literature lacks a comprehensive approach to coordinating and harmonizing the efforts of various stakeholders and drivers to leverage the reactive power capability of IBRs. To bridge this gap, this article thoroughly reviews the reactive power implications for future grids with a considerable share of primary IBRs, comprising distributed and large-scale wind, PV and battery storage plants. This article starts with a summary of the concept, measurement methods, and importance of reactive power for voltage control and how it is managed today utilizing conventional sources. The reactive power transition from current to future grids within the context of the greater energy transition is then discussed by shedding light on its diverse aspects. Afterward, the reactive capability curve of each IBR is derived from the equivalent circuits and

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equations. Various grid codes and integration requirements of IBRs are then analyzed from a reactive power support viewpoint. Also, the concepts related to reactive power and voltage control comprising control extents, modes, and techniques are elaborated. Finally, recommendations are provided to set the stage for leveraging the capabilities of IBRs to address the reactive power requirements of future grids. The presented material sheds light on the pivotal role of reactive power in future grids and provides a roadmap for policymakers, utilities, and grid operators to manage a seamless transition to a decarbonized grid.

**KEYWORDS** | Energy transition; grid code; inverter-based resources (IBRs); reactive power capability; renewable energy and storage integration.

## I. INTRODUCTION

Climate change and energy security have been the two primary triggers of the energy transition during the past decades [1], [2]. In 2022, the energy sector has the largest share of global $CO_2$ emissions at about 38%, contributing to climate change [3]. The power industry emissions reached an unprecedented record of 14.65 Gton of $CO_2$ in 2022, dominating other sectors [4]. Deep decarbonization of the electricity grid is critical to meet a net-zero nationwide economy by paving the way for fully decarbonizing the whole energy system [5]. Furthermore, the power system decarbonization is especially essential when aligned with efficient electrification, leading to the decarbonization of significant parts of other sectors, i.e., heat, transportation, buildings, and industry [6].

Distinguishing the critical role of the power sector in the whole energy system decarbonization, many major economies have set goals to decarbonize this sector first [5]. For instance, the United States and the United Kingdom have set a 100% net-zero power sector goal by 2035 [7], [8]. Following this scenario, a carbon-free whole energy system is set to be achieved by 2050 by the same nations [7], [8]. Achieving net-zero power and whole energy systems translates into relying thoroughly on clean resources. Clean electricity, or equivalently, clean energy includes mainly hydropower, nuclear energy, fossil energy equipped with carbon capture and storage (CCS), and renewable resources [9]. Hydropower plants have constituted the backbone of clean electricity for decades. However, their dependence on natural rainfalls, special geographic topology requirements, and limited access to new suitable sites have limited their further development [10].

On the other hand, nuclear power is associated with system complexity, high construction cost and time overruns, and high risks of nuclear process failures, natural disasters, and sabotage attacks similar to the Chornobyl and Fukushima accidents [11]. The CCS technology can facilitate the energy transition by removing carbon from current equipment, especially in some challenging sectors like industrial heat. However, lifecycle environmental impacts, uncertain costs, low technological maturity, and societal acceptance are obstacles to widespread deployment. Besides, the CCS technology cannot tackle the emissions generated during mining, refining, and transporting fossil fuels [12].

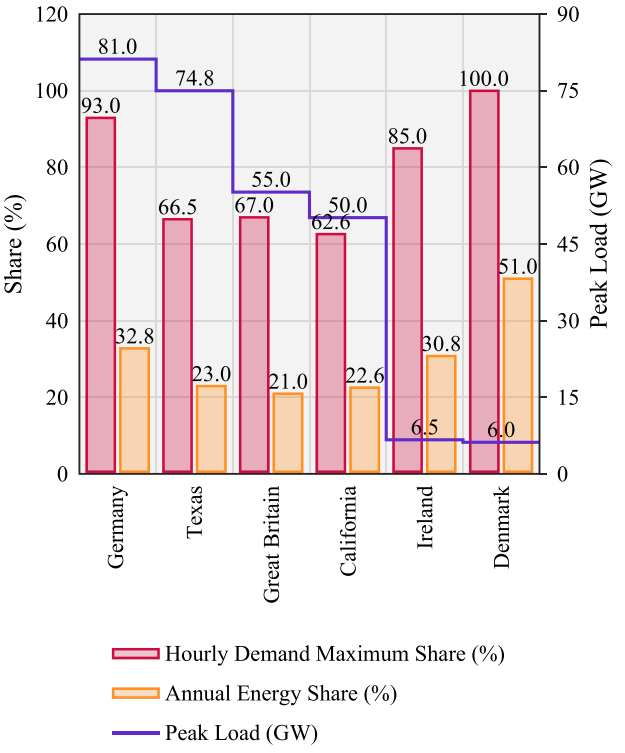



**Fig. 1.** *Current status of renewable penetration in selected countries and states.*

Consequently, 100% clean energy systems will be predominantly based on renewable resources, particularly wind and solar energy. This preference arises from several key factors comprising technological maturity, widespread potential, and declining costs, making these resources increasingly competitive with traditional fossil fuel-burning plants [13], [14]. Fig. 1 shows some key statistics on the penetration of renewables in the leading countries and states. It contains the maximum share of renewables in hourly demand, the share of renewables in total yearly supplied energy, and the peak load for Germany, Texas, Great Britain, California, Ireland, and Denmark [15]. As can be observed, the hourly share of renewables has reached its maximum level in Denmark and will approach that level in Germany soon. Based on the growing penetration rates of renewables, this situation will also be the case for many other countries and states in the near future.

Integrating energy storage systems to address the variability associated with these intermittent renewable resources is essential to smoothing the output power. Energy storage also helps to increase the capacity factor of these resources, which is inherently low [16], [17], [18]. The utilization of renewable-based generation and storage systems is currently at the forefront of energy transition scenarios comprising wind energy conversion

system (WECS), solar photovoltaic (PV), and battery energy storage system (BESS) plants [16], [19]. These assets, known as inverter-based resources (IBRs), are typically integrated into the grid via a power electronics interface. The primary function of the power electronics interface is to regulate and condition the power being exchanged with the grid [20], [21], [22]. However, the impact of IBRs on power systems is significant. Considering reliance on converter integration, significant penetration of IBRs into power systems can lead to dispatchability, stability, balancing, and power quality challenges [23], [24]. Penetrating high shares of IBRs in the future grids will primarily cause two issues due to the decommissioning of conventional synchronous machine-based generation, as discussed next [25].

Currently, power grids rely on the mechanical energy stored in the rotating masses of large conventional generators with rotational inertia to maintain the grid's power balance. Unlike synchronous generators (SGs), IBRs do not utilize heavy mechanical rotating masses since they are connected to the grid via static power converters. As the utilization rate of IBRs, without physical inertia, is steadily increasing, events in the power system occur faster and require new ways to balance the grid. The fist issue with the increasing penetration rates of IBRs is maintaining the frequency of the future low-inertia grids during steady-state operations and emergencies. This issue, inertia-related technical challenges in the IBRs-integrated future grids, has been actively investigated recently [26], [27], and novel real-time inertia monitoring and management methods are proposed [28], [29], [30].

The second issue with the increasing penetration rates of the IBRs is reactive power deficiency, which results from decommissioning synchronous machines, the primary source of reactive power. Despite their essential role and capabilities, an often overlooked feature of converters used for the IBRs integration is their ability to swap reactive power with the grid [31]. By effectively modeling and employing the reactive power capability of IBRs, they can serve as a primary means of voltage regulation in future grids [32]. However, the potential of IBRs to address reactive power requirements in future decarbonized grids still needs to be fully addressed. The existing literature lacks a comprehensive approach to coordinating and harmonizing the efforts of various stakeholders and drivers to leverage the reactive power capability of IBRs. The current research on the reactive power capability of IBRs is focused primarily on control methods or has narrowed to a specific IBR technology, typically wind turbines [33], [34]. Furthermore, in most optimization-based research related to IBR investment and operational planning, the reactive power capability is entirely overlooked or approximated by unrealistic schemes [35], [36], [37]. This practice has resulted in either an underestimation or an overestimation of the actual reactive power capability of IBRs, consequently creating a gap between theoretical studies and field practice. Despite the previous research [38], [39], there are concerns about the scarcity of reactive power, the inefficiency of reactive power markets, and their competitiveness in future net-zero grids [40], [41].

This article aims to address the gap in understanding the reactive power transition in the broader context of energy transformation. It examines the reactive capability of major IBRs comprising WECS, PV, and BESS plants for utilization in future decarbonized grids. This article, after this introduction, begins by explaining the concepts, measurement methods, and importance of reactive power for voltage control in interconnected power networks in Section II. Section III briefly reviews conventional reactive power sources employed for voltage control. Section IV delves into the current to future status of the conventional reactive power sources during the energy transition. Section V presents a classification of various IBRs based on the plant size and connection location while considering reactive power support. Section VI highlights the opportunities and challenges associated with reactive power support from IBRs. The analytical reactive power capability of the WECS, PV, and BESS plants is derived and described using equivalent circuits and relevant equations in Sections VII–IX, respectively. Section X analyzes the grid codes and integration requirements for each IBR from a reactive power capability perspective. Section XI explores the concepts related to reactive power control of IBRs comprising control extent, modes, and techniques. Finally, conclusion and suggestions to set the stage for leveraging the reactive power capability of IBRs in future grids are provided in Section XII.

## II. REACTIVE POWER CONCEPT, MEASUREMENT, AND VOLTAGE CONTROL

Reactive power is a concept that has become critical for understanding and analyzing interconnected ac electric power systems since their early days of evolution [42]. Reactive power, unlike its active power counterpart, oscillates between devices. Continuous exchange occurs between devices that generate, store, and transmit energy in their electric and magnetic fields. In a power system, the dominant load is inductive, which involves alternating reactive power between the magnetic fields and the system. Inductive loads continuously cycle between the storage and return of energy. Capacitive loads also follow a storage/return pattern but store reactive power in their electric fields rather than the magnetic field. It is crucial to understand that the storage and return cycles of inductive and capacitive loads are quite inverse. Utilities can exploit this difference to maintain balance in reactive power [43].

### A. Reactive Power Measurement

Contractually, reactive power generation is associated with capacitors and consumption with reactors, and apparent power is defined as the vector sum of the active (real) and reactive (imaginary) components of power [44]. Consequently, the magnitude of the apparent power is

equal to the square root of the sum of the squares of the active and reactive power. Over the past century, the concept of reactive power has been widely utilized in the electrical engineering society. It has been used efficiently to describe the standard for power transmission and distribution, measurement devices, billing, and power factor compensation for sinusoidal single-phase and balanced three-phase systems. This definition performs well within the industry as long as the current and voltage waveforms remain close to sinusoidal and balanced three-phase systems. Nevertheless, there are considerable challenges in determining and assessing reactive power in systems characterized by periodic yet nonsinusoidal waveforms, such as those impacted by harmonics [45].

Budeanu [46] introduced the first and most recognized power theory in the frequency domain to address the discrepancy between reactive power calculation in nonsinusoidal and sinusoidal voltages and currents. Budeanu [46] noticed that unlike sinusoidal cases where the quadratic summation over active and reactive power equals the apparent power, this relationship does not hold for nonsinusoidal waveforms. In order to address this challenge, Budeanu introduced a novel idea, called distortion (or deformation) power [46], [47]. Since then, Budeanu's power model has gained substantial recognition, appearing in various research works, academic textbooks, and IEEE standards, which primarily deal with power-related phenomena in systems featuring periodic but distorted waveforms [46].

From its early days of presentation, this idea faced opposition from various critics. In the beginning, it took approximately 50 years to produce initial meters that worked based on Budeanu's reactive and distortion power concepts. The main concerns, however, centered around the absence of a physical interpretation for distortion power and the practice of merging the magnitudes of oscillating components from different harmonics. Budeanu's power model eventually faced a significant challenge by Czarnecki [48], who aimed to demonstrate its impracticality in the realm of instrumentation and measurement. Although these arguments did not immediately sway supporters of Budeanu's power model [49], it was ultimately removed from the most recent IEEE standards in 2010 [50].

Czarnecki [51]was the scientist who ground the second power theory in the frequency domain during the 1980s, Currents' Physical Components (CPC), which gained many supporters. Both theories share similarities in describing active power but differ in their approach to reactive power [52]. In recent investigations, it has been demonstrated that Budeanu's method can be advantageous for linear and time-invariant systems [53]. However, other research has indicated that employing Budeanu's approach might lead to inaccurate outcomes when certain types of distortions are present [54]. In 2010, the IEEE working group on nonsinusoidal situations has updated their status on the topic by publishing the standard on "Definitions for the Measurement of Electric Power Quantities Under Sinusoidal, Nonsinusoidal, Balanced, or Unbalanced Conditions" [50].

### B. Reactive Power Imbalance and Impact on System Voltage

In a typical power system, there is always a deficit (under heavy loading) or surplus (under light/no loading) of reactive power in the system, which requires additional reactive power sources or sinks to manage. Imbalances in the system's reactive power can cause voltage issues across the system [55]. The balance of the reactive power between sources and sinks in the system may be disrupted, leading to voltage variations at different levels and time frames, due to [56].

1) Continuous changes in load demand, both active and reactive power. As reactive power parts of loads vary, reactive power requirements of the grid vary clearly. Also, changing active power, even under constant reactive power, will change reactive power requirements of the system due to dependency of the reactive power losses of lines and transformers on the flowing current (the flowing apparent power).
2) Extreme events causing faults, equipment outages (lines, transformers, generators, and reactive power sources), or sudden changes in generation and/or consumption leading to a significant imbalance in the reactive power equilibrium.

The aforementioned phenomena may cause undervoltage or overvoltage in the system due to reactive power imbalance. Factors contributing to undervoltage caused by reactive power deficiency include heavy line loading, power lines' outages, reactive power devices' failures, and inadequate reactive power production by reactive power sources (such as capacitors) during voltage drops. Undervoltage poses risks, such as overload in lines and transformers, reduced load demand supplied below nominal values, instability risks, and motor stalling/starting issues. Furthermore, it leads to increased losses (both active and reactive) in network lines and transformers [2].

In contrast, excess reactive power may lead to overvoltages of various durations and natures. Long-term overvoltages can result from the Ferranti effect, which depends on the equivalent shunt capacitance, nominal voltage, and line loading [43]. Short-term overvoltages can result from generator self-excitation or harmonic overvoltages. Additionally, transient overvoltages can be caused by lightning strikes or power system switching transients. High voltages often lead to insulation failure in equipment such as circuit breakers, transformer bushings, and cables, resulting in internal flashovers, degradation, and reduced lifespan. Furthermore, overvoltage can cause increased transformer magnetic fields, resulting in saturation, excessive heating, and failure. It can also increase the normal system load and damage customers' electronic devices with sensitive circuits [2].

### C. Voltage-Related Phenomena

To comprehend, control, and manage voltage fluctuations, various voltage-related concepts have been proposed in the literature, such as voltage stability, collapse, and security. Voltage stability denotes the ability of a power system to maintain appropriate voltage levels during normal operation and after an event [57]. In a system with stable voltage, an increase in the load results in a corresponding increase in the power delivered, allowing for control over both voltage and power. In contrast, a system with voltage instability has no proper control over voltage magnitudes and power transfers. Voltage collapse occurs when voltage instability leads to a substantial system-wide voltage drop. Before a voltage collapse, the system experiences a period of voltage instability, during which the power system is in peril, and operators have lost control over voltage levels and power flows. In this case, voltage levels may decline to 70%–90% of normal values [2], [58]. Voltage security refers to the ability of the system to maintain voltages within predetermined limits following any system changes [55]. This requires a sufficient margin from the operating point to the voltage instability state. Access to reactive power reserves from designated resources is crucial for maintaining voltage security [55], [56]. In summary, to achieve efficient, reliable, and stable system operation, it is essential to maintain system voltages within appropriate limits by regulating reactive power. This aims to achieve the following goals [56].

1) All system equipment, including utility components and customer devices, operate within their specified voltage ratings. Prolonged operation beyond these limits can negatively impact performance and potentially cause damage.
2) Enhanced system stability leads to greater utilization of the transmission network.
3) Minimizing reactive power flow contributes to reducing both active and reactive power losses, fostering network efficiency.

Maintaining voltage security and stability of the system can be achieved by controlling the generation, absorption, and flow of reactive power within the system [59]. It is crucial to note that both the appropriate levels of reactive power injection and withdrawal must be accessible to avoid overvoltages and undervoltages within the system. This involves utilizing conventional besides the emerging reactive power sources, i.e., IBRs.

### D. Convention on Sign of Reactive Power

Understanding the sign of reactive power and related terms is a critical aspect before delving into the capability curve of diverse sources. This understanding helps to avoid confusion and ensures a common understanding among various sources. Two primary standards are widely used in this context: IEEE-1547 and IEC 61850-7-520. The main difference between these standards lies in the sign convention for power factor. In IEEE-1547, the sign of the power factor is used to denote the capacitive (positive) or inductive (negative) nature of reactive power, regardless of whether the system is generating or consuming active power [60]. In contrast, IEC 61850-7-520 adopts a different approach, where a positive sign of reactive power indicates a generator, while a negative sign denotes a consumer, regardless of the reactive power injection or absorption [61]. When working with these reference frames, a primary challenge is the change in sign convention when crossing from the right to the left side of the plane and vice versa. This was not an issue historically due to the limited availability of the four-quadrant devices. However, with the integration of BESSs into the power systems as an integral part of the IBRs, it becomes essential to define and standardize the sign convention. To clarify the terminology, active power generators in the right side of the plane, or BESSs in the discharging mode, can be described as "overexcited" when producing reactive power (increasing voltage) and "underexcited" when absorbing reactive power (decreasing voltage), as shown in Fig. 2. Conversely, BESSs in the charging mode (left side of the plane) can be characterized as either capacitive (sourcing reactive power and increasing voltage) or inductive (sinking reactive power and decreasing voltage). As illustrated in Fig. 2, the upper side of the capability curve represents an overexcited, capacitive, leading, or equivalently exporting reactive power state, while the lower side denotes an underexcited, inductive, lagging, or importing reactive power state.

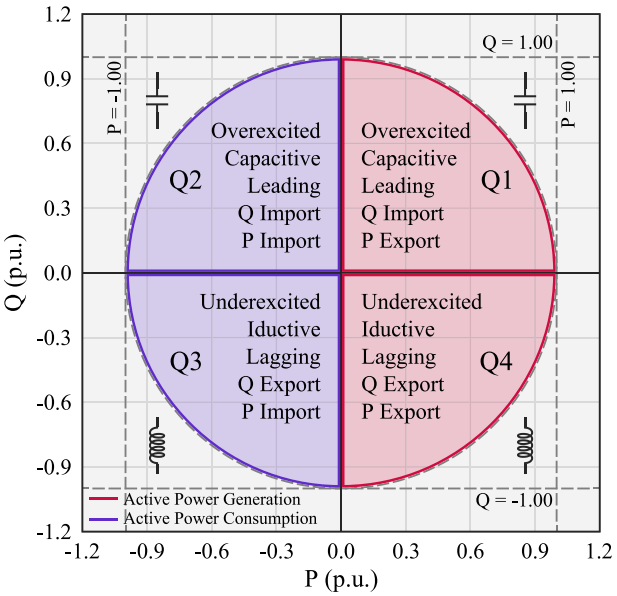



**Fig. 2.** *Convention on reactive power sign.*

## III. CONVENTIONAL REACTIVE POWER SOURCES

Since the early development days of interconnected power grids, conventional reactive power sources have been used

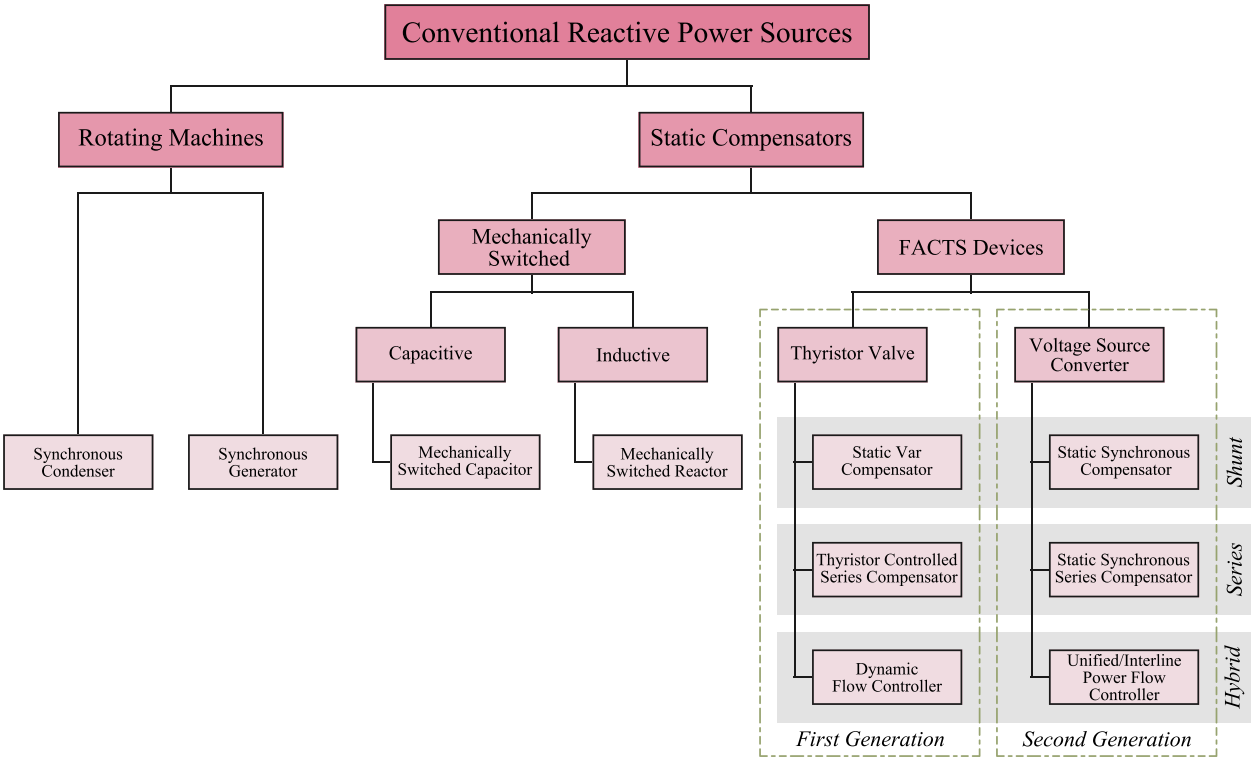



**Fig. 3.** ***Classification of conventional reactive power sources.***

for voltage control as the primary regulation tool. The common characteristic of these sources is that they are equivalent to a nonzero and variable reactance from the point of common coupling (PCC) view [62]. Fig. 3 presents a general classification of these sources. As in the figure, the conventional sources can be generally categorized into rotating machines and static equipment. Rotating machines, including SGs and condensers, as the backbone of the reactive power support of the system, provide continuous and full-range reactive support.

In contrast, there are static compensators that are nonrotating and stationary circuits based on the electric and power electronics components. This category is typically classified into the half-range discrete mechanically switched compensators (capacitors and reactors) and full-range continuous flexible ac transmission systems (FACTSs) [62], [63], [64]. The following briefly describes these devices, focusing on their merits and demerits. It is worth remembering that reactive power transmitters like transformer tap changers and auto-boosters and the inherent capacitance of long high-voltage transmission lines and cables are not included because of their limited extent and relatively lower controllability.

## A. Synchronous Generators

SG is the grid's first and foremost reactive power control tool [65]. Reactive power control in an SG can be performed simply by adjusting the excitation current in a closed-loop control system known as automatic voltage regulator (AVR). Fig. 4(a) demonstrates a typical SG's reactive power capability curve. As can be observed from the figure, the SG provides the operator with a flexible range of reactive power control from leading to lagging power factors. However, some limitations exist on achieving low power factors enforced by the stator's current limit. Besides, the location of the exchanged reactive power will be limited to the location of the power plant and cannot be decided independently and optimally. On the contrary, the delivered reactive power does not depend on the connection point voltage and acts independently. Additionally, the machine produces no switching transients and is unaffected by system harmonics [65]. To the extent of today's power grid history, there is a prolonged experience in utilizing the AVR of synchronous machines for reactive power control [65].

## B. Synchronous Condensers

A synchronous machine operating without a prime-mover force or a mechanical load is known as a synchronous condenser (SC). By managing the field excitation of the machine, it can be designed to absorb or inject reactive power. Furthermore, it automatically adjusts the reactive power output to maintain a steady terminal voltage using a voltage regulator. This device draws a minimal amount of active power from the power system to cover its internal energy losses, which is shown by the negative $P_0$ in Fig. 4(a). The range of reactive power support is higher compared to an equivalent SG regarding

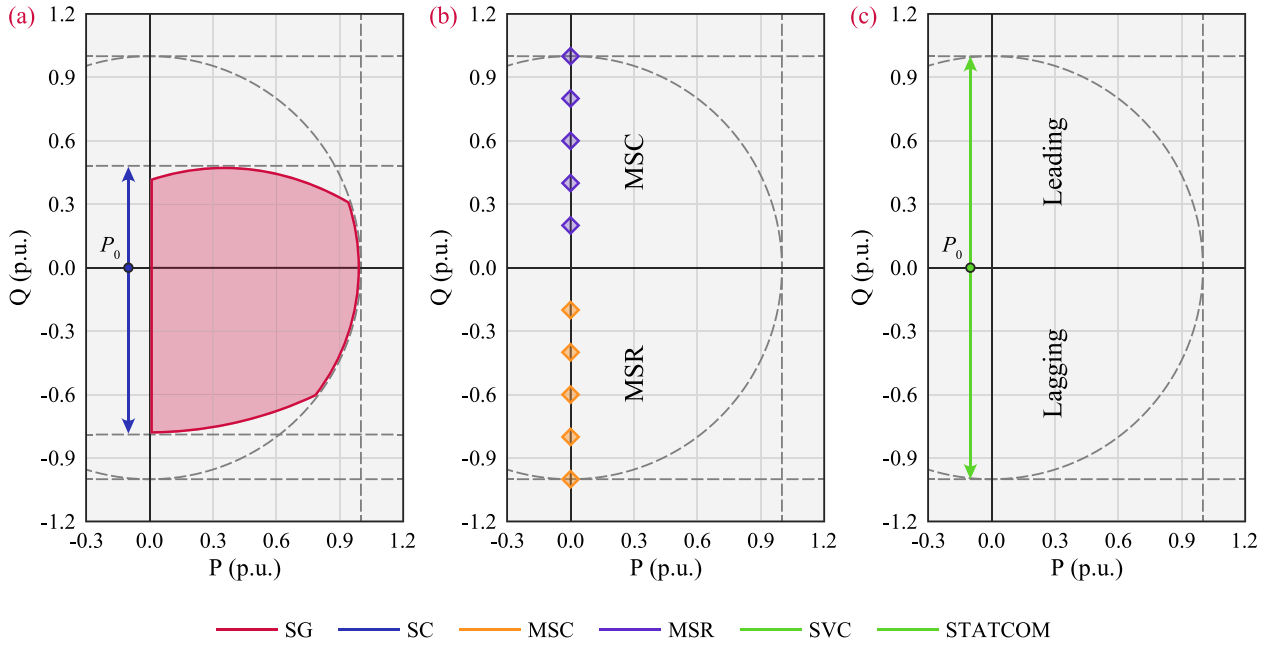



**Fig. 4.** ***Reactive power capability of conventional sources. (a) SG and SC. (b) MSR and MSC. (c) SVC and STATCOM.***

not having limitations of active power generation. In this case, the entire apparent power capacity of the machine is devoted to the reactive power, ignoring a minor share of the internal losses. SCs offer continuous and controllable reactive power support compared to mechanically switched shunt compensators (capacitors and reactors) without concerns about grid harmonics and connected bus voltage fluctuations. Since the 1930s, they have been employed for voltage and reactive power regulation across transmission and subtransmission networks [66]. However, the initial investment cost and routine operation and maintenance expenses are comparatively higher than the shunt compensators considering the reliance on rotating parts. On the other hand, regarding the required space for FACTS installation, SCs may be of interest with a lower space requirement since SCs do not need harmonic filter circuits. Furthermore, the reactive power produced by SCs is considered superior to static VAR options. SCs increase reactive power when the system voltage drops, simultaneously providing the local system with substantial short-circuit support [67].

Currently, many conventional fossil fuel-burning generation assets based on SGs are on the list of decommissioning, including fossil fuel- and nuclear-based generation equipped with SGs. The retired generators are valuable options to retrofit to an SC by applying minor changes to the system. Usually, through this conversion process, an approximately 30% increase in reactive power production is achievable, as shown in Fig. 5. This transformation to SC mode allows the machine to be overexcited, a situation that would not happen during power generation mode. There are two critical issues with the generator-to-condenser conversion besides minor control, protection, cooling, and lubrication subsystem changes. The first issue to be addressed is supporting the generator shaft on the previous turbine side. The second is starting the SG in

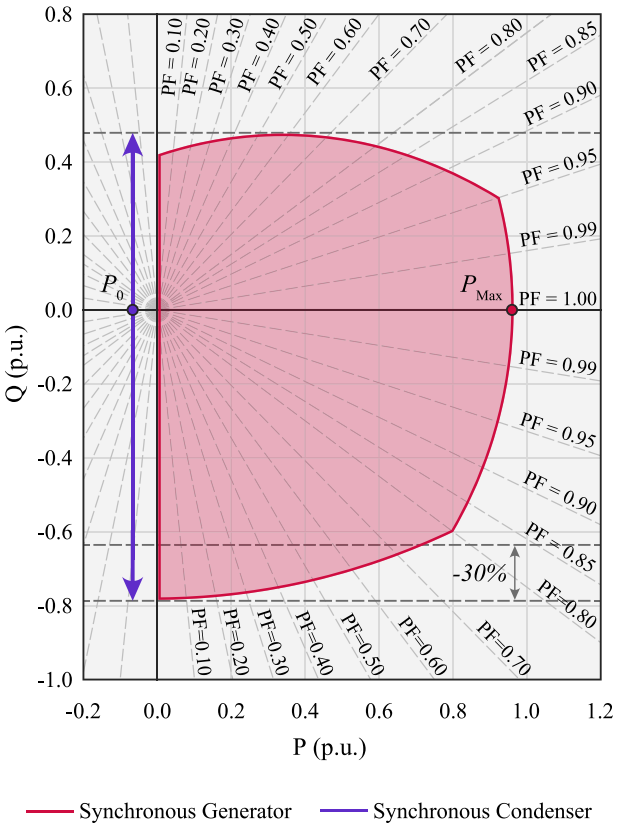



**Fig. 5.** ***Repurposing SG to condenser.***

motor mode and synchronizing it to the grid with a variable frequency drive or, a cheaper option, a pony motor [68], [69].

### C. Mechanically Switched Capacitors and Reactors

These devices comprise noncontinuous and switchable shunts, including capacitors and inductors (reactors). The mechanically switched capacitor (MSC) and mechanically switched reactor (MSR) are simple in structure and are static devices used for reactive power control at the lowest initial and lifetime expenses. These devices can be installed anywhere needed and provide decoupled reactive power control. As shown in Fig. 4(b), the main drawback of these devices is that it is not possible to continuously control reactive power due to the discrete nature of capacity changes. Another drawback of these devices is that the exchanged reactive power is proportional to the square of the voltage at the connection point, causing operational challenges during severe voltage drops. Furthermore, some problems with the harmonics of the system affect MSCs and transients induced during their switching [70]. Generally, MSCs and MSRs constitute the simplest and most economical way to control reactive power, especially when fast dynamics is not the main focus [70].

### D. Flexible AC Transmission Systems

With the progress in power electronics, it became possible to control reactive power through these devices continuously and quickly. This progress led to the emergence of FACTS devices in the late 1980s. The term FACTS was first used by the Electric Power Research Institute (EPRI) and publicly introduced by Hingorani [71] in 1985. Construction of the FACTS is based on power semiconductors enabling a continuous and faster response than conventional mechanically switched compensators. Some FACTS devices can control the active power for system damping and flow control. Furthermore, they can provide conventional reactive power control used for voltage stability and control. As defined by IEEE, FACTS is "a power electronics-based system and other static equipment that provide control of one or more ac transmission system parameters to enhance controllability and increase power transfer capability" [72]. Fig. 3 demonstrates a general classification of FACTS devices. As in the figure, there are three types of FACTS devices based on the connection to the grid: series, shunt, and hybrid. The FACTS connection defines its role and functionality in the grid [72].

Additionally, there are two generations of these devices. The first generation was built based on thyristor-controlled switches as control devices. The thyristor-controlled series capacitor (TCSC), static var compensator (SVC), and dynamic flow controller (DFC) are series, shunt, and hybrid first-generation FACTS devices, respectively. On the contrary, the second-generation FACTS is based on a fast and controllable voltage source converter (VSC) configuration. Static synchronous series compensator (SSSC), static synchronous compensator (STATCOM), and unified power flow controller–interline power flow controller (UPFC-IPFC) constitute the series, shunt, and hybrid second-generation FACTS, respectively. Currently, the shunt FACTS devices comprising SVC and STATCOM constitute the highest market share and are the most utilized for reactive power support [73]. These two devices, with the capability curves shown in Fig. 4(c), have also been used in Type 2 WECSs to comply with grid connection requirements for years [13].

Numerous SVC arrangements are available, yet the majority incorporate thyristor-switched capacitor (TSC), thyristor-controlled reactor (TCR), harmonic filters, and either breaker-switched or fixed capacitors as fundamental components. The TCRs are consistently regulated to allow for a reactive power exchange with phase angles ranging from leading to lagging. STATCOM, another prominent FACTS technology, operates similar to SVC but is based on a VSC, providing more flexibility. The limitation of gate turn-off (GTO) thyristors in SVCs is overcome by introducing insulated-gate bipolar transistors (IGBTs) in STATCOM, which play a crucial role in transient voltage regulation rather than SVC.

The STATCOM offers almost a constant reactive current output, even at low voltage levels. In contrast, an SVC cannot achieve this capability, as its reactive current is contingent on fluctuations in terminal voltage. The reactive power support by SVC is contingent on the system voltage, and its capability is diminished at lower voltages. Nonetheless, in many transmission network installations, the effectiveness of shunt FACTS devices is degraded when operating below 80% of the nominal voltage [73]. In specific use cases, STATCOMs exhibit faster response times compared to SVCs. The primary challenge with FACTS is the relatively high investment costs compared to other compensation methods [71], [72], [73].

## IV. FUTURE STATUS OF CONVENTIONAL REACTIVE POWER SOURCES

Currently, conventional SGs utilized in fossil fuel-burning, nuclear, and hydropower plants constitute the primary source for meeting grid requirements on reactive power. Additionally, mechanically switched shunts (MSC and MSR), SCs, and FACTS (primarily SVC and STATCOM) support the role of SGs in the reactive power supply. The conventional power plants based on the SG are steadily phasing out as the generation portfolio changes toward higher shares of renewable and storage resources. Fig. 6 represents a snapshot of the transition from the present to future grids, focusing on the reactive power supply. As depicted in the figure, there will be substantial changes in the paradigm of reactive power supply, as described next.

SGs in fossil fuel-burning plants will be decommissioned primarily except those equipped with CC systems. Some of the decommissioned plants will be repurposed as SCs.

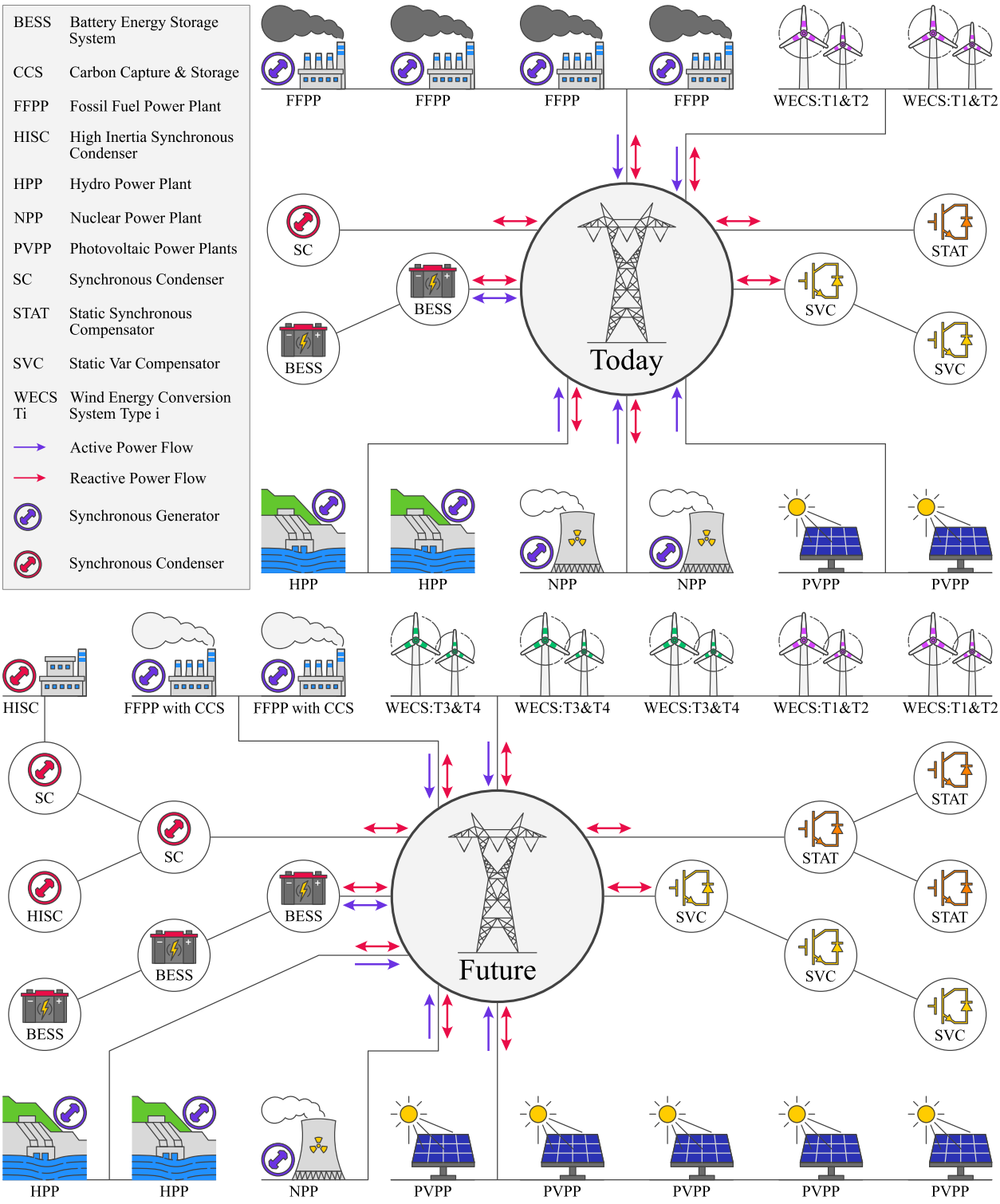



**Fig. 6.** ***Reactive power transition from today to future grids.***

As mentioned earlier, many retired SGs in fossil fuel-based plants are currently on the list of phasing out. These assets will be valuable options for retrofitting to SCs, especially new high-inertia versions, after minor modifications. This conversion is especially valuable as retiring power generation units can produce a system-wide reactive power deficit, considering the absence of previous central sources of reactive support. Significant advantages can be achieved from this transformation. These include reusing the current generator with enhanced reactive power capability and its foundation, structure, auxiliary systems, and connection equipment [68], [69].

**Table 1** Recent SC Installations

| Year | Country | Project | Capacity | Services |
|---|---|---|---|---|
| 2021 | Australia | Davenport project | 2×575 MVA | Inertia |
| 2021 | Australia | Robertstown project | 2×575 MVA | Inertia |
| 2022 | United Kingdon | Phoenix project in Neilson | 70 MVA + 70 MVA STATCOM | Blackstart |
| 2022 | Denmark | Faroe Islands | 8 MVA + 7.5 MVA Battery | 100% renewable penetration |
| 2022 | Ireland | Green Atlantic (Moneypoint) | 400 MVA + 130 Tones flywheel | Increasing wind penetration |
| 2023 | United Kingdon | Lister Driver in Liverpool | 2×67 MVar + 40 Tones flywheel | Inertia |
| 2024 | Australia | EnergyConnect | 4×120 MVA | Inertia |
| 2024 | United Kingdon | Rassau project in Wales | 100 MVA + 239 Tones flywheel | System stability |

Although the advent of FACTS devices, thanks to static power electronic merits, has relatively decreased the application of SCs for reactive power applications, they have recently been reintroduced to support the grid by offering inertia and enhanced short-circuit current in addition to reactive power support. Inertia provision and improved short-circuit capability comply with the future heavily IBR-integrated power grids lacking both features. The new high-inertia SC (HISC) designed for frequency regulation applications, in addition to reactive power support, is equipped with a built-in flywheel compared to conventional condensers. The HISCs' numerous merits comprise continuous reactive power control, enhanced overload capability, low-voltage ride through (VRT), inertial response, enhanced dynamic response, short-circuit contribution, minimal harmonic generation, the possibility to make them by repurposing retired SGs, and less space requirements (compared to FACTS) guarantee their deployment in future IBR-based grids [74], [75]. In February 2021, a leading manufacturer signed a contract with Statkraft, Europe's largest renewable energy producer, to create and install two high-inertia SCs at the Lister Drive Greener Grid Park in Liverpool, United Kingdom [76]. This project combines two 67 MVAr SCs with a 40-ton flywheel, enhancing instantaneous inertia by 3.5 times. The installed HISC provides more than 900 MW · s of inertia, accounting for approximately 1% of the United Kingdom's projected minimum total inertia requirement for 2025. The condensers were commissioned in March 2023. Table 1 shows a list of recent SC projects commissioned from 2021, denoting the increasing market share of this repurposed conventional but efficient technology, reintroduced for multiple grid services more than a reactive power source. As it can be observed from the table, the SCs are installed individually or hybridized with the other equipment. In the hybrid form, they are combined with a heavy flywheel to constitute an HISC or are combined with a battery or STATCOM to enhance the capabilities beyond the reactive power support [77].

The FACTS market has witnessed significant growth in recent years, with SVCs having the largest market share among other technologies. SVCs are widely recognized as critical power quality enhancement tools, providing reactive power and voltage control. Their applications are diverse, ranging from steady-state and dynamic voltage stabilization, oscillation damping, and load balancing. As a result, they are commonly deployed at critical substations, large industrial plants, and traction load centers. In contrast, STATCOMs, utilizing VSCs, are increasingly gaining popularity and surpassing SVCs in market share. The development of advanced controllable static switches, comprising GTO thyristors and IGBTs, has contributed to this trend. European market has seen a significant increase in the installation of STATCOMs over the past decade, with grid operators increasingly adopting this technology. This shift is attributed to the advantages of STATCOMs, including their ability to provide faster response times, higher power ratings, and improved flexibility compared to SVCs [78].

Amidst these changes, the role of mechanically switched devices, MSC and MSR, remains steadfast in future grids. Their simplicity and straightforward operation ensure low initial investment costs and reduced routine maintenance requirements, making them a cost-effective solution for grid voltage and reactive power control. As a result, these devices will continue to support grid stability and reliability without requiring significant modifications or upgrades. Table 2 shows a summary of the current and future status of the reactive power sources within the energy transition.

**Table 2** Current and Future Status of Reactive Power Sources

| Reactive Power Source | | Current Status | Future Status |
|---|---|---|---|
| Synchronous Generator | | Under decommissioning | Partly repurposed to SC or HISC |
| Synchronous Condenser | | Conventional SC | Mostly HISC and hybrid SC |
| FACTS Devices | SVC<br>STATCOM | Mostly SVC | STATCOM will dominate the market |
| Mechanically-Switch | MSC<br>MSR | Conventional | Conventional |
| Inverter-Based Resources | Wind Turbines<br>PV Panels<br>Battery Storage | Limited reactive power exchange | Primary source of reactive power |

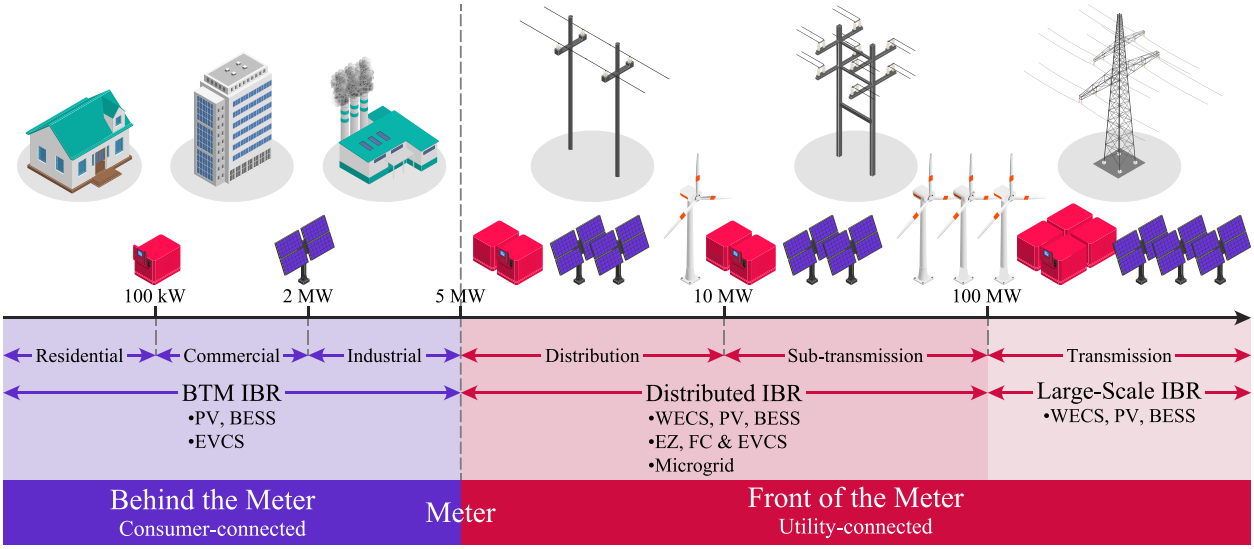



**Fig. 7.** ***Various IBRs integrated into the grid.***

## V. EXTENT AND SCALE OF IBR PENETRATION FROM A REACTIVE POWER VIEWPOINT

The future grids will undoubtedly be penetrated by a considerable share of renewable and storage resources in different scales and connection locations. Steadily increasing installation rates of these new resources during the past decade evidence the transition from today's fossil fuel-based generation to future grids relying primarily on IBRs. Although the reactive power share of these resources is currently inconsiderable, they will be the primary source of reactive power in future grids given their increasing dominant share [20]. Fig. 7 describes a classification of IBRs based on the meter connection side, grid connection location, plant scale, and asset technology [24], [77], [79], [80].

As illustrated in Fig. 7, the IBRs can be broadly categorized as behind-the-meter (BTM) and front-of-the-meter (FTM) resources. BTM resources primarily encompass relatively small-to-medium-sized PV and battery storage facilities connected on the customer side in residential, commercial, or industrial settings to deliver various services [81], [82]. They may also encompass electric vehicle charging stations (EVCSs) used for private charging purposes. The inverters used to integrate these resources are typically designed for active power transfer, especially in smaller plants. However, they are also capable of providing reactive power during idle periods of active power transfer or even at any time in larger plants. Leveraging the reactive power capabilities of BTM resources necessitates aggregation due to their small individual sizes and large numbers.

Distributed renewable and storage resources that are linked to the distribution or subtransmission network are positioned in the middle of the IBRs scale axis. These distributed IBRs typically have a capacity range in the hundreds of kilowatts to the tens of megawatts. WECSs, PVs, and BESSs are the key components of distributed IBRs and are primarily connected to medium-voltage distribution feeders and substations [83], [84]. Additionally, new types of distributed IBRs have emerged, comprising public EVCSs [85], water electrolyzers in power-to-gas plants [86], and fuel cells [87], which are now connected to the distribution grid. The control of inverters in these emerging technologies can offer grid operators a fresh source of reactive power, necessitating increased attention and detailed modeling studies. Charging stations are similar in nature to storage batteries. However, key differences such as control objectives, network connection, operating modes, and voltage levels make them offer different capabilities and dynamics. Some grid codes and requirements have considered the reactive power capability of EVCSs at different levels, for example, Energinet, the network operator of Denmark [88]. These requirements are similar to those set for battery storage systems. As a result of utilizing a full-rate inverter, the electrolyzers can exchange reactive power with the grid on both sides. In some cases, the grid operators consider them as a source of reactive power. However, in practice, these devices have mostly been used on the scales of 0.3–100 MW in order to provide frequency-based services in different modes. The functionality of the inverters used for integrating the fuel cells enables them to contribute to the reactive power support in the grid similar to the electrolyzers. In the IEEE-1547 standard, these resources can be placed in the distributed energy resource (DER) category, similar to a WECS or PV, and present the same behavioral characteristics in terms of reactive power capability [89].

At the same voltage and connection level, there may be some microgrids or community energy systems connected to the grid. Microgrids are self-contained systems

with a clearly defined electrical boundary consisting of loads, distributed generators (renewable and nonrenewable sources), and energy storage systems. These systems can disconnect from or connect to the main grid at a specific point of connection, known as the PCC, allowing them to operate independently or in coordination with the grid [90]. In the isolated mode, the microgrid must achieve a balance between sources and sinks separately for active and reactive power using the available devices within the system. In islanded ac microgrids, there can be issues with reactive power sharing due to the mismatched impedances of the feeders connecting the inverters and the load bus. This issue is particularly evident when utilizing the conventional frequency and voltage droop control (DRC) approach. It is worth noting that the high R/X ratio in microgrids makes it challenging to fully decouple frequency and voltage controls. In the context of IBR-based reactive power control in the utility grid, it is important to consider the reactive power capabilities of a grid-connected microgrid at the PCC. In the grid-connected mode, the microgrid can act aggregately as either a source or sink of reactive power at the connection point, enabling it to inject or absorb the necessary reactive power as required [91]. Depending on the size, a microgrid may be connected to the utility grid at the distribution or subtransmission level.

In recent years, notable strides have been made in the field of large-scale IBRs in the form of WECSs, PV installations, and BESSs primarily connected to the transmission network. These installations typically have a capacity range from tens to hundreds of megawatts, making them comparable to conventional power plants in terms of their reactive power capability. Both distributed and large-scale IBRs, comprising WECS, PV, and BESS plants, form a crucial foundation for providing reactive power in future grids, as they are becoming the dominant sources of the grid. Therefore, the subsequent sections will focus on these forms of IBRs. Specifically, these sections will explore the opportunities and challenges of deploying IBRs for reactive power support, analyze the analytical capability curves, examine grid codes, and address control concepts for distributed and large-scale WECS, PV, and BESS plants.

## VI. OPPORTUNITIES AND CHALLENGES OF REACTIVE POWER SUPPORT BY IBRs

Although leveraging the reactive power capability of IBRs is deemed crucial for future decarbonized grids, it presents a dual aspect by showcasing a range of challenges that must be addressed in conjunction with the abundant opportunities it provides [92], [93]. It is important to highlight that the challenges and opportunities vary when deploying a distributed IBR at the distribution network as opposed to a large-scale plant in the transmission network based on the structural differences and requirements of these two grids [94], [95]. The analysis conducted below delves into the benefits and drawbacks of utilizing these evolving reactive power sources in contrast to conventional ones, irrespective of the network type. Subsequently, an examination of the unique considerations, advantages, and limitations of harnessing reactive power of the IBRs within transmission and distribution networks is presented.

### A. Challenges

In addition to the necessary software and control modifications, the hardware adaptations needed in IBRs to utilize their reactive power capability are closely tied to the interface converter. Simply put, tapping into the reactive power potential of these sources mandates specific upgrades in both hardware and software within their converters. In essence, grid-connected converters can control reactive power without requiring extra investments owing to the predefined power electronic features. However, it is crucial to account for additional investment costs if the converter's capacity needs to be increased for enhanced reactive power capability. Typically, the manufacturers oversize converters to meet the requirements, especially grid codes. At the bottom of these upgrades, the apparent power rating of the converter must be enhanced. The incurred cost increase is becoming more relevant to the total system cost as a result of steadily decreasing other system equipment prices, especially PV panels, and increasing the share of power converters' costs [96], [97], [98], [99], [100], [101].

Currently, the power converters constitute the least reliable subsystem of renewable energy and storage systems. In PV plants, the converters possess the highest rates of failures, outages, and service calls for operation and maintenance [98], [100]. The reliability issues for wind turbines are also crucial, especially when moving from the onshore to the offshore sites. In wind farms, the power converter constitutes the highest failure rates compared to other system parts, i.e., turbine, gearbox, generator, and controls [99]. Accordingly, the increased converter size could translate into a growth in the total system failure rates and unavailability, besides an increased investment cost. Despite the lack of dedicated research, the above-mentioned adverse effects when supporting reactive power will also be the case for the converters utilized to integrate BESSs, i.e., increased converter cost and vulnerability [96], [97], [102], [103], [104], [105].

To supply reactive power, the converter's current and power losses will increase due to increased apparent power. The increased losses equate to the increased temperature of the semiconductor switches, which, in turn, diminishes the converter's lifetime and reliability. In an industrial PV-based system study, it has been shown that the reactive power supply by the inverter reduces the system reliability by 24.1% compared to the operation with a unity power factor [105]. These adverse effects will also be the case for the WECSs and BESSs. In other words, if the wind or battery conversion system supplies the reactive power, then the increased thermal stress as a result of increasing apparent power, current, and Joule

losses will decrease the converter's and the whole system's lifetime and reliability, besides increasing operation expenses. Finally, it is worth mentioning that a dedicated reactive power source is always installed at the desired location based on the system needs [96]. However, an IBR that can potentially function as a reactive power source may not be located at the desired site. This issue, which is a grid-type-dependent problem, is discussed in transmission and distribution network considerations.

## B. Opportunities

Although supporting reactive power by IBRs increases their converter cost and vulnerability besides some installation location limitations, they are far more competitive than installing individual reactive support devices. The superiorities of exploiting reactive power support of the IBRs compared to the conventional counterparts, especially SVCs and STATCOMs, are elaborated in the following.

*1) Installation Cost:* The increased investment, operation, and maintenance costs are negligible compared to the cost of new reactive support installations [101], [103]. It is shown that transforming an existing 5–100-MW solar PV system into a PV-STATCOM plant is about ten times more economical when compared to a new installation of equivalent-sized STATCOM [96]. Typically, the FACTS devices have higher overvoltage withstand capability and overcurrent rating than an IBR converter designed for extra reactive power support. If an IBR is not able to match the overcurrent criteria of the same-sized SVC or STATCOM, a derating factor may be considered for the IBR. A derated reactive-capable IBR will still be four times less costly than the STATCOM providing equivalent overload capability [96], [97], [98].

*2) Installation Time:* The time needed to transform an existing IBR into a reactive-capable plant is also significantly smaller than that needed for constructing a fresh equivalent-sized compensator, including the time required to obtain all regulatory and building permissions [96], [97], [98].

*3) Installation Footprint and Permissions:* The reactive support by IBRs is performed internally without additional need for footprint and installation time, which is a pivotal factor when planning reactive power sources. Avoiding new construction also prevents deforestation and eliminates associated greenhouse gas emissions (no vehicle rolls for moving people, equipment, and so on) besides various permissions, for example, right of way [96], [97], [98].

*4) Dynamic Reactive Power:* In addition to providing steady-state reactive power support, IBRs can also offer dynamic reactive power support during disturbances or faults. This dynamic support is crucial in grid applications where the system is vulnerable to transient events such as faults, lightning strikes, sudden fluctuations in load, or generation variability. The dynamic reactive power support by the IBRs could enhance VRT and fault ride-through (FRT) capabilities, enabling operators to effectively maintain grid stability even during challenging conditions [59].

## C. Transmission and Distribution Network Considerations

The communication infrastructure and advanced automation systems in transmission networks have significantly evolved, exemplified by the implementation of technologies such as supervisory control and data acquisition (SCADA), phasor measurement units (PMUs), and wide-area measurement systems (WAMSs). These facilities enable the development of centralized reactive power and voltage control systems, enhancing the accuracy of status monitoring across the grid [106], [107]. However, large-scale IBRs, increasingly connected at the transmission level, often do not occupy the exact physical locations as traditional fossil fuel power plants. This location shift complicates the provision of reactive power due to a new challenge: needing to be at the center of gravity of grid loads. This issue means increased reactive power losses in the grid lines due to the higher distance to the central point of reactive power demand. Furthermore, due to operational constraints, it may be challenging to fully utilize the reactive power capability of large IBRs, mainly when curtailment is necessary.

In contrast, the grids with lower voltages, especially distribution networks, typically have less sophisticated communication and automation infrastructure, making the control and monitoring of distributed IBRs more complex [108]. Despite these challenges, the distribution network benefits from a more decentralized approach to IBR installation and location. The small-scale nature of distributed IBRs allows for a wider distribution of reactive power resources, potentially fostering a more balanced voltage profile across the network. This situation would also be beneficial in terms of reducing losses and loading of the network equipment, i.e., lines and transformers.

Typically, the term "strength" or "weakness" at a given point in a power system is related to the ability to transfer power in a steady state while maintaining an adequate level of voltage and frequency stability. Nevertheless, it is a relative term that should be discussed together with the system characteristics at the given point of connection and the size of the generating system(s) to be connected. Therefore, traditionally, several terms such as short-circuit ratio (SCR) and X/R ratio have been used to quantify the strength of a connection point [109]. A higher X/R ratio indicates that the line is more inductive than resistive, which is typically the case for long-distance transmission networks. Typically, transmission lines have high X/R ratios (5)–(15) in which reactive power flows have a significant impact on the voltage control. Contrarily, distribution networks tend to have lower X/R ratios (e.g., 1–5) since they are shorter and have higher resistive components relative

**Table 3** Advantages and Disadvantages of Reactive Power Support From Distributed IBRs

| Advantages | | Disadvantages |
|---|---|---|
| **System Wide** | **Converter Level** | **Converter Level** |
| Lower system losses | No additional installation time | Higher investment cost |
| Lower equipment loading | No additional installation footprint | Higher internal losses |
| Smoother grid voltage profile | Lower total reactive support costs | Lower lifetime and reliability |

to their reactance. In distribution networks, voltage drops are more sensitive to active power than reactive power [110]. In transmission systems, especially in high loading conditions, voltages can drop significantly, making reactive power support vital. In distribution systems, loads can be highly variable (especially with the integration of BTM renewable resources, such as PV generations), necessitating more localized and real-time reactive power adjustments to maintain voltage. In transmission networks the primary objective of reactive power control is to ensure system-wide voltage stability and reliability across a large footprint; this is especially critical when power is transferred across long distances. In contrast, the primary goal of reactive power control in the distribution network is to maintain voltages at consumer connection points within regulatory limits and support local load variations with cost-effective solutions [110].

A significant aspect of reactive power supply through IBRs is their decentralized installation. This enhances their competitiveness as providing reactive power closer to the consumers offers various advantages over a more centralized approach employed by conventional reactive power suppliers [95]. Considering that reactive power travels long distances poorly, the geographically dispersed reactive support provided by IBRs offers a distinctive and flexible grid solution when effectively coordinated and optimized. This merit is also the case in distribution grids with shorter and radial feeders and with higher rates of IBR installations, e.g., distributed IBRs. Fig. 8(a) shows a typical passive radial distribution grid without embedded sources. According to the figure, the network has only one source, the upstream substation. The reactive power flow in this network is similar to active power and from top to bottom. The voltage will also vary according to the reactive power transferred from the substation to the last feeder buses. In other words, the bus voltages near the substation may experience an overvoltage. In contrast, the voltage decreases by moving away from the substation and closer to the feeders' end. Compensating the voltage drop in the buses at the end of feeders is quite challenging when increasing the transformer tap ratio due to the voltage rise limitation in the buses near the substation. As a result, the upstream part of the network will often have a rise in voltage while the downstream part voltage will drop. In this network, due to the radial nature of the reactive power flow, the resulting power losses may also be high because there is only one path for the reactive power. Also, the substation transformer may be overloaded since it supplies the total reactive power demanded. The condition for an active network with distributed IBRs supporting the reactive power is shown in Fig. 8(b). Accordingly, voltage control will be distributed by employing and optimizing the reactive power of distributed IBRs, which has several advantages. The first merit is the reduction of the reactive flow received from the substation and, as a result, the corresponding decrease in power losses. Reducing reactive power flow both in the substation and the network will also defer the expansion requirements. Additionally, the local exchange of reactive power in the network will flatten the voltage profile in the network without a significant increase or decrease in the bus voltages. As a result, the distribution grid would benefit from the IBRs from a reactive power support viewpoint. Table 3 summarizes the advantages and disadvantages of deploying distributed IBRs for reactive power support in a typical distribution grid.

## VII. REACTIVE POWER CAPABILITY OF WECS

Although there are various WECSs, they are mainly different regarding the two primary equipment: generator and power electronic converter. A wide range of WECS types can be achieved based on the combination of these two

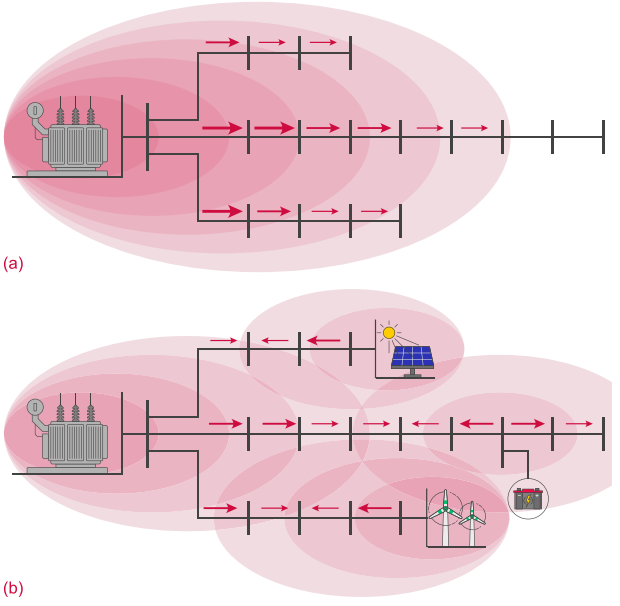



**Fig. 8.** *Reactive power flow in a radial distribution system. (a) Passive network and (b) active network integrating IBRs.*

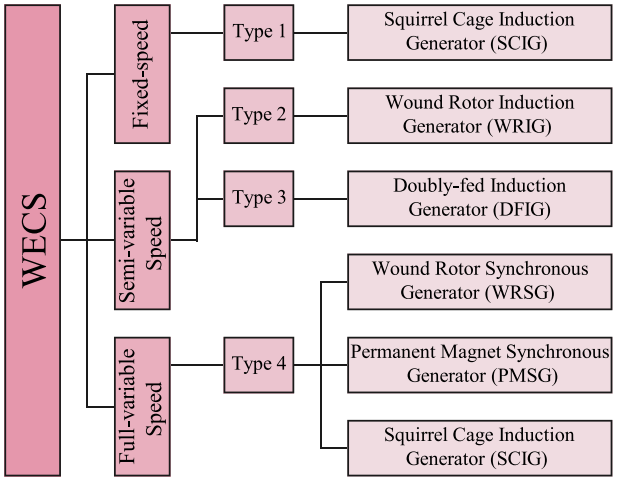



**Fig. 9.** ***Various types of WECSs.***

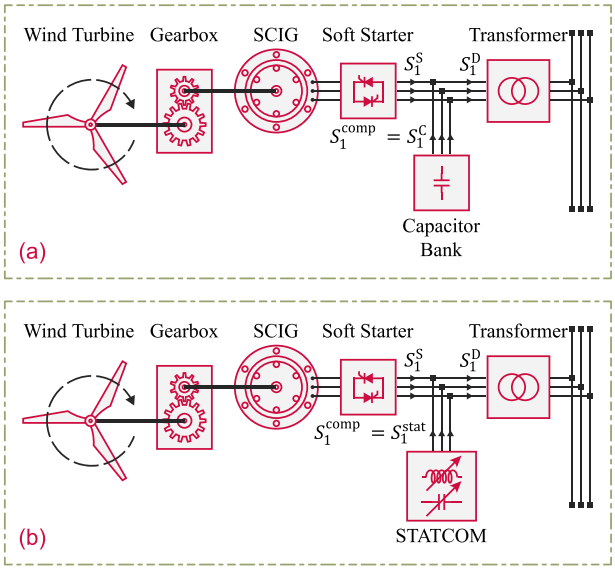



**Fig. 10.** ***WECS type 1. (a) With capacitor bank. (b) With STATCOM.***

parts. As shown in Fig. 9, the following four classes have been defined and commercialized to date [111], [112]. First, the general configuration of each WECS is introduced in the following. Then, the reactive power capability is elucidated from the corresponding equivalent circuits.

## A. WECS Type 1

This type is the oldest and possesses the simplest structure among WECSs. As shown in Fig. 10, in this type of WECS, a squirrel cage induction generator (SCIG), connected to the wind turbine through a gearbox, is the sole device used. The speed of the generator is allowed to change by about 1% of the rated synchronous speed, which is the reason behind calling this WECS a fixed speed [111]. The SCIG can produce active power when the slip is negative. However, there is always a reactive power consumption since it has no independent dc excitation. Since there is no power converter at the generator's output, there is practically no control over the reactive power. As a result, to comply with the grid codes, a shunt capacitor, or STATCOM, must be used in the output. Also, the absence of a power converter, in addition to reducing the efficiency, causes injecting the negative impacts of wind speed changes into the grid. Conversely, anomalies on the grid side will result in mechanical stress on the turbine components. A soft starter may be used at the generator's output to prevent a high transient starting current. This WECS type currently accounts for a high share of installed turbines and continues to operate. However, they are no longer being manufactured or installed due to the above-mentioned drawbacks. These drawbacks have prompted a shift in the industry to newer turbine types designed to overcome the limitations of this type of WECS, discussed in the following [112], [113].

In the following, subscript (i) stands for "WECS type $i$." Also, superscripts (S), (D), and (comp) denote, in turn, stator, total delivered, and compensator power, respectively. As mentioned above, for Type 1 WECS, for any value of produced active power ($P_1^S$), the produced reactive power ($Q_1^S$) is always negative. Therefore, shunt compensating equipment must be installed in the SCIG output terminal to reduce the reactive power received from the grid. According to Fig. 10, the delivered apparent power ($S_1^D$) will be considered as the sum of the power of the stator side ($S_1^S$) and the power of the reactive power compensator ($S_1^{comp}$) [112]

$$S_1^D = S_1^S + S_1^{comp}. \tag{1}$$

Also, according to the SCIG equivalent circuit [Fig. 11(a)], the transferred power from the stator terminals will be defined as a function of the stator terminal voltage ($V_1^S$) and the stator current ($I_1^S$) [113], [114]

$$S_1^S = P_1^S - jQ_1^S = V_1^S I_1^S. \tag{2}$$

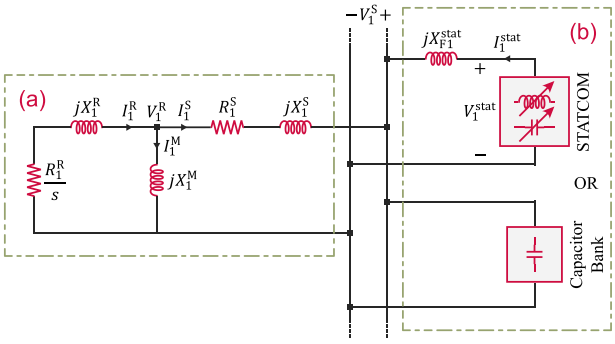



**Fig. 11.** ***(a) SCIG equivalent circuit. (b) Compensators (only one of them is used).***

Equation (2) expresses the limitation of the stator current. Additionally, in order to express the rotor current limitation, it can be written as follows:

$$-R_1^{\mathrm{S}} I_1^{\mathrm{S}} - jX_1^{\mathrm{S}} I_1^{\mathrm{S}} - V_1^{\mathrm{S}} + jX_1^{\mathrm{M}} I_1^{\mathrm{M}} = 0 \tag{3}$$

$$I_1^{\mathrm{M}} = I_1^{\mathrm{R}} - I_1^{\mathrm{S}} \tag{4}$$

where $X_1^{\mathrm{M}}$, $X_1^{\mathrm{S}}$, and $R_1^{\mathrm{S}}$ are magnetization reactance, leakage reactance, and SCIG stator resistance, respectively. Also, $I_1^{\mathrm{R}}$ and $I_1^{\mathrm{M}}$ represent, in turn, the generator's rotor current and magnetizing current [113], [114].

By substituting $I_1^{\mathrm{M}}$ from (4) in (3)

$$I_1^{\mathrm{S}} = \frac{jX_1^{\mathrm{M}} I_1^{\mathrm{R}} - V_1^{\mathrm{S}}}{R_1^{\mathrm{S}} + j\left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)}. \tag{5}$$

By substituting (5) in (2)

$$P_1^{\mathrm{S}} - jQ_1^{\mathrm{S}} = V_1^{\mathrm{S}} \left( \frac{jX_1^{\mathrm{M}} I_1^{\mathrm{R}} - V_1^{\mathrm{S}}}{R_1^{\mathrm{S}} + j\left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)} \right). \tag{6}$$

By simple mathematical manipulations, (6) results in the following equation:

$$P_1^{\mathrm{S}} - j\left( Q_1^{\mathrm{S}} + V_1^{\mathrm{S}} \left( \frac{R_1^{\mathrm{S}} X_1^{\mathrm{M}} I_1^{\mathrm{R}} + V_1^{\mathrm{S}} \left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)}{\left(R_1^{\mathrm{S}}\right)^2 + \left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)^2} \right) \right) = V_1^{\mathrm{S}} \left( \frac{-V_1^{\mathrm{S}} R_1^{\mathrm{S}} + X_1^{\mathrm{M}} I_1^{\mathrm{R}} \left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)}{\left(R_1^{\mathrm{S}}\right)^2 + \left(X_1^{\mathrm{S}} + X_1^{\mathrm{M}}\right)^2} \right). \tag{7}$$

The stator resistance ($R_1^{\mathrm{S}}$) can be ignored to reduce the complexity of (7) as follows [113]:

$$P_1^{\mathrm{S}} - j\left( Q_1^{\mathrm{S}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}} \right) = \frac{X_1^{\mathrm{M}} V_1^{\mathrm{S}} I_1^{\mathrm{R}}}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}}. \tag{8}$$

In order to achieve the rotor voltage limit, (3) can be rewritten as follows:

$$-R_1^{\mathrm{S}} I_1^{\mathrm{S}} - jX_1^{\mathrm{S}} I_1^{\mathrm{S}} - V_1^{\mathrm{S}} + V_1^{\mathrm{R}} = 0. \tag{9}$$

Thus, it can be written that

$$I_1^{\mathrm{S}} = \frac{V_1^{\mathrm{R}} - V_1^{\mathrm{S}}}{R_1^{\mathrm{S}} + jX_1^{\mathrm{S}}}. \tag{10}$$

By substituting (10) in (2)

$$P_1^{\mathrm{S}} - jQ_1^{\mathrm{S}} = V_1^{\mathrm{S}} \left( \frac{V_1^{\mathrm{R}} - V_1^{\mathrm{S}}}{R_1^{\mathrm{S}} + jX_1^{\mathrm{S}}} \right). \tag{11}$$

By ignoring the stator resistance ($R_1^{\mathrm{S}}$)

$$P_1^{\mathrm{S}} - j\left( Q_1^{\mathrm{S}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}}} \right) = j\frac{V_1^{\mathrm{S}} V_1^{\mathrm{R}}}{X_1^{\mathrm{S}}}. \tag{12}$$

If a capacitor bank is used in the output terminal of the generator, the total injected reactive power will be

$$S_1^{\mathrm{comp}} = -jQ_1^{\mathrm{C}} = j\omega C \left(V_1^{\mathrm{S}}\right)^2 \tag{13}$$

where $C$ is the capacitor bank capacitance and $\omega$ is the angular frequency. Accordingly, in the case of using a capacitor bank, the total exchanged active/reactive power of a WECS Type 1 will be obtained by using (1) and substituting (13) in (2), (8), and (12) as follows:

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + \omega C \left(V_1^{\mathrm{D}}\right)^2 \right) = V_1^{\mathrm{S}} I_1^{\mathrm{S}} \tag{14}$$

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}} + \omega C \left(V_1^{\mathrm{S}}\right)^2 \right) = \frac{X_1^{\mathrm{M}} V_1^{\mathrm{S}} I_1^{\mathrm{R}}}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}} \tag{15}$$

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}}} + \omega C \left(V_1^{\mathrm{S}}\right)^2 \right) = j\frac{V_1^{\mathrm{S}} V_1^{\mathrm{R}}}{X_1^{\mathrm{S}}}. \tag{16}$$

Fig. 12 (top left) shows the capability curve based on (14)–(16) for a generic 2-MW turbine compensated by a capacitor bank. As the red dashed area shows, reactive power capability will remain in the negative range before compensation. After using the capacitor bank, this area shifts upward as a function of the added reactive compensation.

If the STATCOM is used in the SCIG terminals instead of a capacitor bank, the power delivered to the grid is the sum of the SCIG and the STATCOM power. The STATCOM connection configuration is shown in Fig. 11(b), and its reactive power is limited by the following relation [114]:

$$S_1^{\mathrm{comp}} = -jQ_1^{\mathrm{stat}} = V_1^{\mathrm{stat}} I_1^{\mathrm{stat}} \tag{17}$$

where $Q_1^{\mathrm{stat}}$, $V_1^{\mathrm{stat}}$, and $I_1^{\mathrm{stat}}$ are reactive power, terminal voltage, and current of STATCOM, respectively. Also

$$-jQ_1^{\mathrm{stat}} = \frac{m_a V_1^{\mathrm{dc}}}{2} I_1^{\mathrm{stat}} \tag{18}$$

where $V_1^{\mathrm{dc}}$ and $m_a$ are the dc-link voltage and controller modulation factor, respectively. According to Fig. 10(b)

$$V_1^{\mathrm{stat}} - jX_{\mathrm{F1}}^{\mathrm{stat}} I_1^{\mathrm{stat}} - V_1^{\mathrm{S}} = 0 \tag{19}$$

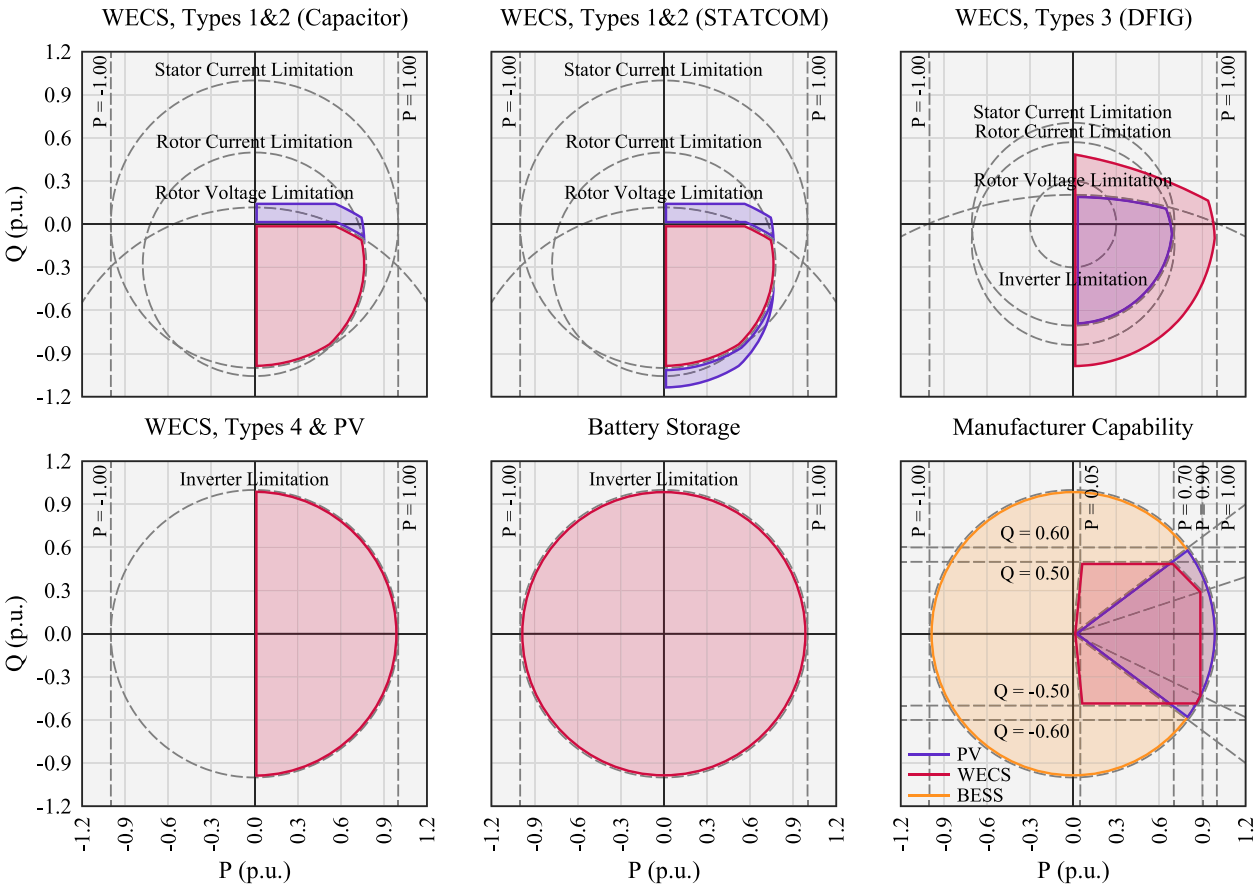



**Fig. 12.** ***Analytical capability curve of IBRs and a sample of manufacturer capability.***

where $X_{\mathrm{F1}}^{\mathrm{stat}}$ is the reactance of the STATCOM output filter. From (18), it can be written as follows:

$$I_1^{\mathrm{stat}} = \frac{V_1^{\mathrm{stat}} - V_1^{\mathrm{S}}}{jX_{\mathrm{F1}}^{\mathrm{stat}}}. \tag{20}$$

By replacing (20) in (18), the reactive power absorbed or injected by STATCOM is obtained as follows:

$$Q_1^{\mathrm{stat}} = m_a V_1^{\mathrm{dc}} \left( \frac{V_1^{\mathrm{stat}} - V_1^{\mathrm{S}}}{2X_{\mathrm{F1}}^{\mathrm{stat}}} \right). \tag{21}$$

Accordingly, in the case of compensation utilizing the STATCOM, the total exchanged active/reactive power of a WECS Type 1 will be obtained by using (1) and substituting (21) in (2), (8), and (12) as follows:

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + m_a V_1^{\mathrm{dc}} \left( \frac{V_1^{\mathrm{stat}} - V_1^{\mathrm{S}}}{2X_{\mathrm{F1}}^{\mathrm{stat}}} \right) \right) = V_1^{\mathrm{S}} I_1^{\mathrm{S}} \tag{22}$$

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}} + m_a V_1^{\mathrm{dc}} \left( \frac{V_1^{\mathrm{stat}} - V_1^{\mathrm{S}}}{2X_{\mathrm{F1}}^{\mathrm{stat}}} \right) \right) = \frac{X_1^{\mathrm{M}} V_1^{\mathrm{S}} I_1^{\mathrm{R}}}{X_1^{\mathrm{S}} + X_1^{\mathrm{M}}} \tag{23}$$

$$P_1^{\mathrm{D}} - j\left( Q_1^{\mathrm{D}} + \frac{\left(V_1^{\mathrm{S}}\right)^2}{X_1^{\mathrm{S}}} + m_a V_1^{\mathrm{dc}} \left( \frac{V_1^{\mathrm{stat}} - V_1^{\mathrm{S}}}{2X_{\mathrm{F1}}^{\mathrm{stat}}} \right) \right) = j\frac{V_1^{\mathrm{S}} V_1^{\mathrm{R}}}{X_1^{\mathrm{S}}}. \tag{24}$$

Fig. 12 (top middle) shows the capability curve based on (22)–(24) for a generic 2-MW turbine compensated by the STATCOM. If the STATCOM is used, this area is shifted in both negative and positive directions of the vertical axis since the STATCOM compensates for the reactive power in both directions. The reactive power capability for this compensation mode includes the purple region resulting from the shift in two directions.

## B. WECS Type 2

In this type of WECS, to increase flexibility in power conversion, a wound rotor induction generator (WRIG) is used instead of a generator with a squirrel cage rotor, as depicted in Fig. 13. The WRIG enables changing the generator torque-speed characteristic curve using the control of the rotor winding resistance, and the possibility of extracting energy at a variable speed is provided in a limited way. The resistance of the rotor circuit is typically manipulated by a power converter consisting of a chopper and a diode [114]. It should be noted that the speed

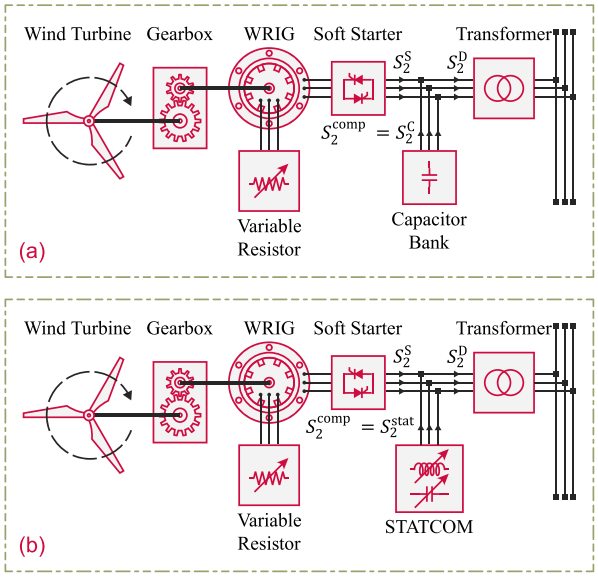



**Fig. 13.** *WCES type 2. (a) With parallel capacitor. (b) With STATCOM.*

change range is 10% of the nominal speed [115]. With this capability, the generator can set its output power in the same range of 10%, and more active power can be extracted from wind energy compared to a WECS Type 1.

However, losses in winding resistance will be much higher. This type of WECS can improve the dynamic response during grid disturbances. From Fig. 13, it is evident that this setup requires the use of a reactive power compensation device (shunt capacitor or STATCOM) in addition to a gearbox and soft starter. This WECS type is also becoming obsolete due to limitations in power control, the need for reactive power compensating equipment, and low efficiency in wind energy conversion. Since the equivalent circuits of WRIG and SCIG are considered the same, (18)–(21) will be expressed without change. Therefore, to present the reactive power capability of this WECS, Fig. 13 is sufficient. If using a capacitor bank [114], [115]

$$P_2^D - j\left(Q_2^D + \omega C\left(V_2^S\right)^2\right) = V_2^S I_2^S \tag{25}$$

$$P_2^D - j\left(Q_2^D + \frac{\left(V_2^S\right)^2}{X_2^S + X_2^M} + \omega C\left(V_2^S\right)^2\right) = \frac{X_2^M V_2^S I_2^R}{X_2^S + X_2^M} \tag{26}$$

$$P_2^D - j\left(Q_2^D + \frac{\left(V_2^S\right)^2}{X_2^S} + \omega C\left(V_2^S\right)^2\right) = j\frac{V_2^S V_2^R}{X_2^S}. \tag{27}$$

Similarly, by utilizing the STATCOM [116]

$$P_2^D - j\left(Q_2^D + m_a V_2^{dc}\left(\frac{V_2^{stat} - V_2^S}{2X_{F2}^{stat}}\right)\right) = V_2^S I_2^S \tag{28}$$

$$P_2^D - j\left(Q_2^D + \frac{\left(V_2^S\right)^2}{X_2^S + X_2^M} + m_a V_2^{dc}\left(\frac{V_2^{stat} - V_2^S}{2X_{F2}^{stat}}\right)\right) = \frac{X_2^M V_2^S I_2^R}{X_2^S + X_2^M} \tag{29}$$

$$P_2^D - j\left(Q_2^D + \frac{\left(V_2^S\right)^2}{X_2^S} + m_a V_2^{dc}\left(\frac{V_2^{stat} - V_2^S}{2X_{F2}^{stat}}\right)\right) = j\frac{V_2^S V_2^R}{X_2^S}. \tag{30}$$

As a result, the resulting capability curve for this type will be quite similar to the previous one depicted in Fig. 12 (top left and middle).

## C. WECS Type 3

In a Type 3 WECS, which is another variant of semi-variable speed WECSs, a doubly fed induction generator (DFIG) is employed. In DFIG, as illustrated in Fig. 14, both the stator and the rotor are connected to the grid through a converter. However, the rotor connection is partial, with a capacity of 30% of the rated power of the generator. This configuration contributes to the dominance of WECS Type 3 in the wind industry today, holding a market share of around 50%. However, the FRT capability of DFIG is limited due to the small-scale power converter [102], [115], [117].

As mentioned, about 30% of the total output power of WECS Type 3 passes via the converter. As in Fig. 14, the delivered apparent power ($S_3^D$) is equal to the sum of the power of the stator side ($S_3^S$) and the power of the grid-side converter (GSC), i.e., ($S_3^{GSC}$) [118], [119]:

$$S_3^D = S_3^S + S_3^{GSC}. \tag{31}$$

Also, according to the DFIG equivalent circuit [Fig. 15(a)], the transmitted power from the stator terminals will be defined as follows:

$$S_3^S = P_3^S - jQ_3^S = V_3^S I_3^S. \tag{32}$$

This equation expresses the limitation of the stator current, where $V_3^S$ and $I_3^S$ represent the voltage and current of the DFIG stator line, respectively. By applying Kirchhoff's voltage law (KVL) and Kirchhoff's current law (KCL)

$$-R_3^S I_3^S - jX_3^S I_3^S - V_3^S + jX_3^M I_3^M = 0 \tag{33}$$

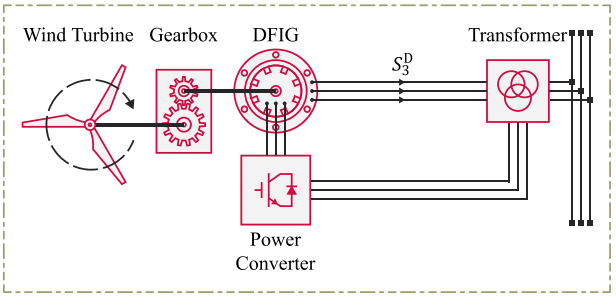



**Fig. 14.** *WCES type 3.*

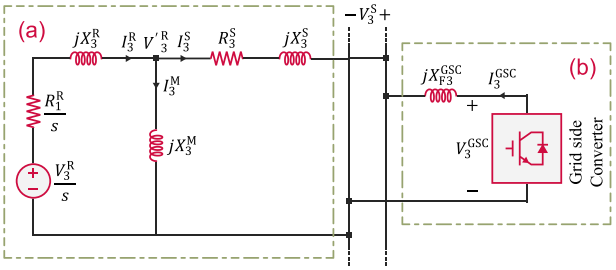



**Fig. 15.** *(a) DFIG equivalent circuit. (b) Grid-side converter.*

$$I_3^{\mathrm{M}} = I_3^{\mathrm{R}} - I_3^{\mathrm{S}} \tag{34}$$

where $X_3^{\mathrm{M}}$, $X_3^{\mathrm{S}}$, and $R_3^{\mathrm{S}}$ are the magnetization reactance, leakage reactance, and stator resistance of DFIG, respectively. Also, $I_3^{\mathrm{R}}$ and $I_3^{\mathrm{M}}$ represent the generator's rotor current and magnetizing current, respectively. By substituting the value of $I_1^{\mathrm{M}}$ from (34) in (33) [116], [118], [119]

$$I_3^{\mathrm{S}} = \frac{jX_3^{\mathrm{M}} I_3^{\mathrm{R}} - V_3^{\mathrm{S}}}{R_3^{\mathrm{S}} + j\left(X_3^{\mathrm{S}} + X_3^{\mathrm{M}}\right)}. \tag{35}$$

By substituting (35) in (32)

$$P_3^{\mathrm{S}} - jQ_3^{\mathrm{S}} = V_3^{\mathrm{S}}\left(\frac{jX_3^{\mathrm{M}} I_3^{\mathrm{R}} - V_3^{\mathrm{S}}}{R_3^{\mathrm{S}} + j\left(X_3^{\mathrm{S}} + X_3^{\mathrm{M}}\right)}\right). \tag{36}$$

To avoid the complexity of equations, the stator resistance ($R_3^{\mathrm{S}}$) is ignored

$$P_3^{\mathrm{S}} - j\left(Q_3^{\mathrm{S}} + \frac{\left(V_3^{\mathrm{S}}\right)^2}{X_3^{\mathrm{S}} + X_3^{\mathrm{M}}}\right) = \frac{X_3^{\mathrm{M}} V_3^{\mathrm{S}} I_3^{\mathrm{R}}}{X_3^{\mathrm{S}} + X_3^{\mathrm{M}}}. \tag{37}$$

Equation (37) expresses the limitation imposed by the rotor current. In order to express the rotor voltage limit

$$-R_3^{\mathrm{S}} I_3^{\mathrm{S}} - jX_3^{\mathrm{S}} I_3^{\mathrm{S}} - V_3^{\mathrm{S}} + V_3^{'\mathrm{R}} = 0 \tag{38}$$

and

$$I_3^{\mathrm{S}} = \frac{V_3^{\mathrm{R}} - V_3^{\mathrm{S}}}{R_3^{\mathrm{S}} + jX_3^{\mathrm{S}}}. \tag{39}$$

By substituting (39) in (32) and ignoring the rotor resistance ($R_3^{\mathrm{S}}$), it can finally be written as

$$P_3^{\mathrm{S}} - jQ_3^{\mathrm{S}} = V_3^{\mathrm{S}}\left(\frac{V_3^{\mathrm{R}} - V_3^{\mathrm{S}}}{R_3^{\mathrm{S}} + jX_3^{\mathrm{S}}}\right) \tag{40}$$

and

$$P_3^{\mathrm{S}} - j\left(Q_3^{\mathrm{S}} + \frac{\left(V_3^{\mathrm{S}}\right)^2}{X_3^{\mathrm{S}}}\right) = \frac{V_3^{\mathrm{S}} V_3^{\mathrm{R}}}{X_3^{\mathrm{S}}}. \tag{41}$$

Therefore, the output power on the generator side is limited by (32), (37), and (40) [120]. Furthermore, the power on the grid-side converter depends on its capacity ($S_3^{\mathrm{GSC}}$) and the transmission capability of the filter installed on GSC

$$S_3^{\mathrm{GSC}} = P_3^{\mathrm{GSC}} - jQ_3^{\mathrm{GSC}} = V_3^{\mathrm{GSC}} I_3^{\mathrm{GSC}} \tag{42}$$

where $P_3^{\mathrm{GSC}}$, $Q_3^{\mathrm{GSC}}$, $V_3^{\mathrm{GSC}}$, and $I_3^{\mathrm{GSC}}$ are active power, reactive power, terminal voltage, and GSC current, respectively. Therefore, the reactive power limit of WECS Type 3 will be expressed by (32), (37), (40), and (42), as depicted in Fig. 12 (top right). As can be seen, the sharing of the ranges defined by (32)–(40) is determined by the red dashed line. By adding the power range of the inverter (yellow line) to this area, the power capability of the DFIG, which is limited to the purple area, is determined.

### D. WECS Type 4

In this type of WECS, the total power produced by the generator is delivered to the grid through an electronic power converter. Therefore, the converter's capacity is 100% of the generator's rated power, which increases the overall system's size, cost, and complexity. With this converter, the generator is completely disconnected from the grid and can work in the entire speed range (0%–100%). The power converter allows the system to perform reactive power compensation and ensures a seamless connection to the grid. Wound rotor SG (WRSG), permanent magnet SG (PMSG), and SCIG are used as the generators in this WECS type (Fig. 16(a)–(c), respectively). The wind energy conversion efficiency in this WECS is higher than that of other types. High levels of FRT capability are also accessible without requiring any additional equipment. Additionally, utilizing a WRSG or PMSG with more winding poles can eliminate the need for a gearbox. This configuration demonstrates enhanced robustness against power system faults compared to other alternatives [102], [112], [114].

As mentioned before, the generator used in this WECS can be one of the WRSG, PMSG, or SCIG. Regardless of the generator type, these generators' stator is connected to the grid through a full-scale power converter. All the generated power is transferred through this converter, and its capacity is at least equal to the generator's capacity. The delivered power of WECS Type 4 to the grid ($S_4^{\mathrm{D}}$) or the power of the grid-side converter ($S_4^{\mathrm{GSC}}$) is limited by the rated capacity of the grid-side converter

$$S_4^{\mathrm{D}} = S_4^{\mathrm{GSC}} = P_4^{\mathrm{GSC}} - jQ_4^{\mathrm{GSC}} = V_4^{\mathrm{GSC}} I_4^{\mathrm{GSC}} \tag{43}$$

where $P_4^{\mathrm{GSC}}$, $Q_4^{\mathrm{GSC}}$, $V_4^{\mathrm{GSC}}$, and $I_4^{\mathrm{GSC}}$ are active power, reactive power, terminal voltage, and GSC current in WESC Type 4, respectively. The corresponding reactive power capability curve is demonstrated in Fig. 12 (bottom left),

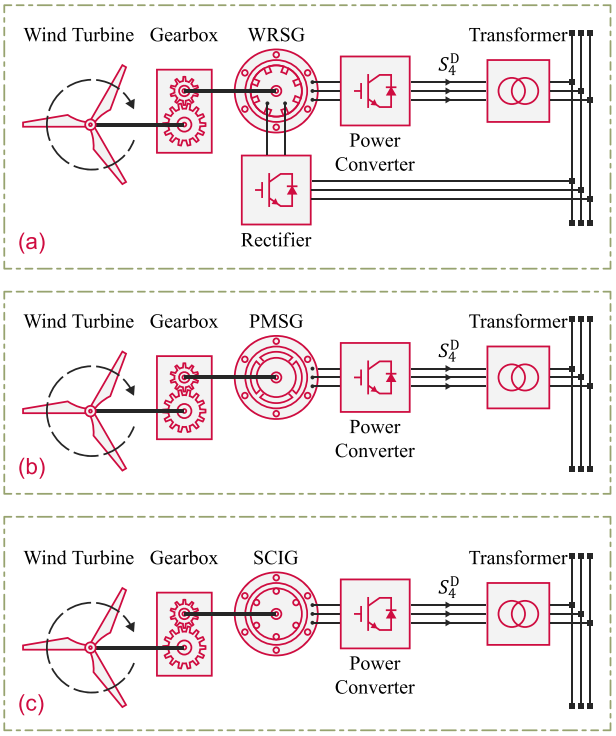



**Fig. 16.** *WCES Type 4. (a) WRSG. (b) PMSG. (c) SCIG.*

considering that the entire turbine output power is delivered via the converter.

## VIII. REACTIVE POWER CAPABILITY OF PVs

The PV inverters have the same technological design as WECS Type 4 and are increasingly employed with a similar reactive power capability. Accordingly, this feature is provided to consumers, even in low-capacity inverters. Like WECSs, PV inverters are designed to operate between 90% and 110% of the rated terminal voltage. The inverter's reactive power capability differs depending on the terminal voltage [121]. In addition, the dc input voltage of the inverter generated by PV panels can also affect the reactive power capability. Most PV inverters can absorb or inject reactive power when needed if they do not exceed the rated current and terminal voltage of the inverter. Like STATCOM, PV inverters may support reactive power exchange even at zero active power production, i.e., at night [122]. In practice, delivering power to the grid ($S_{\mathrm{PV}}^{\mathrm{D}}$) is limited by the rated capacity of the grid-side converter ($S_{\mathrm{PV}}^{\mathrm{GSC}}$)

$$S_{\mathrm{PV}}^{D} = S_{\mathrm{PV}}^{\mathrm{GSC}} = P_{\mathrm{PV}}^{\mathrm{GSC}} - jQ_{\mathrm{PV}}^{\mathrm{GSC}} = V_{PV}^{GSC} I_{PV}^{GSC}. \quad (44)$$

The plant uses $P_{\mathrm{PV}}^{\mathrm{GSC}}$, $Q_{\mathrm{PV}}^{\mathrm{GSC}}$, $V_{\mathrm{PV}}^{\mathrm{GSC}}$, and $I_{\mathrm{PV}}^{\mathrm{GSC}}$ as active power, reactive power, terminal voltage, and GSC current, respectively. The corresponding reactive power capability curve for a generic plant is shown in Fig. 12 (bottom left).

## IX. REACTIVE POWER CAPABILITY OF BESSs

A BESS shares similarities with other inverter-based power sources, particularly PV. Their controllers typically consist of current regulation loops, phase-locked loops, dc bus voltage regulation loops, battery charge/discharge control loops, active power–frequency controls, reactive power–voltage controls, and plant-level controls [123]. Unlike other inverter-based generation resources, such as WECSs and PVs, BESS serves as equipment capable of transmitting active power in both directions [17], [123]. Consequently, a BESS's power capability curve forms a complete four-quadrant curve or an entire circle, as demonstrated in Fig. 12 (bottom middle) [124]. It is essential to note that in practical applications, a BESS might possess varying maximum active power ranges for charging and discharging modes. The reactive power range may also be different when working in inductive or capacitive modes. In practice, the power delivered by BESS to the grid ($S_{\mathrm{BESS}}^{\mathrm{D}}$) is limited by the nominal capacity of the grid-side converter ($S_{\mathrm{BESS}}^{\mathrm{GSC}}$):

$$S_{\mathrm{BESS}}^{\mathrm{D}} = S_{\mathrm{BESS}}^{\mathrm{GSC}} = P_{\mathrm{BESS}}^{\mathrm{GSC}} - jQ_{\mathrm{BESS}}^{\mathrm{GSC}} = V_{\mathrm{BESS}}^{\mathrm{GSC}} I_{\mathrm{BESS}}^{\mathrm{GSC}} \quad (45)$$

where $P_{\mathrm{BESS}}^{\mathrm{GSC}}$, $Q_{\mathrm{BESS}}^{\mathrm{GSC}}$, $V_{\mathrm{BESS}}^{\mathrm{GSC}}$, and $I_{\mathrm{BESS}}^{\mathrm{GSC}}$ are active power, reactive power, terminal voltage, and converter current in BESS, respectively.

The bottom-right part of Fig. 12 demonstrates the capability curves provided by the sample manufacturers comparable to the analytically derived curves for various IBRs, including WECS, PV, and BESS. As illustrated, the actual capability curve of the battery is characterized by a complete four-quadrant area or, equivalently, a full circular shape [125]. Notably, this curve closely resembles the analytical curve in the middle lower plot, indicating a strong correlation between the analytical modeling and the actual setup. In contrast, the capability curve of the PV inverter, as supplied by the manufacturer, is shaped like a sector, revealing limitations in providing reactive power when the active power output is zero. This observation denotes that this PV inverter necessitates either supplementary reactive power compensation or reconfiguring its inverter circuits to enhance its reactive power capability at zero active power output [126]. Finally, the figure also features the reactive power capability of a Type 3 WECS, as provided by a sample manufacturer. This polygonal curve indicates that the turbine does not support reactive power when the active power output falls below 0.05 of its rated capacity [127].

## X. GRID CODES AND CONNECTION REQUIREMENTS

A grid code provides a set of guidelines that govern the participation of various stakeholders, including power generators, consumers, energy storage facilities, and other assets. By implementing these codes, system operators can ensure

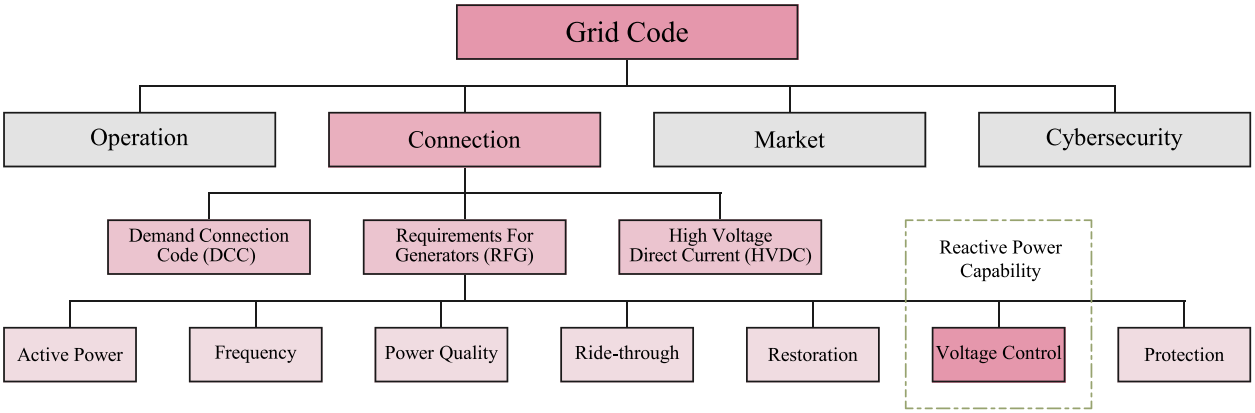



**Fig. 17.** ***Sample of a comprehensive grid code (ENTSO-E) and the role of reactive power capability.***

that all connected assets operate in a way that maintains the integrity and reliability of the electricity supply [128]. Developing a grid code is essential for encouraging private sector involvement in the power industry and facilitating the integration of IBRs into the system. The purpose of grid codes is multifaceted, encompassing aspects such as coordination among stakeholders, increased transparency, grid security, reliability, and seamless integration of emerging energy solutions [129]. Grid codes cover various power system aspects, including market structures, operational procedures, planning frameworks, and connection protocols. A notable example of a well-developed grid code is European Union network codes [15]. The code, a collection of documents, was developed by European Network of Transmission System Operators (ENTSO-E) to promote the coordination, integration, and optimization of European electricity market. Fig. 17 shows the ENTSO-E grid code categorized into operation, connection, market, and cybersecurity [130].

As per the scope of this article, the focus is on the network connection grid codes or equivalently connection requirements. As can be observed from Fig. 17, the connection codes for ENTSO-E, as an example, include demand connection code (DCC), requirements for generators (RFGs), and high-voltage direct current (HVdc) codes [130]. The IBRs connection to the grid falls into the RFG category based on their role in the grid. The connection grid codes cover a wide range of technical requirements, including active power, frequency, ride through, power quality, protection, restoration, and voltage control. The primary tool for voltage control in the connection grid code is reactive power capability provided by the generators' capability curve. This section focuses on grid connection codes concerning an IBR's reactive power capability and elaborates on the technical criteria an IBR must meet to be granted permission to connect to the grid. The capability of an IBR to provide reactive support is defined in almost all codes or the codes have been updated recently to include this feature. It is worth noting that the codes are different for various jurisdictions and connected grids and are updating continuously because of differences in [15], [77], [128], and [131].

1) *Regulatory framework:* Each country or area has its regulatory framework, which influences the development of grid codes. For example, some countries may have more stringent regulations or incentives for renewable energy, leading to more stringent grid code requirements.
2) *Grid infrastructure:* The size and scale of the grid in each country can affect grid code requirements. Larger grids may require more complex and sophisticated grid codes to ensure reliable operation. Countries with older or more complex grids may also require different grid codes to ensure reliable and stable operation.
3) *Renewable energy penetration:* The level of renewable energy penetration in a country's power mix affects grid code requirements. Countries with higher shares of renewable energy may require more stringent grid codes to ensure reliable and stable operation.
4) *Grid operator's expertise and experience:* The experience and expertise of the grid operator, including TSO and DSO, can influence the development of grid code requirements. For example, a TSO with extensive experience integrating renewable energy sources may develop more stringent grid codes.
5) *Economic and environmental considerations:* Grid code requirements can also be influenced by economic and environmental considerations, such as the cost of implementing new technologies or greenhouse gas reduction expenses.
6) *Cultural and historical context:* It can also influence grid code requirements. For example, some countries may prioritize reliability over flexibility or have specific requirements for decentralized or off-grid systems.
7) *National priorities and policy goals:* National priorities and policy goals, such as energy security, sustainability, or economic development, can influence the development of grid code requirements.

The following elaborates on area-based grid codes with a range of more than a country first. Then, the grid connection requirements for the selected countries with a dominant IBR penetration and integration are discussed. These codes are the most essential means for network operators to exploit the reactive power support of the IBRs.

### A. IEEE-1547

IEEE standards are industry guidelines that are not mandatory, and their implementation requires adoption by the relevant regional authority responsible for interconnection requirements. The IEEE has issued two grid codes related to IBRs connection. The first, IEEE-1547, focuses on the interconnection and interoperability of DERs with associated electric power systems interfaces [132]. A guideline for interconnecting distributed energy storage systems with the distribution grid has also been developed based on this standard [60]. IEEE-1547 was developed by the IEEE Standards Coordinating Committee 21 on fuel cells, PVs, dispersed generation, and energy storage. The standard outlines and covers guidelines for integrating various DERs comprising synchronous and induction machines besides the power converters connected to the grid's primary or secondary distribution voltages. The code defines a reactive power capability for different categories of DERs, including the following.

1) *Category A:* The DER's impact and penetration in the distribution network are low or the aggregate DER output power does not undergo significant frequent changes.
2) *Category B:* The DER's impact and penetration in the distribution network are high or the aggregate DER power output does undergo significant frequent changes.

It should be noted that there are no measures to distinguish the above categories quantitatively defined by the code. Code developers have acknowledged that the effect of DERs on the system performance may vary significantly. Hence, the regulatory authority overseeing interconnection is responsible for determining the optimal levels of DER penetration. The reactive power capability for categories A and B are depicted in Fig. 18. As can be observed, there is no need for reactive power support from both categories for an active power operating at a rate lower than 5% of the rated apparent power capacity. For category A, the DER must be able to inject 44% of the rated reactive power to absorb 25% of the rated apparent power for operating points higher than 20% of the rated active power capacity. In the 5%–20% active power operation range, the category A DER must inject 11%–44% and absorb 6%–25% of reactive power regarding the rated capacity. Besides category A, category B DER has a symmetrical capability curve for capacitive (injection) and inductive (absorption) sides. It denotes a zero reactive power for active powers below 5% of the rated capacity, 2.2 times (11%–44%) the operating active power from 5% to 20% of the rating and a constant 44% reactive support for loadings above 20%. Also, Fig. 18 illustrates the reactive power capabilities of the distributed energy storage system. The curve denotes a constant 44% reactive support from capacitive to inductive power factor for both charging and discharging states.

### B. IEEE-2800

The second grid code issued by the IEEE, IEEE-2800, was published in 2022. It discusses the interconnection and interoperability of IBRs connected to transmission grids. The code was developed by the IEEE Power and Energy Society's committees of Electric Machinery, Energy Development and Power Generation, and Power System Relaying and Control [133]. It is the first effort to establish a harmonized framework for performance and connection requirements. It can help the system participants, especially grid operators, optimize the facilities' performance, ultimately contributing to a more stable and resilient grid. This standard establishes the required interconnection capability and performance criteria of the IBRs interconnected with transmission and subtransmission systems, including solar PV, wind, and energy storage systems or a combination of them (hybrid system). The Type III wind turbine generators have been given different specifications and requirements in this code due to utilizing DFIGs. There are two definitions regarding the nominal capacity of an IBR in this code, including the following.

1) *IBR continuous rating (ICR):* The maximum amount of continuous active power an IBR can deliver to the grid in a steady state.
2) *IBR continuous absorption rating (ICAR):* The maximum amount of continuous active power an IBR can absorb from the grid in a steady-state situation.

The ICR definition denotes renewable-based generators, including wind and solar PV. The ICAR definition applies to an energy storage system or a hybrid storage-generator system. Based on the code, a generator IBR should be capable of injecting and absorbing a minimum amount of reactive power equal to 0.3287 times ICR, as illustrated in Fig. 18. Notably, this minimum reactive power capability must be maintained across all active power output levels, including zero, regardless of whether the primary energy source is available or unavailable, and during the transition periods between these two states. For an IBR plant comprising type III wind turbines, which do not utilize a VSC-HVdc line for grid connection, the minimum reactive power capacity requirement is modified, as shown in the figure. As in the figure, for the active power output levels below 10% of nominal value (ICR), the reactive power support decreases linearly from 0.3287 times ICR to zero (or the purple region can be ignored). This reduction is necessitated by the technical limitations of the technology used in this turbine type, a DFIG. Also, a storage or hybrid storage-generator IBR plant shall be able to inject and absorb minimum reactive power equal to 0.3287 $\times$ ICAR

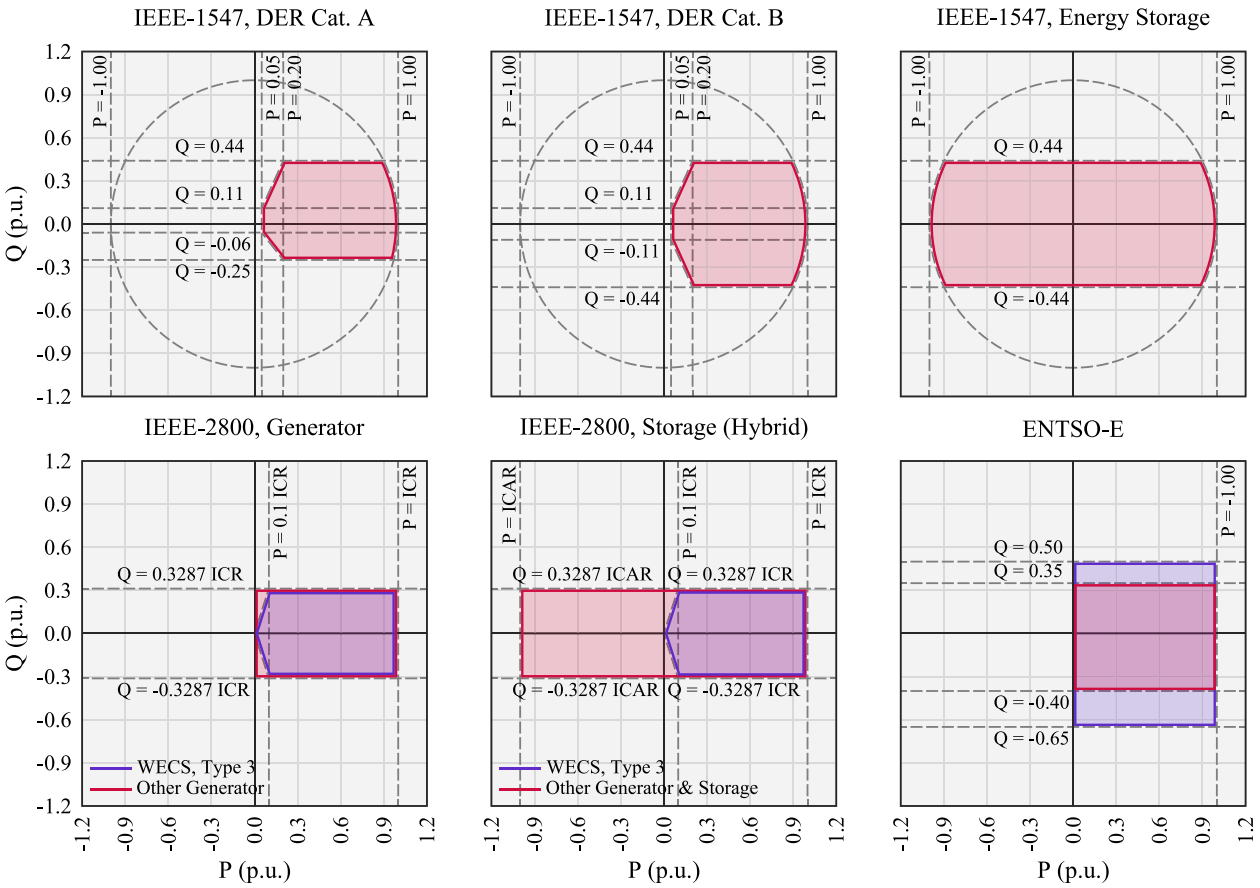



**Fig. 18.** ***Reactive power capability requirements of IEEE and ENTSO-E.***

when absorbing active power from and injecting it to the transmission grid, as shown in Fig. 18. Similar to the previous case, there is also a situation when a storage system is hybridized with a Wind-Type III, denoted by the purple region in the figure.

### C. European Union (ENTSO-E)

The ENTSO-E is a collaborative organization that unites transmission system operators from European countries and even extends beyond the EU borders. Established in 2009 under the EU's Third Package for the Internal Energy Market, ENTSO-E aims to foster cooperation and harmonization among its members [15]. ENTSO-E acts as a central hub, facilitating the exchange of information and expertise on technical, market, and policy matters affecting transmission system operators and European power grid. Through collaboration with various stakeholders, including power consumers, EU authorities, regulatory bodies, and national governments, ENTSO-E plays a crucial role in shaping the future of European power network. To facilitate integration and optimize European electricity market's performance, ENTSO-E cooperates with the Agency for the Cooperation of Energy Regulators (ACER) to establish and develop a set of standardized guidelines or grid codes [130]. These codes provide guidelines for integration and harmonization, in addition to optimizing European electricity market's performance. The grid codes are essential for European Union's goal of creating a fully integrated internal energy market. This ambitious objective is part of a broader effort to realize the EU's energy strategy, which aims to the following.

1) Reducing greenhouse gas emissions to 55% compared to 1990.
2) Increase to 32% of renewable energy consumption from the total energy demand.
3) Achieving 32.5% energy savings compared to the business-as-usual scenario.

As discussed before, the ENTSO-E network codes comprise four categories: market, operation, connection, and cybersecurity. The connection code comprises the DCC, RFG, and HVdc. As per the scope of this article, the reactive power requirements part of RFG for non-SGs or IBRs is focused. The ENTSO-E has five distinct synchronous areas, each with varying capacities and sizes. To address the challenge of ensuring compatibility across these systems, the RFG has established a generator classification system based on their size and type, as described in Table 4. This categorization system consists of four classes, ranging from A to D, each with specific requirements that must be met. Class A possesses the most basic, and Class D has the most

**Table 4** Types A–D Generator Sizes for Synchronous Areas of ENTSO-E (kW)

| Type | Baltic | Continental | Great Britain | Ireland | Nordic |
|---|---|---|---|---|---|
| A | 0.8 | 0.8 | 0.8 | 0.8 | 0.8 |
| B | 500 | 1,000 | 1,000 | 100 | 1,500 |
| C | 10,000 | 50,000 | 50,000 | 5,000 | 10,000 |
| D | 15,000 | 75,000 | 75,000 | 10,000 | 30,000 |

stringent requirements. It should be noted that the units connected to 110 kV and grids with the higher voltages are always considered as Type D regardless of capacity. The maximum size thresholds for each class are determined by the specific power system to which the generator is connected.

Regarding reactive power capability, Type A generators are exempted. For Type B generators, reactive power requirements are only imposed on the synchronous machines, not the IBRs. There are identical reactive power capabilities that are only imposed on the Types C and D non-SGs, or IBRs, as depicted in Fig. 18. As in the figure, the capability curve is defined by an inner and outer envelope. The range of the inner envelope for the Baltic, Continental, Great Britain, Ireland, and Nordic areas is equal to 0.8, 0.75, 0.66, 0.66, and 0.95, respectively. For offshore wind turbines connected to the grid in Great Britain, the reactive power range is ignored for offshore connections type 1 (ac connection to single onshore point) and equal to 0.33 for offshore connections type 2 (Meshed ac connection).

### D. United States (Federal Energy Regulatory Commission)

The Federal Energy Regulatory Commission (FERC) is an independent agency that regulates and oversees the interstate transmission of electricity and critical fuels such as natural gas and oil in the United States. The commission has a broad regulatory reach, including reviewing and approving plans for building liquefied natural gas (LNG) terminals and interstate pipelines transporting natural gas across the country. Additionally, FERC is responsible for issuing licenses for hydropower projects and ensuring that they operate safely and efficiently. Under the Large Generator Interconnection Agreement (LGIA), FERC requires that all generators operating under its jurisdiction with a generating capacity of more than 20 MW adhere to standardized procedures for connecting with the electricity grid [134]. FERC's Order 827 sets forth requirements for reactive power support from non-SGs, i.e., IBRs. Announced in June 2016, this order has been updated to include wind turbines due to technological advancements, which were previously exempted. As a result, all non-synchronized generators, including wind turbines, must be capable of providing reactive power support to the transmission grid. Specifically, generators must provide 0.95 leading to 0.95 lagging reactive power support when connecting to the transmission network, as shown in Fig. 19 [134].

### E. Germany (VDE-AR-N)

In Germany, grid codes impose specific requirements on DERs connected to the grid. Specifically, DERs must be able to provide reactive power control only during their feed-in periods. This allows distribution system operators to tap into the reactive power capabilities of DERs within a specified range to achieve operational objectives and offer additional grid services, such as voltage support and stability [131]. These requirements are tailored to the specific voltage level of the connected grid, as outlined in the following.

*1) Low-Voltage Grid (VDE-AR-N 4105):* The technical requirements for connecting DERs to the low-voltage grid are outlined in the VDE-AR-N 4105 standard. A graphical representation of the minimum reactive power requirements for DERs is shown in Fig. 19, which are dependent on both the active power fed into the system and the installed capacity. The curve consists of two sections for low-voltage DERs with installed capacities of less than and greater than 4.6 kVA, represented by LV-V1 and LV-V2. According to the figure, LV-V1 and LV-V2 DERs must provide reactive power with a power factor of 0.9 and 0.95, in turn, in underexcited and overexcited conditions [135].

*2) Medium-Voltage Grid (VDE-AR-N 4110):* The guidelines for technical connections in VDE-AR-N 4110 outline the requirements for designing, constructing, operating, and modifying customer facilities connected to a medium-voltage grid at the connection point. This includes facilities directly connected to the medium-voltage network or connected to a private low-voltage grid and subsequently tied to the medium-voltage network via transformers and connection lines. Fig. 19 illustrates that reactive power support is not mandatory for active powers below 10%. It remains constant at 0.312 p.u., capacitive to inductive, reactive power support for active powers above 20%. The reactive power limits decrease linearly from 0.312 p.u. (at 20% loading) to 0.1 p.u. (at 10% loading) from leading to lagging power factor [136].

*3) High-Voltage Grid (VDE-AR-N 4120:* The technical requirements for the reactive power provision capability of generators connected to the high-voltage level are outlined in VDE-AR-N 4120. The curves depicted in Fig. 19 define three distinct options, each with specific requirements for delivering reactive power. The available options guarantee that generators connected to the high-voltage level can have the following.

1) *Variant 1:* Reactive power range from 0.48-p.u. leading to 0.23-p.u. lagging, equivalent to a power factor of 0.9 underexcited to 0.975 overexcited.
2) *Variant 2:* Reactive power range from 0.41-p.u. leading to 0.33-p.u. lagging, equivalent to a power factor of 0.925 overexcited to 0.95 underexcited.
3) *Variant 3:* Reactive power range from 0.33-p.u. leading to 0.41-p.u. lagging, equivalent to a power factor of 0.95 overexcited to 0.925 underexcited.

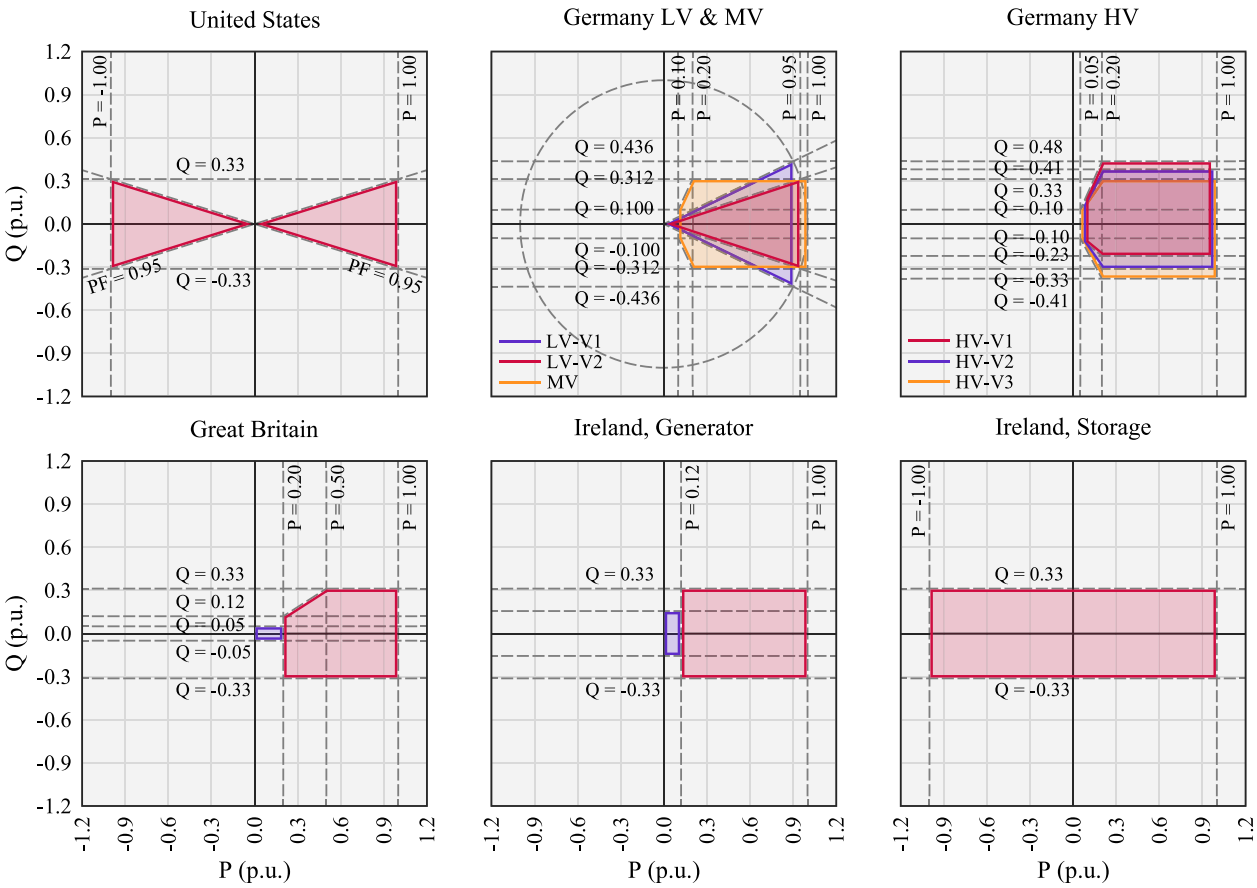



**Fig. 19.** ***Reactive power capability requirements of the United States, Germany, Great Britain, and Ireland.***

The specific characteristics of the generator's connection point, including its location and voltage level, enable system operators to select a suitable option for providing reactive power based on their operational requirements [137].

## F. Great Britain (National Grid-ESO)

National Grid is a leading international energy company that operates high-voltage electricity and gas transmission systems across Great Britain. There is a separate business entity within the national grid called the electricity system operator (ESO), which oversees and manages the electricity supply and demand in Great Britain. It establishes the guidelines and protocols for interacting with its various stakeholders, including power generators, suppliers, and customers. The grid code developed by the ESO sets standardized rules and guidelines for planning and operational activities, ensuring consistent and efficient handling of normal and exceptional circumstances. Additionally, the grid code specifies the obligatory reactive power service (ORPS), which requires generators to provide or absorb reactive power to maintain system voltage stability at their connection point. According to the grid code, all transmission-connected generators with a capacity of 47 MW or higher are required to be able to provide this service. The minimum technical requirements for reactive power from a non-SG defined as the power park module in the code are shown in Fig. 19. As shown in the figure, a reactive power limit of 33% applies at active power levels above 20% of nominal MW. Above 50% of the rated MW, the 33% reactive power limits will apply. The reactive power limits decrease linearly from 33% to 12% leading when moving from 50% to 20% of the rated active power. Also, there is no mandate to provide reactive power at operating points lower than 20% rated active power [138].

## G. Ireland (EirGrid)

EirGrid, the state-owned electric power transmission operator in Ireland, has been responsible for operating and developing the national high-voltage electricity grid since 2006. The company's primary focus is ensuring reliable and efficient electricity transmission nationwide while adhering to strict guidelines and regulations. The grid code, which EirGrid has established, serves as a comprehensive framework for the transmission system's operation, maintenance, and development. This code outlines the procedures for managing the actions of all parties involved in the transmission system, including users, operators, and maintainers. The grid code categorizes

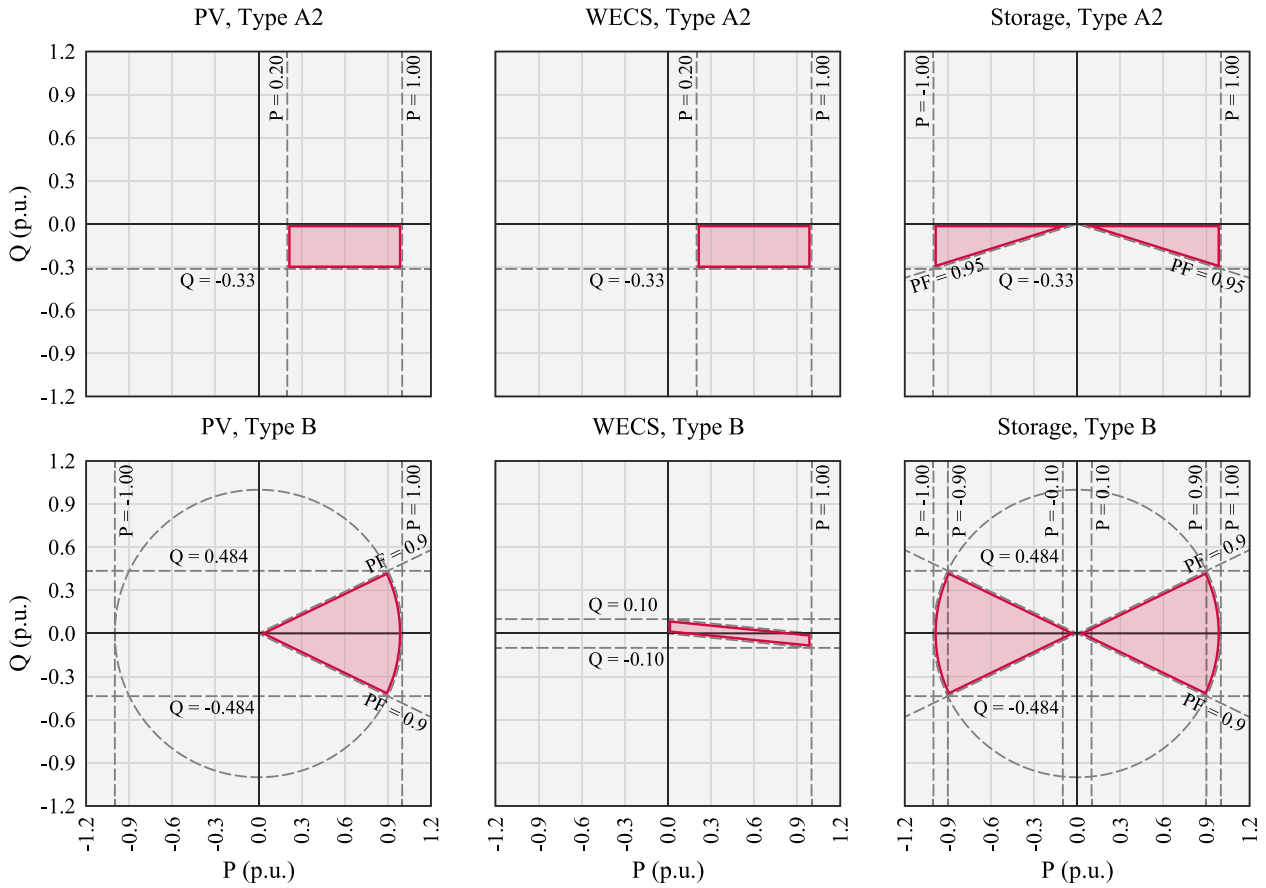



**Fig. 20.** ***Reactive power capability requirements of Denmark—Types A2 and B.***

the transmission system voltages into four primary levels, including 400, 220, 110, and 66 kV, all of which an IBR may be connected. The corresponding grid code for generators and storage systems is depicted in Fig. 19. As can be observed, the generators have no mandate to support reactive power for active powers below 12%. Also, there is a constant 33% reactive support for operating points above this point. Additionally, there is a constant 33% leading to lagging reactive power support for a storage system operated at any charging and discharging state [139].

## H. Denmark (Energinet)

Energinet is a public enterprise owned by the Danish government that manages the country's national transmission systems for electricity and natural gas. The organization operates independently under the guidance of the Ministry of Climate and Energy. As the national transmission system operator, Energinet plays a crucial role in Denmark's energy infrastructure, overseeing the operation of the electricity transmission grid and ensuring the system's reliability. At its core, Energinet's mission is to guarantee a stable and efficient supply of electricity to meet the demands of Denmark's citizens. Denmark has established different guidelines for wind, PV, and battery-based IBRs. Each IBR is categorized based on its capacity, ranging from Type A2 to Type D, with specific requirements and regulations applying to each classification. Table 5 shows each type's installed capacity range separately for generators (wind and PV) and storage systems [88], [140], [141].

It should be noted that the storage systems under this code include the following facilities.

1) Electric energy storage systems capable of synchronizing with the grid (synchronous energy storage systems).
2) Nonsynchronously energy storage systems connected to the grid via a power electronic inverter (a storage IBR), like batteries.
3) Storage systems in electrified transport systems, such as electric vehicles and ferries, used to transport passengers or goods and connected to the grid via a charging station in the vehicle-to-grid (V2G) mode.

Figs. 20 and 21 show Denmark's grid code for wind, PV, and BESS for each installation type provided in Table 5. As can be observed, there are substantial differences in grid codes based on the primary IBR type and installation size (from A2 to D).

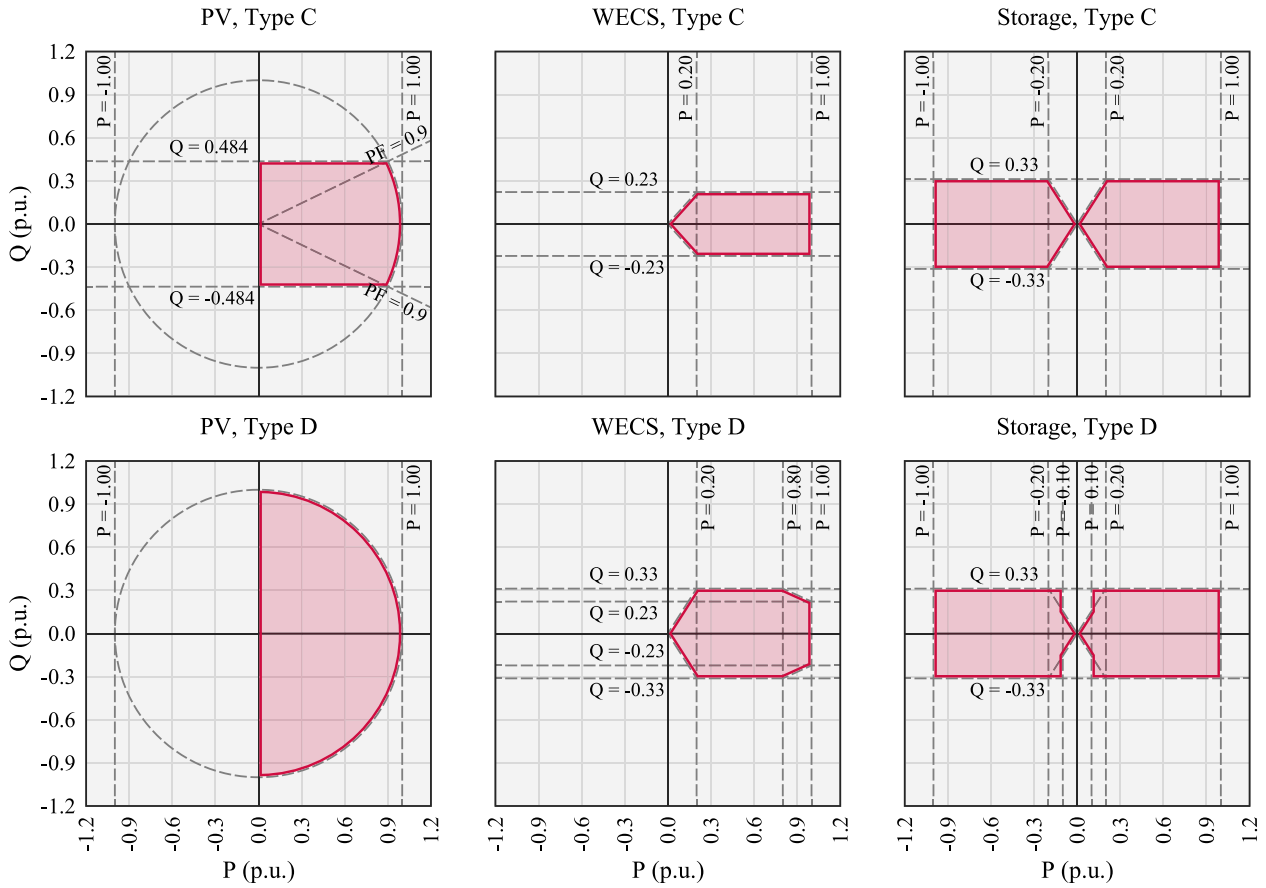



**Fig. 21.** ***Reactive power capability requirements of Denmark—Types C and D.***

## XI. ROLE OF CONVERTER TYPE IN REACTIVE POWER EXCHANGE OF IBRs

The type of intermediate power converter plays a critical role in managing the reactive power exchange between an IBR and the grid. These power converters are broadly classified into two main categories: grid-following converters (GFLC) and grid-forming power converters (GFMCs). Each type has distinct characteristics influencing grid stability, reactive power control, and voltage regulation. However, a more granular classification is also commonly used to describe their operational capabilities in adjusting power output in response to grid conditions and maintaining voltage stability [20].

In this refined classification, GFLCs are further divided into grid-feeding converters (GFECs) and grid-supporting converters (GSUCs). The GFECs maintain constant active and reactive power output, while the GSUCs adjust both active and reactive power to varying degrees in response to grid conditions. On the other hand, grid-forming converters, which are designed to operate autonomously, setting the frequency and voltage of the grid, are crucial for forming independent or islanded grids. This autonomy instills confidence in their reliability, providing both active and reactive power support without relying on an external voltage reference [142].

This section explores each converter type's specific roles and capabilities in enhancing the reactive power exchange mechanisms of IBRs. It focuses on how these converters interact with the grid under different operating conditions, highlighting their performance during steady-state and transient scenarios. Particular attention is given to their ability to support grid stability, enhance voltage regulation, and mitigate potential disturbances, such as voltage dips or frequency fluctuations, that may arise in real-world operating environments.

**Table 5** Denmark Type A2–D IBRs (kW)

| Type | Wind and PV | Energy Storage |
|---|---|---|
| A2 | 11-50 | Up to 125 |
| B | 50-1,500 | 125-3,000 |
| C | 1,500-25,000 | 3,000-25,000 |
| D | Above 25,000 or connected to over 100 kV | |

### A. Grid-Feeding Converters

GFECs primarily aim to transfer active power from renewable energy and battery storage resources to the grid. They do not directly manage voltage or frequency. Their role in reactive power exchange is secondary, as they

typically focus on active power control through maximum power point tracking (MPPT) [143]. However, during steady-state operation, the voltage and reactive power are generally determined by the grid conditions since these converters synchronize with the grid's ac voltage. The reactive power flow is limited and typically passive, with the converter adjusting to the grid's reference parameters rather than influencing them [144].

1) *Steady state:* The GFEC operates parallel with the grid, with minimal influence on voltage or reactive power regulation.
2) *Transient conditions:* During transients, these converters may struggle with sudden voltage and frequency fluctuations due to the lack of intrinsic voltage or frequency control, leading to potential instability. This highlights the need for systems with higher inertia or additional control from the subsequent converter types or SGs [143], [144], [145].

### B. Grid-Forming Power Converter (GFMC)

GFMCs are crucial in managing both active and reactive power, playing a vital role in grid stability. These converters are designed to control the network's voltage and frequency, allowing them to operate independently of the grid and form a new grid in island mode [143]. The ability to manage both active and reactive power in a coupled manner is a defining feature of GFMCs, making them effective in stabilizing voltage levels, mitigating frequency fluctuations, and ensuring the network's resilience during transients [146].

1) *Steady state:* The GFMC ensures steady-state voltage and frequency control in isolated networks (e.g., microgrids), providing reactive power support to maintain grid balance [146], [147].
2) *Transient conditions:* GFMCs can respond quickly to transients by adjusting both active and reactive power, providing damping and enhancing frequency stability. They do not rely on mechanical inertia, offering oscillation damping through active power control. This makes these converters well suited for power systems with high IBR penetration, where traditional inertia-based solutions are limited [145], [147].

### C. Grid-Supporting Converter

GSUCs offer a flexible solution between GFECs and GFMCs, providing an intermediate level of control over both active and reactive power. While GSUCs synchronize with the grid, they can adjust power flow to support voltage and frequency regulation, making them versatile in grid-connected and islanded modes [145]. These converters are especially useful in systems with high renewable energy penetration, where they help manage voltage levels and improve overall grid stability [145], [148].

1) *Steady state:* In steady-state conditions, GSUCs support the grid by improving power quality, stabilizing voltage, and ensuring effective reactive power exchange. They coordinate with other converters to maintain grid balance and assist in maintaining voltage profiles within acceptable limits [147].
2) *Transient conditions:* During transients, GSUCs play an important role in stabilizing the grid by adjusting reactive power exchange, thus supporting voltage regulation and reducing frequency fluctuations [149]. However, the limited inertia provided by GSUCs can result in vulnerability to rapid changes, which may lead to instability if not supported by other elements like GFMCs or SGs. The GSUCs are often equipped with advanced control strategies to address these limitations, such as virtual inertia, to enhance their reactive power capabilities and improve the system's response to disturbances [148], [149].

### D. Integration and Challenges in Grids With High Shares of IBRs

While GFECs primarily focus on transferring active power and have limited reactive power management capabilities, GFMCs, and GSUCs provide more functional solutions by managing both active and reactive power, thereby improving voltage and frequency control. However, the presence of IBRs, particularly in large-scale power systems, can lead to instability issues, especially in scenarios with sudden changes in load, renewable energy generation, or grid faults. GFMCs, with their ability to operate independently and actively control voltage and frequency, can significantly mitigate these problems by providing both damping and support for reactive power regulation. GSUCs, while helpful in providing flexibility and supporting grid dynamics, still face challenges related to inertia and require additional measures like virtual inertia or advanced control strategies to stabilize the grid effectively. Combining GFECs, GFMCs, and GSUCs is essential for a modern grid that relies heavily on IBRs. These converters must coordinate for grid stability, with GFMCs and GSUCs providing support for reactive power exchange and voltage regulation, particularly during transients. Advanced control schemes, including virtual inertia and enhanced reactive power management techniques, will be critical in future grid designs to accommodate higher levels of renewable energy penetration and ensure reliable and stable grid operation. In conclusion, reactive power exchange capabilities are crucial for maintaining the stability and efficiency of future grids, particularly as they transition toward greater deployment of IBRs.

## XII. TRANSIENT DYNAMICS OF REACTIVE POWER PROVISION BY IBRs

While steady-state control of reactive power has been extensively investigated, the transient dynamics associated with reactive power provision by IBRs during disturbances remain insufficiently discovered. However, analyzing these transient processes is vital for preserving voltage stability, preventing cascading failures, and ensuring overall system

reliability. This is especially critical in low-inertia systems, where the absence of traditional SGs exacerbates vulnerability to instability. Addressing the transient response of IBRs, therefore, represents a crucial step in enhancing grid resilience during faults, frequency deviations, and other perturbations. Recent research has increasingly highlighted the importance of understanding the transient behavior of reactive power in IBRs [150].

Unlike steady-state conditions, transient events demand rapid and adaptive responses from IBRs. Their ability to inject or absorb reactive power within milliseconds can be pivotal in mitigating grid disturbances. The speed and precision of this response are particularly critical during severe faults and abrupt load changes [150]. Furthermore, as power systems evolve to include multiterminal dc configurations and hybrid ac/dc networks, the complexity of managing transient dynamics increases, necessitating advanced modeling techniques and simulation methodologies. Addressing these challenges is essential for achieving system-wide stability and enabling seamless renewable integration. The transient provision of reactive power by IBRs has attracted growing attention due to its immediate and critical implications for grid stability during disturbances [151].

Several factors influence the transient behavior of IBRs, including the control algorithm design, grid topology, and nature of disturbances. For instance, the design and configuration of inverter control loops, ranging from basic DRC to more advanced approaches like model predictive control (MPC), directly affect the speed and effectiveness of reactive power support. DRC schemes, in particular, have demonstrated potential for scalability and responsiveness in high-variability systems [152]. Similarly, grid topology plays a vital role; highly interconnected systems with significant IBR penetration present unique challenges in coordinating the responses of multiple IBRs during transient events. Nevertheless, tradeoffs between response speed, computational complexity, and hardware constraints remain active areas of investigation. Future research must address these issues by exploring hybrid approaches that integrate local control schemes with centralized supervisory architectures, ensuring optimal coordination and performance.

## XIII. REACTIVE POWER CONTROL OF IBRs

In reactive power control of IBRs, the primary objective is to maintain the voltage of all system buses within the permissible bounds to ensure system reliability, stability, and efficient operation. This objective is achieved primarily via controlling the available reactive power sources, including IBRs. This practice involves a multifaceted approach that considers various aspects, including capability, loading, and reserve margin of the available reactive power sources, as well as the losses, flows, and loading of the grid equipment, especially lines. There are three governing factors

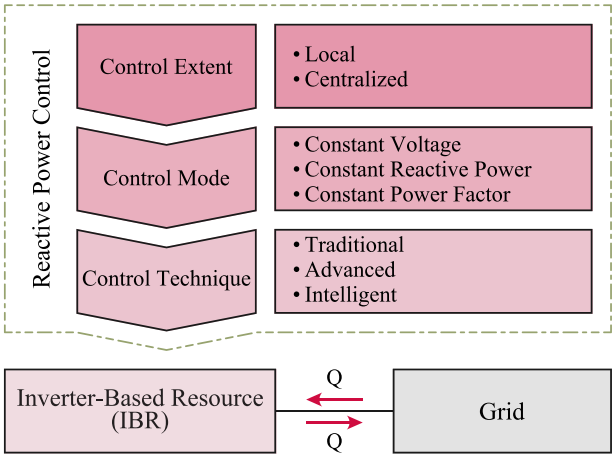



**Fig. 22.** *Governing factors in reactive power control of IBRs.*

that must be considered when controlling an IBR reactive power to conserve system voltages, as shown in Fig. 22.

Before using the IBRs' reactive power exchange capability, it is necessary to answer the following questions. First, the extent of the voltage control within the system by employing the IBR's reactive power should be determined. In other words, according to the requirements and conditions of the network, as well as the available voltage control resources and tools, it is necessary to use one of the local or centralized schemes. After determining this factor and, in the second step, the relevant control mode, intended to employ the IBR's reactive power, should be determined. At this stage, it is necessary to choose one of the control modes, which includes constant voltage, constant reactive power, or constant power factor. Finally, the desired control technique should be selected after determining the control extent and mode according to the primary source of energy and the network requirements [153], [154].

### A. Control Extent

The existing control methods regarding the extent of the controlled area have primarily been divided into local and centralized categories. Local control methods involve controlling the voltage at a specific location in the system utilizing available reactive power sources, typically a system bus representing a substation or a load center. In this case, there is no view or information about the system's operating point than the controlled location. This means that there is no coordination between reactive power sources and controlled voltage locations. In contrast, centralized methods involve maintaining the voltage level across the entire system or a significant portion via a centralized optimization algorithm. This method requires advanced algorithms and communication systems to monitor and maintain the voltage levels across multiple buses or locations. Various operation functions and control mechanisms, such as optimal power flow (OPF), economic

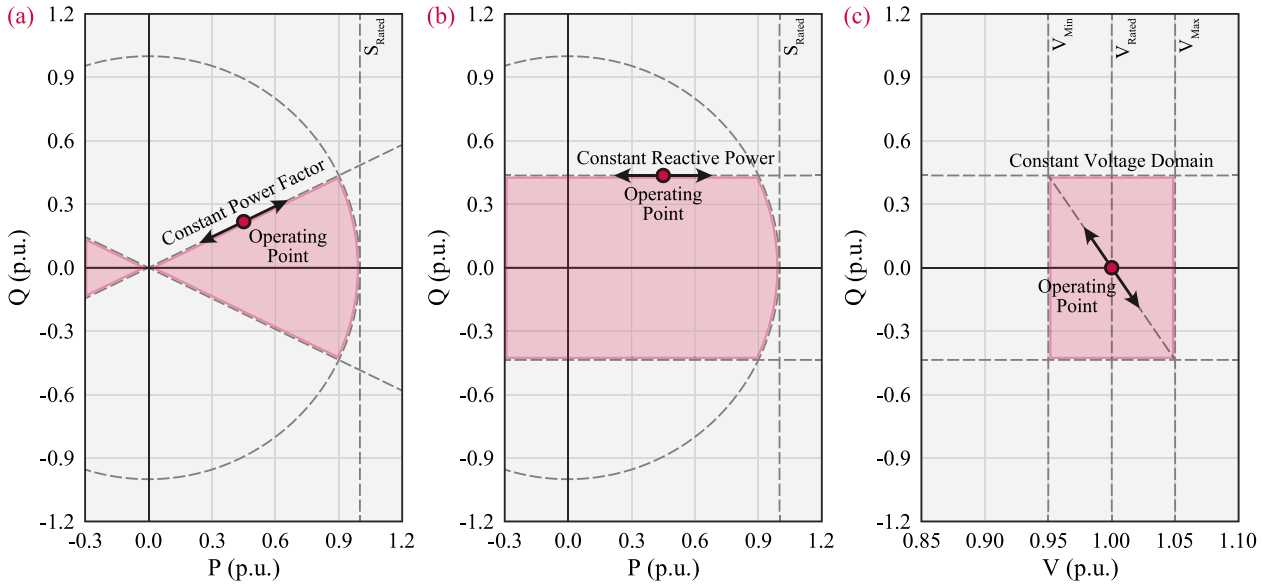



**Fig. 23.** *Control modes of IBR's reactive power. (a) Constant power factor. (b) Constant reactive power. (c) Constant voltage.*

dispatch (ED), state estimation (SE), and MPC, may be used to achieve this goal [153]. In the centralized method, there is harmony between available reactive power sources in the whole system, and the voltage profile is maintained across the entire region. While centralized voltage control systems may offer improved performance, they also come with significant investment requirements in communication infrastructure as well as in sensing and measurement technologies, leading to potential vulnerabilities arising from reliance on this equipment. Furthermore, the latency associated with communication and control can compromise performance during sudden changes in the system. In contrast, local voltage control schemes that operate on local variables are less susceptible to latency and vulnerabilities as they do not rely on communication. However, these schemes are limited by their lack of access to area-wide flow data, and thus, they cannot guarantee optimal control [154].

## B. Control Modes

The IBR's control mode determines how reactive power is utilized to maintain a specific variable constant. In other words, the control mode defines the primary goal of setting the reactive power of the IBR, which is to regulate a particular parameter. Fig. 23 illustrates the various control modes of IBR reactive power. When selecting the desired control mode, system operators must consider several factors, including the following [154].

1) *Load conditions:* The type and size of loads connected to the grid can affect the choice of control mode.
2) *Grid topology:* The grid's topology, including the number of buses, lines, and transformers, can influence the choice of control mode.
3) *Renewable energy penetration:* The level of renewable energy integration can affect the choice of control mode, as inverters may need to adjust their output to maintain stability.
4) *Inverter capabilities:* The network operator considers the capabilities and limitations of the inverters used in the IBRs. For example, some inverters may be limited in adjusting reactive power or voltage output.
5) *Grid code compliance:* The network operator must comply with relevant grid codes and regulations, which may specify requirements for reactive power control or power factor correction.
6) *Operator preference:* The network operator may prefer a particular control mode based on their experience, operational practices, or other factors.

In the constant voltage control mode, the right plot in Fig. 23, the IBR is mandated to operate in a closed-loop automatic voltage control mode, maintaining the steady-state voltage at the point of connection within the allowed bound. The grid operator must determine the permissible domain of voltage variation so the control system can use the IBR reactive power to provide a stable response. In the left plot in Fig. 23, the constant power factor mode obliges the IBR to deliver the reactive power to the grid in proportion to the output active power. In other words, the output reactive power must be maintained constant in proportion to the output active power. The power factor control mode must operate at all active power levels, from the minimum to maximum active power output. The operator must determine the desired power factor in accordance with the corresponding grid code. Finally, in the constant reactive power mode, the middle plot in Fig. 23, the IBR consistently delivers a specified amount of reactive power at the connection point. The operator

determines the target reactive power level in addition to its sign. Maximum deliverable reactive power has to be consistent with the corresponding grid code and within the IBR capability.

### C. Control Techniques

Various reactive power control techniques of IBRs are categorized and presented in Fig. 24, showcasing the diversity of methods used to manage reactive power in power systems. As can be observed, the techniques can be broadly classified into three categories: traditional, advanced, and intelligent control methods. It is worth mentioning that traditional control methods are based on classical control theories. These methods are widely used in many linear systems due to their simplicity and effectiveness. However, they have limitations when dealing with complex systems or nonlinear dynamics. Advanced control methods are based on recent and modern control theory and are designed to handle more complex systems and provide improved performance and stability. Finally, intelligent control methods incorporate artificial intelligence (AI) and machine learning (ML) techniques to improve control system performance and adaptability. Intelligent control methods are beneficial in applications where uncertainty is high or system dynamics are complex or nonlinear. Each abovementioned category encompasses a range of specific control approaches described below.

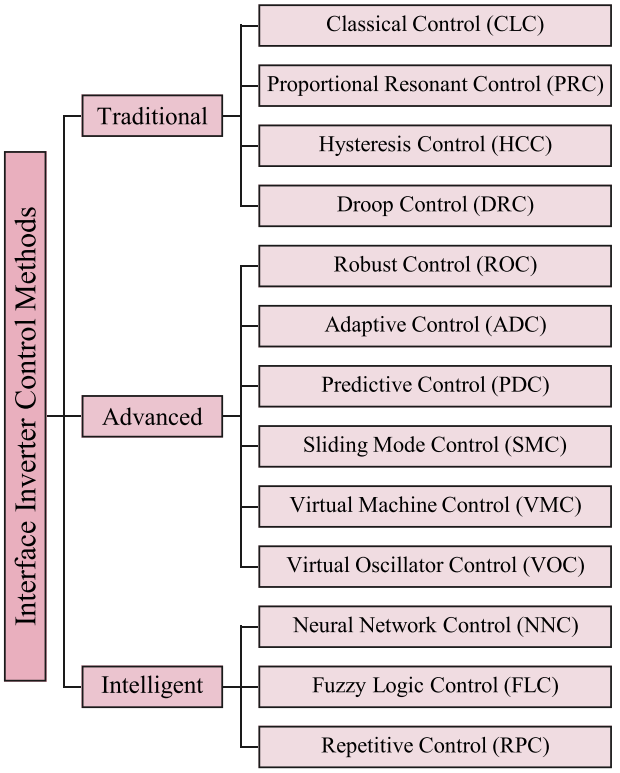



**Fig. 24.** ***Control techniques for IBR's reactive power control.***

*1) Classic Control:* Classic control (CLC) is based on the idea that the control system should be designed to achieve a specific desired behavior rather than simply trying to minimize or optimize a performance criterion [155]. Therefore, it uses simple feedback control to adjust the system under control. Despite the simplicity, they have a limited ability to handle complex nonlinear dynamics and reject disturbances. They are also sensitive to changes in system parameters [156]. The three common types of CLC are given as follows.

1) *Proportional (P):* In this type of controller, only the error signal is used to perform the control action. As a result, the control output is directly proportional to the error signal.
2) *Proportional–Integral (PI):* In PI, an integral term is added to the proportional term that accumulates the error over time. As a result, the elimination of steady-state errors and disturbance rejection are improved.
3) *Proportional–Integral–Derivative (PID):* A derivative term is added to the PI terms in PID. The derivative term measures the error signal's rate of change and helps improve the transient response.

*2) Proportional Resonance Control:* Proportional resonance control (PRC) and PIs have many features in common. The main difference is in the integrator unit [157]. In PRC, the integrator primarily focuses on and sees frequencies nearly identical to the resonant frequency. Therefore, they do not include a static error or phase shift. Thus, PIs allow efficient tracking of dc signals, while PRCs track a sinusoidal signal with a sinusoidal frequency as its center frequency [158]. Hence, PRCs are able to be employed in both $abc$ and $\alpha\beta$ reference frames [158]. It should be noted that PRCs can provide selective harmonic compensation without requiring any computational resources. This control method can quickly eliminate the steady state error of electrical quantities due to its high gain near the resonance frequency. However, this control method is strongly influenced by the selected frequency. Therefore, it has a limited bandwidth and modulation range, requiring precise parameters to be set, which is complicated since it is a complex structure [158].

*3) Hysteresis Control:* Hysteresis control (HCC) is a nonlinear control with a straightforward approach based on the error signal. In other words, the output control signals change status by crossing the reference and the measured value difference from a particular band, the hysteresis band. This control method offers advantages such as simplicity of structure, fast dynamic response, independence from load parameters, and effective control strategy, ensuring high stability of the system [159]. Also, this method has an inherent current protection system. However, HCC has two main drawbacks. First, it has a wide frequency range and a relatively large current ripple since having no fixed switching frequency. Therefore, designing the output filter due to the randomness of the output and the inability to control the output ripple is challenging.

Also, changing frequency can lead to resonance problems and high total harmonic distortion (THD). Second, the controlled systems cannot maintain the measured current in the bandwidth appropriately, leading to unwanted fluctuations in the switching frequency [160].

*4) Droop Control:* The main idea of DRC is based on creating a linear compromise between network parameters, including frequency and voltage, with active and reactive power delivered by a power exchange unit. This approach is reminiscent of an SG's steady-state operation [161]. This control method inherently leads to synchronizing all network components and achieving the reference frequency. Also, each component contributes to the power demand response according to its ability or the slope determined by the specific droop characteristic. In other words, load sharing and prioritization among generators or inverters is possible with this method. With inherent simplicity and quick response, this control method helps maintain voltage and frequency in the network [162]. Although some of the presented methods are based on DRC, inertia is also supplied to the network, and it primarily guarantees the stability of the network without the need for inertia. This control method may be deployed in the conventional form or combined with other control schemes (virtual impedance loop-based droop, adaptive droop, and resistant droop) [20].

*5) Robust Control:* Conventional control methods are ineffective in overcoming disturbances caused by internal and external factors, and as a result, the stability of the controlled system is affected. Robust control (ROC) is one of the most efficient solutions for overcoming system uncertainties [163]. ROC is based on a control theory with the control system design approach considering the uncertainties, disturbances, and nonlinear nature of RES electronic equipment and the system's load. Despite limited modeling errors, this method aims to achieve robust performance and/or stability. This control technique, even in multivariable systems, can guarantee the stability and strong performance of closed-loop systems and be efficient [164]. ROC includes two main subsets under the exponents of $\mu$-synthesis control and H-infinity ($H\infty$) control.

*6) Adaptive Control:* Adaptive control (ADC) is based on the theory that the power converter's optimal control parameters may change depending on the operating conditions [165]. It can adapt to changes in the controlled system by adjusting control actions in real time. Unlike conventional control methods relying on predefined set points and fixed parameters, ADC allows the system to exhibit proper performance even when subjected to nonlinear, unbalanced, and sudden load changes, maintaining optimal system stability [166]. In this technique, the system parameters do not need to be known with high accuracy, which improves the flexibility and performance. However, this controller's complexity and high computational requirements are some of its key drawbacks, requiring significant computing resources and processing power. Accordingly, the computational complexity makes troubleshooting and initial settings of the control system challenging [167].

*7) Predictive Control:* Predictive control (PDC) uses a system model based on a mathematical approach to predict the controlled parameters' future behavior [168]. The PDC can control a nonlinear system besides reducing the switching frequency of high-power converters. This method can be used in various multivariable systems with fast, nonlinear dynamic responses and easily applied constraints. This control method can be easily implemented, and its fast dynamic response makes it suitable for applications that require quick adjustments. It can also effectively reduce the occurrence of resonant behavior when operating under unbalanced conditions. Furthermore, it ensures robust current control with minimal harmonics. However, PDC requires filtering and is sensitive to parameter changes. Besides, the implementation of PDC is more challenging than classical control methods. It requires extensive calculations, and the specific load must be matched. The PDC is classified into deadbeat and MPC [169].

*8) Sliding Mode Control:* Sliding mode control (SMC) is a nonlinear control mechanism that changes the system dynamics by appropriately selecting the sliding function and switching between control laws based on the state vector [170]. Insensitivity to load disturbances and parameter changes, disturbance rejection, and simple implementation are the advantages of this technique. As a result, it is adaptive and robust to parameter changes and noise. Therefore, the steady-state response can be ideally obtained using this control method. However, this method has some drawbacks. The effectiveness of SMC varies depending on the sliding surface, and choosing a suitable sliding surface is complicated. Choosing the appropriate sampling time is critical since insufficient sampling time leads to high distortion and reduced effectiveness of SMC. Finally, the critical drawback of SMC is the chattering effect during the tracking process [171].

*9) Virtual Machine Control:* The concept of virtual machine control (VMC) is centered around manipulating converter power flows constructed from an electrical machine's mechanical and electrical characteristics. The real-time parameters at the converter terminals serve as input to a simulated rotating machine model [172]. The level of detail in the model directly affects the complexity of the virtual machine. More advanced models that accurately replicate the specifications of induction and synchronous machines require greater complexity, whereas simpler approaches, such as virtual inertia methods, focus on simulating rotor dynamics only [173]. By simulating the characteristics of real machines in the network, VMCs ensure the stability of the power system. VMC can be broadly categorized into two types: virtual induction

machines and virtual synchronous machines. The inverter is a specific type of VMC based on the principles of induction machines. VMCs based on SGs can be further divided into three categories: synchronverter, synchronous power controller, and virtual synchronous machine [174].

*10) Virtual Oscillator Control:* In contrast to VMC, virtual oscillator control (VOC) uses the equations of a natural frequency synchronous oscillator circuit with the ac grid frequency [175]. This integrator-based control technique relies on the current–voltage dynamics of inductors, capacitors, their combination, or the voltage or current equations of resistors exhibiting droop-like behavior. Due to power–frequency equations, VOC does not require a PLL or external reference for voltage and frequency. Based on time-domain control and without low-pass filters, this approach enables immediate response, fast synchronization, and accurate power sharing. In addition, the computational load is significantly reduced [176]. VOCs are categorized into two types: Liénard-based and consensus-based approaches. Liénard oscillator control includes van der Pol oscillator and dead zone oscillator. At the same time, consensus-based approaches are divided into three control subcategories: distributed virtual oscillator, Andronov–Hopf oscillator, and integrated virtual oscillator [20].

*11) Repetitive Control:* The basic idea of repetitive control (RPC) is derived from the iterative learning control (ILC) concept and is used for periodic nonlinear loads [177]. Parallel combinations of PI, PRC, and P are considered mathematical equivalents of RPC [178]. The internal low-pass filter at the RPC reduces high-frequency resonance peaks in the controller's gain. This control technique performs accurately under periodic disturbances and achieves zero steady-state error in high-order harmonics. RPC also provides a frequency adaptive mechanism to overcome network frequency variations. The primary advantage of this control method is its ability to enhance microgrid stability by coordinating the system's voltage and current. Despite the simple structure and the minimal adjustment parameters, RPC does not have a favorable dynamic response since RPC is dependent on the PLL performance and has a slow dynamic response. To solve this problem, RPC can be combined with a parallel or cascade structure, and the other controller methods can be combined with a fast dynamic response [177].

*12) Neural Network Control:* Neural network control (NNC) is an interconnected computational approach to control data systems inspired by the biological system of the human brain and neural network mimicking the connections between neurons with a time delay [179]. The layers (input, hidden, and output), weights, activation functions, and closed-loop methods to transfer information without executing an error in the desired function are the elements of the NNC system architecture chain [180]. As an adaptive, robust, intelligent, and self-built controller, NNC offers a flexible control system with few designs and operating conditions. The self-learning feature of the NNC algorithm allows easy feasibility and design for various operating conditions and network disturbances. It also ensures stable behavior and a fast control process contributing to the system stability. NNC can be used online or offline to transfer data through control system processes [179], [180].

*13) Fuzzy Logic Control:* Fuzzy logic is a form of multiple-value logic that deals with reality. Instead of explicit values, it deals with linguistic values, where the range is one for absolutely true and zero for absolutely false. In fuzzy logic control (FLC), two key concepts are applied: the idea of fuzzy set membership from fuzzy set theory and subjective probability from probability theory [181]. The architecture of the FLC method includes fuzzification, basic rules, inference mechanisms, and defuzzification. In fuzzification, a set of information is converted into fuzzy data; in the rule bases, certain rules must be controlled according to the system requirements. In the inference mechanism, the rules are evaluated, and decisions are made based on them. According to the defined rules, in defuzzification, the fuzzified data are converted back into precise data, and as a result, an appropriate control action is obtained. This method is widely used in recent control applications due to its robustness, ability to handle imprecise inputs, nonlinearity, and ability to work without a precise mathematical model [182].

### D. Comparative Analysis of Control Techniques

Table 6 provides a structured comparison of control methods for reactive power management, categorized into traditional (classic, proportional resonance, and hysteresis), advanced (droop, robust, adaptive, predictive, sliding mode, virtual machine, and virtual oscillator), and intelligent approaches (repetitive, neural network, and fuzzy logic). Each category highlights specific strengths and limitations tailored to different levels of complexity and system requirements.

Traditional methods, such as PI, PID, and DRC, are simple and widely used in linear systems due to their ease of design and low computational demands. They perform well in predictable, stable environments while facing challenges in handling nonlinear dynamics, system uncertainties, and rapidly changing conditions. While effective for basic power systems, their sensitivity to parameter variations and limited adaptability make them less suitable for renewable-dominated or complex networks.

Advanced methods leverage modern control theories to address the shortcomings of traditional techniques. Robust, predictive, and ADCs can manage nonlinearities, uncertainties, and disturbances with greater precision and stability. For instance, PDC anticipates future system behavior for dynamic adjustments, while ADC realigns parameters in real time to maintain optimal performance. These methods provide improved control but

**Table 6** Comparing Control Techniques

| Control Technique | Key Characteristics | Advantages | Disadvantages |
|---|---|---|---|
| Classic Control | Based on classical control theory with simple feedback mechanisms. | Simple, cost-effective, suitable for linear systems. | Limited in handling nonlinear dynamics, sensitive to parameter changes. |
| Proportional Resonance Control | Focuses on frequencies near the resonance frequency. | Selective harmonic compensation, quick elimination of steady-state error. | Limited bandwidth and modulation range, requires precise parameter tuning. |
| Hysteresis Control | Nonlinear control method based on the hysteresis band of error signals. | Simple structure, fast dynamic response, high stability, inherent current protection. | Wide frequency range, high current ripple, challenges in designing output filters. |
| Droop Control | Adjusts system parameters (frequency, voltage) linearly with active and reactive power. | Quick response, effective load sharing, maintains voltage and frequency stability. | Sensitive to system changes, often requires integration with other methods. |
| Robust Control | Designed to handle uncertainties and disturbances. | Strong performance in multivariable systems, ensures stability under disturbances. | Complex design, requires detailed knowledge of system dynamics. |
| Adaptive Control | Real-time adjustment of control parameters based on operating conditions. | Effective under nonlinear dynamics and varying loads, maintains stability. | High complexity, requires significant computational resources. |
| Predictive Control | Uses mathematical models to predict system behavior. | Fast response, reduces switching frequency, effective control under unbalanced conditions. | Sensitive to parameter variations, requires extensive calculations. |
| Sliding Mode Control | Nonlinear mechanism that adjusts control laws based on system state. | Robust to disturbances and parameter changes, ideal steady-state performance. | Chattering effect, dependent on sliding surface design, critical sampling time adjustments required. |
| Virtual Machine Control | Simulates characteristics of real electrical machines. | Ensures system stability, flexible in various modeling approaches. | High complexity in detailed models, requires real-time processing. |
| Virtual Oscillator Control | Uses natural oscillator dynamics for grid synchronization. | Immediate response, reduced computational burden, fast synchronization. | Limited to specific applications, requires precise design. |
| Repetitive Control | Derived from iterative learning control, designed for periodic nonlinear loads. | Accurate under periodic disturbances, enhances microgrid voltage and current stability. | Slow dynamic response, dependent on Phase-Locked Loop (PLL) performance. |
| Neural Network Control | Inspired by biological neural networks for adaptive and intelligent control. | Self-learning, robust performance under complex dynamics, highly adaptive. | High computational requirements, complex design and implementation. |
| Fuzzy Logic Control | Uses fuzzy logic principles to manage imprecise inputs and nonlinear systems. | Robust to imprecise inputs, no need for a precise mathematical model. | Challenging to design rules and parameters, performance depends on the quality of the rule base. |

are computationally demanding, requiring accurate system modeling and significant processing resources. This makes them ideal for complex systems like renewable-integrated grids or microgrids.

Intelligent methods, such as fuzzy logic, neural networks, and RPC, incorporate AI and ML to enable adaptability, learning, and optimization in uncertain, dynamic environments. They excel in managing nonlinearities without requiring explicit mathematical models, making them particularly suitable for decentralized and renewable-heavy grids. However, the complexity of their design, reliance on computational resources, and need for extensive training data can present challenges. As technology advances, these methods are becoming more practical and are expected to play a pivotal role in future smart grids.

Control techniques reflect the evolution of power systems from simple, static networks to dynamic, IBR-integrated grids. Traditional methods remain cost-effective for simpler systems, while advanced methods offer robustness and precision for complex scenarios. Intelligent methods, with their adaptability and self-learning capabilities, are poised to address the uncertainties of modern power networks. The choice of control method depends on the system complexity, operational requirements, and desired performance, ensuring that control systems are tailored to the challenges of sustainable and decentralized energy systems.

## XIV. CONCLUSION AND FUTURE DIRECTIONS

The future of power grids will undoubtedly rely on a significant integration of IBRs, primarily wind, PV, and batteries. The transition from current to future decarbonized grids will bring about structural changes, particularly in the realm of reactive power support. As a significant share of SGs is decommissioned and replaced with IBRs, reactive power support will undergo a substantial transformation. This review provides an overview of the current state of reactive power supply, the impacts of the ongoing transition, and future trends in the context of the reactive power capability of IBRs. It has been shown that the upgrading cost to enable the reactive power capability of an inverter deployed in an IBR is significantly less than the installation cost of an equivalent grid-connected reactive power resource, even with imposed consequences on the inverter's lifetime and reliability. However, the potential

of IBRs to address reactive power requirements in future decarbonized grids still needs to be fully addressed. The existing literature lacks a comprehensive approach to coordinating and harmonizing the efforts of various stakeholders and drivers to leverage the reactive power capability of IBRs. A multifaceted strategy is needed to set the stage for leveraging the capabilities of IBRs to address the reactive power requirements of future grids. The recommendations to address this gap, enabling policymakers, utilities, and grid operators to manage a seamless transition to a decarbonized grid, are given as follows.

1) Improving existing models of IBRs to accurately represent the steady-state and dynamic behaviors of active and reactive power while incorporating the state-of-the-art technology, for ensuring consistency with the future decarbonized grid with high penetrations of IBRs.
2) Extension of available system operation software packages responsible for market clearing and ancillary service pricing to promote and incentivize reactive power support from IBRs.
3) Developing modern planning and allocation tools that focus on reactive power requirements of future decarbonized grids when sizing and sitting of IBRs at different capacity and connection scales.
4) Inclusive consideration of all IBRs' technologies, comprising primary wind, PV, and batteries to less developed V2G services (EVCSs), power-to-gas facilities (electrolyzers), and fuel cells.
5) Taking all system levels of IBRs' integration into account, ranging from BTM resources requiring aggregation to larger distributed resources and large-scale plants.
6) Updating current grid codes to fully appreciate the reactive power capabilities of the state-of-the-art technologies used for IBRs' interconnection.
7) Developing guidelines to benefit from reactive support of IBRs at the transmission and distribution interface as well as microgrids' aggregate reactive capability at the PCC.
8) Deploying automated wide-area monitoring, protection, and control (WAMPAC) systems, using fast detection and control devices as well as high-speed reliable communication infrastructures to control and fully appreciate IBRs' reactive capabilities. ■

## REFERENCES


[1] K. J. Chalvatzis and E. Hooper, "Energy security vs. Climate change: Theoretical framework development and experience in selected EU electricity markets," *Renew. Sustain. Energy Rev.*, vol. 13, no. 9, pp. 2703–2709, Dec. 2009.

[2] G. Bang, "Energy security and climate change concerns: Triggers for energy policy change in the United States?" *Energy Policy*, vol. 38, no. 4, pp. 1645–1653, Apr. 2010.

[3] P. Dechamps, "The IEA world energy outlook 2022—A brief analysis and implications," *Eur. Energy Climate J.*, vol. 11, no. 3, pp. 100–103, Sep. 2023.

[4] *$CO_2$ Emissions in 2022*, IEA Paris, France, 2022. [Online]. Available: https://www.iea.org/reports/co2-emissions-in-2022

[5] J. E. T. Bistline and G. J. Blanford, "The role of the power sector in net-zero energy systems," *Energy Climate Change*, vol. 2, Dec. 2021, Art. no. 100045.

[6] M. Child, C. Kemfert, D. Bogdanov, and C. Breyer, "Flexible electricity generation, grid exchange and storage for the transition to a 100% renewable energy system in Europe," *Renew. Energy*, vol. 139, pp. 80–101, Aug. 2019.

[7] The United States Department of State and The United States Executive Office of the President. (2021). *The Long-Term Strategy of the United States: Pathways to Net-Zero Greenhouse Gas Emissions by 2050*. Whitehouse, USA. [Online]. Available: https://www.whitehouse.gov/wp-content/uploads/2021/10/us-long-term-strategy.pdf

[8] *Future Energy Scenarios*. National Grid ESO, Electricity System Operator, Faraday House, U.K., 2019, pp. 1–166. [Online]. Available: https://www.nationalgrideso.com/document/170756/download

[9] B. Kroposki et al., "Achieving a 100% renewable grid: Operating electric power systems with extremely high levels of variable renewable energy," *IEEE Power Energy Mag.*, vol. 15, no. 2, pp. 61–73, Mar. 2017.

[10] C. S. Kaunda, C. Z. Kimambo, and T. K. Nielsen, "Hydropower in the context of sustainable energy supply: A review of technologies and challenges," *ISRN Renew. Energy*, vol. 2012, pp. 1–15, Dec. 2012.

[11] S. Wheatley, B. Sovacool, and D. Sornette, "Of disasters and dragon kings: A statistical analysis of nuclear power incidents and accidents," *Risk Anal.*, vol. 37, no. 1, pp. 99–115, Jan. 2017.

[12] T. Wilberforce, A. Baroutaji, B. Soudan, A. H. Al-Alami, and A. G. Olabi, "Outlook of carbon capture technology and challenges," *Sci. Total Environ.*, vol. 657, pp. 56–72, Mar. 2019.

[13] F. Blaabjerg and K. Ma, "Wind energy systems," *Proc. IEEE*, vol. 105, no. 11, pp. 2116–2131, Nov. 2017.

[14] A. O. M. Maka and J. M. Alabid, "Solar energy technology and its roles in sustainable development," *Clean Energy*, vol. 6, no. 3, pp. 476–483, Jun. 2022.

[15] *Grid Codes for Renewable Powered Systems*, M. IRENA, Abu Dhabi, United Arab Emirates, 2022.

[16] W. A. Braff, J. M. Mueller, and J. E. Trancik, "Value of storage technologies for wind and solar energy," *Nature Climate Change*, vol. 6, no. 10, pp. 964–969, Oct. 2016.

[17] H. Saboori, R. Hemmati, S. M. S. Ghiasi, and S. Dehghan, "Energy storage planning in electric power distribution networks—A state-of-the-art review," *Renew. Sustain. Energy Rev.*, vol. 79, pp. 1108–1121, Nov. 2017.

[18] R. Hemmati and H. Saboori, "Emergence of hybrid energy storage systems in renewable energy and transport applications—A review," *Renew. Sustain. Energy Rev.*, vol. 65, pp. 11–23, Nov. 2016.

[19] V. Sivaram, J. O. Dabiri, and D. M. Hart, "The need for continued innovation in solar, wind, and energy storage," *Joule*, vol. 2, no. 9, pp. 1639–1642, Sep. 2018.

[20] H. Pishbahar, F. Blaabjerg, and H. Saboori, "Emerging grid-forming power converters for renewable energy and storage resources integration—A review," *Sustain. Energy Technol. Assessments*, vol. 60, Dec. 2023, Art. no. 103538.

[21] A. Cabrera-Tobar, E. Bullich-Massagué, M. Aragüés-Peñalba, and O. Gomis-Bellmunt, "Review of advanced grid requirements for the integration of large scale photovoltaic power plants in the transmission system," *Renew. Sustain. Energy Rev.*, vol. 62, pp. 971–987, Sep. 2016.

[22] Z. Tang, Y. Yang, and F. Blaabjerg, "Power electronics: The enabling technology for renewable energy integration," *CSEE J. Power Energy Syst.*, vol. 8, no. 1, pp. 39–52, Jan. 2022.

[23] B. S. Hodge et al., "Addressing technical challenges in 100% variable inverter-based renewable energy power systems," *WIREs Energy Environ.*, vol. 9, no. 5, p. 376, Sep. 2020.

[24] Y. Lin et al., "Pathways to the next-generation power system with inverter-based resources: Challenges and recommendations," *IEEE Electrific. Mag.*, vol. 10, no. 1, pp. 10–21, Mar. 2022.

[25] S. Wogrin, D. Tejada-Arango, S. Delikaraoglou, and A. Botterud, "Assessing the impact of inertia and reactive power constraints in generation expansion planning," *Appl. Energy*, vol. 280, Dec. 2020, Art. no. 115925.

[26] T. Ackermann, T. Prevost, V. Vittal, A. J. Roscoe, J. Matevosyan, and N. Miller, "Paving the way: A future without inertia is closer than you think," *IEEE Power Energy Mag.*, vol. 15, no. 6, pp. 61–69, Nov. 2017.

[27] P. Tielens and D. Van Hertem, "The relevance of inertia in power systems," *Renew. Sustain. Energy Rev.*, vol. 55, pp. 999–1009, Mar. 2016.

[28] Y. Cheng, R. Azizipanah-Abarghooee, S. Azizi, L. Ding, and V. Terzija, "Smart frequency control in low inertia energy systems based on frequency response techniques: A review," *Appl. Energy*, vol. 279, Dec. 2020, Art. no. 115798.

[29] W. Bao et al., "Analytically derived fixed termination time for stepwise inertial control of wind turbines—Part I: Analytical derivation," *Int. J. Electr. Power Energy Syst.*, vol. 121, Oct. 2020, Art. no. 106120.

[30] Y. Guo et al., "Analytically derived fixed termination time for stepwise inertial control of wind turbines—Part II: Application strategy," *Int. J. Electr. Power Energy Syst.*, vol. 121, Oct. 2020, Art. no. 106106.

[31] M. N. I. Sarkar, L. G. Meegahapola, and M. Datta, "Reactive power management in renewable rich power grids: A review of grid-codes, renewable generators, support devices, control strategies and optimization algorithms," *IEEE Access*, vol. 6, pp. 41458–41489, 2018.

[32] A. Baviskar, A. D. Hansen, K. Das, and F. Ul Nazir, "Reactive power potential of converter-connected renewables using convex power flow optimization," *Int. J. Electr. Power Energy Syst.*, vol. 152, Oct. 2023, Art. no. 109193.

[33] S. Engelhardt, I. Erlich, C. Feltes, J. Kretschmann, and F. Shewarega, "Reactive power capability of wind turbines based on doubly fed induction generators," *IEEE Trans. Energy Convers.*, vol. 26, no. 1, pp. 364–372, Mar. 2011.

[34] A. K. Pathak, M. P. Sharma, and M. Bundele, "A critical review of voltage and reactive power management of wind farms," *Renew. Sustain. Energy Rev.*, vol. 51, pp. 460–471, Nov. 2015.

[35] S. Dehghan and N. Amjady, "Robust transmission and energy storage expansion planning in wind farm-integrated power systems considering transmission switching," *IEEE Trans. Sustain. Energy*, vol. 7, no. 2, pp. 765–774, Apr. 2015.

[36] H. Saboori and Y. Heshmati, "Novel load partitioning and prioritization for distribution restoration with storage integrated-soft open point and energy-constrained resources," *Electr. Power Syst. Res.*, vol. 214, Jan. 2023, Art. no. 108947.

[37] A. M. Nakiganda, S. Dehghan, U. Markovic, G. Hug, and P. Aristidou, "A stochastic-robust approach for resilient microgrid investment planning under static and transient Islanding security constraints," *IEEE Trans. Smart Grid*, vol. 13, no. 3, pp. 1774–1788, May 2022.

[38] A. Rabiee, H. Shayanfar, and N. Amjady, "Multiobjective clearing of reactive power market in deregulated power systems," *Appl. Energy*, vol. 86, no. 9, pp. 1555–1564, Sep. 2009.

[39] A. Rabiee, H. A. Shayanfar, and N. Amjady, "Reactive power pricing [the business scene]," *IEEE Power Mag.*, vol. 7, no. 1, pp. 18–32, Jan./Feb. 2009.

[40] F. Capitanescu, "Evaluating reactive power reserves scarcity during the energy transition toward 100% renewable supply," *Electric Power Syst. Res.*, vol. 190, Jan. 2021, Art. no. 106672.

[41] K. L. Anaya and M. G. Pollitt, "Reactive power procurement: A review of current trends," *Appl. Energy*, vol. 270, Jul. 2020, Art. no. 114939.

[42] J. H. Chow, F. F. Wu, and J. A. Momoh, *Applied Mathematics for Restructured Electric Power Systems*. Cham, Switzerland: Springer, 2005.

[43] EPRI. (2009). *Power System Dynamics Tutorial*. [Online]. Available: https://www.epri.com/research/products/000000000001016042

[44] H. Saadat et al., *Power System Analysis*, vol. 2. New York, NY, USA: McGraw-Hill, 1999.

[45] J. L. Willems, "Budeanu's reactive power and related concepts revisited," *IEEE Trans. Instrum. Meas.*, vol. 60, no. 4, pp. 1182–1186, Apr. 2011.

[46] C. Budeanu, *Puissances Reactives et Fictives*, no. 2. Bucarest, Romania: Impr. Cultura Naional, 1927.

[47] D. Yildirim and E. F. Fuchs, "Commentary on various formulations of distortion power D," *IEEE Power Eng. Rev.*, vol. 19, no. 5, pp. 50–52, May 1999.

[48] L. S. Czarnecki, "What is wrong with the budeanu concept of reactive and distortion power and why it should be abandoned," *IEEE Trans. Instrum. Meas.*, vol. IM-36, no. 3, pp. 834–837, Sep. 1987.

[49] L. S. Czarnecki, "Considerations on the reactive power in nonsinusoidal situations," *IEEE Trans. Instrum. Meas.*, vol. IM-34, no. 3, pp. 399–404, Sep. 1985.

[50] *IEEE Standard Definitions for the Measurement of Electric Power Quantities Under Sinusoidal, Nonsinusoidal, Balanced, or Unbalanced Conditions*, IEEE Standard 1459-2010 (Revision of IEEE Std 1459-2000), pp. 1–50, Mar. 2010, doi: 10.1109/IEEESTD.2010.5439063.

[51] L. S. Czarnecki, "Currents' physical components (CPC) concept: A fundamental of power theory," in *Proc. Int. School Nonsinusoidal Currents Compensation*, Jun. 2008, pp. 1–11.

[52] L. S. Czarnecki, "Physical interpretation of the reactive power in terms of the CPC power theory," *Electr. Power Qual. Utilisation. J.*, vol. 13, pp. 89–95, Jan. 2007.

[53] D. Jeltsema, "Budeanu's concept of reactive and distortion power revisited," in *Proc. Int. School Nonsinusoidal Currents Compensation (ISNCC)*, Jun. 2015, pp. 1–6.

[54] J.-H.-C. Pretorius, J. D. van Wyk, and P. H. Swart, "An evaluation of some alternative methods of power resolution in a large industrial plant," *IEEE Trans. Power Del.*, vol. 15, no. 3, pp. 1052–1059, Jul. 2000.

[55] O. A. Mousavi and R. Cherkaoui, *Literature Survey on Fundamental Issues of Voltage and Reactive Power Control*. Lausanne, Switzerland: Ecole Polytechnique Fédérale de Lausanne, 2011.

[56] L. L. Grigsby, Ed., *Power System Stability and Control*, 3rd ed., USA: CRC Press, 2012.

[57] N. Hatziargyriou et al., "Definition and classification of power system stability—Revisited & extended," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3271–3281, Dec. 2021.

[58] M. Derafshian, N. Amjady, and S. Dehghan, "Special protection scheme against voltage collapse," *IET Gener., Transmiss. Distrib.*, vol. 10, no. 2, pp. 341–351, Feb. 2016.

[59] S. Imai, D. Novosel, D. Karlsson, and A. Apostolov, "Unexpected consequences: Global blackout experiences and preventive solutions," *IEEE Power Energy Mag.*, vol. 21, no. 3, pp. 16–29, May 2023.

[60] *Interconnection of Energy Storage Distributed Energy Resources With Electric Power Systems*, Standard 1547.9-2022, 2022, pp. 1–87.

[61] *IEC Technical Report 61850-7-5*, Standard 1850-7-520-2021, 2021, pp. 1–33.

[62] J. Dixon, L. Moran, J. Rodriguez, and R. Domke, "Reactive power compensation technologies: State-of-the-art review," *Proc. IEEE*, vol. 93, no. 12, pp. 2144–2164, Dec. 2005.

[63] W. Zhang, F. Li, and L. M. Tolbert, "Review of reactive power planning: Objectives, constraints, and algorithms," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 2177–2186, Nov. 2007.

[64] M. T. L. Gayatri, A. M. Parimi, and A. V. P. Kumar, "A review of reactive power compensation techniques in microgrids," *Renew. Sustain. Energy Rev.*, vol. 81, pp. 1030–1036, Jan. 2018.

[65] O. Homaee and S. Jadid, "Investigation of synchronous generator in reactive power market—An accurate view," *IET Gener., Transmiss. Distrib.*, vol. 8, no. 11, pp. 1881–1890, Nov. 2014.

[66] A. Ipakchi and F. Albuyeh, "Grid of the future," *IEEE Power Energy Mag.*, vol. 7, no. 2, pp. 52–62, Feb. 2009.

[67] G. Zhou et al., "Synchronous condenser applications: Under significant resource portfolio changes," *IEEE Power Energy Mag.*, vol. 17, no. 4, pp. 35–46, Jul. 2019.

[68] A. Jindal and G. Shrimali, "Cost–benefit analysis of coal plant repurposing in developing countries: A case study of India," *Energy Policy*, vol. 164, May 2022, Art. no. 112911.

[69] N. Masood, R. Yan, T. K. Saha, and S. Bartlett, "Post-retirement utilisation of synchronous generators to enhance security performances in a wind dominated power system," *IET Gener., Transmiss. Distrib.*, vol. 10, no. 13, pp. 3314–3321, Oct. 2016.

[70] T. J. E. Miller, *Reactive Power Control in Electric Systems*. New York, NY, USA: Wiley, 1982.

[71] N. G. Hingorani, "FACTS technology-state of the art, current challenges and the future prospects," in *Proc. IEEE Power Eng. Soc. Gen. Meeting*, Jun. 2007, vol. 2, no. 3, p. 11.

[72] A.-A. Edris, "Proposed terms and definitions for flexible ac transmission system (FACTS)," *JEEE Trans. Power Del.*, vol. 12, no. 4, pp. 1848–1853, Jan. 1997.

[73] F. Li et al., "Smart transmission grid: Vision and framework," *IEEE Trans. Smart Grid*, vol. 1, no. 2, pp. 168–177, Sep. 2010.

[74] S. Hadavi, M. Z. Mansour, and B. Bahrani, "Optimal allocation and sizing of synchronous condensers in weak grids with increased penetration of wind and solar farms," *IEEE J. Emerg. Sel. Topics Circuits Syst.*, vol. 11, no. 1, pp. 199–209, Mar. 2021.

[75] S. Hadavi, J. Saunderson, A. Mehrizi-Sani, and B. Bahrani, "A planning method for synchronous condensers in weak grids using semi-definite optimization," *IEEE Trans. Power Syst.*, vol. 38, no. 2, pp. 1632–1641, Mar. 2023.

[76] *ABB Synchronous Condensers*. Accessed: Mar. 10, 2024. [Online]. Available: https://new.abb.com/motors-generators/synchronous-condensers

[77] D. Al Kez et al., "A critical evaluation of grid stability and codes, energy storage and smart loads in power systems with wind generation," *Energy*, vol. 205, Aug. 2020, Art. no. 117671.

[78] F. Z. Peng, "Flexible AC transmission systems (FACTS) and resilient AC distribution systems (RACDS) in smart grid," *Proc. IEEE*, vol. 105, no. 11, pp. 2099–2115, Nov. 2017.

[79] S. Xu, Y. Xue, and L. Chang, "Review of power system support functions for inverter-based distributed energy resources-standards, control algorithms, and trends," *IEEE Open J. Power Electron.*, vol. 2, pp. 88–105, 2021.

[80] J. Matevosyan et al., "A future with inverter-based resources: Finding strength from traditional weakness," *IEEE Power Energy Mag.*, vol. 19, no. 6, pp. 18–28, Nov. 2021.

[81] I. S. Bayram and T. S. Ustun, "A survey on behind the meter energy management systems in smart grid," *Renew. Sustain. Energy Rev.*, vol. 72, pp. 1208–1232, May 2017.

[82] M. Rezaeimozafar, R. F. D. Monaghan, E. Barrett, and M. Duffy, "A review of behind-the-meter energy storage systems in smart grids," *Renew. Sustain. Energy Rev.*, vol. 164, Aug. 2022, Art. no. 112573.

[83] V. Yaramasu, B. Wu, and J. Chen, "Model-predictive control of grid-tied four-level diode-clamped inverters for high-power wind energy conversion systems," *IEEE Trans. Power Electron.*, vol. 29, no. 6, pp. 2861–2873, Jun. 2014.

[84] I. Kamwa and B. Mather, "Taming wind and solar resources: Smart inverters are key [Editor's voice]," *IEEE Power Energy Mag.*, vol. 22, no. 2, pp. 4–11, Mar. 2024.

[85] A. Khazali, N. Rezaei, H. Saboori, and J. M. Guerrero, "Using PV systems and parking lots to provide virtual inertia and frequency regulation provision in low inertia grids," *Electr. Power Syst. Res.*, vol. 207, Jun. 2022, Art. no. 107859.

[86] H. Mehrjerdi, H. Saboori, and S. Jadid, "Power-to-gas utilization in optimal sizing of hybrid power, water, and hydrogen microgrids with energy and gas storage," *J. Energy Storage*, vol. 45, Jan. 2022, Art. no. 103745.

[87] H. Saboori and H. Pishbahar, "Water-conscious green hydrogen production and conversion for harnessing long-term storage in remote renewable microgrids," *Energy Convers. Manage.*, vol. 318, Oct. 2024, Art. no. 118870.

[88] *Technical Regulation 3.3.1 Requirements for Energy Storage Facilities*, Energinet, Erritsø, Denmark, 2016.

[89] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources With Associated Electric Power Systems Interfaces*, Standard 1547-2018, BOG/SC21 Distributed Generation and I. S. Committee, 2018.

[90] A. Naderipour, H. Saboori, H. Mehrjerdi, S. Jadid, and Z. Abdul-Malek, "Sustainable and reliable hybrid AC/DC microgrid planning considering technology choice of equipment," *Sustain. Energy, Grids Netw.*, vol. 23, Sep. 2020, Art. no. 100386.

[91] S. Hasan, "Microgrid-enabled reactive power support to enhance grid economics," Ph.D. dissertation, Dept. Elect. Eng., Univ. Denver, USA, 2020.

[92] Poyry and Imperial College London. (2017). *Roadmap for Flexibility Services to 2030*. [Online]. Available: https://www.theccc.org.uk/publication/roadmap-for-flexibility-services-to-2030-poyry-and-imperial-college-london/

[93] Vivideconomic and Imperial College London. (2019). *Accelerated Electrification and the Gb*

*Electricity System*. [Online]. Available: https://www.theccc.org.uk/wp-content/uploads/2019/05/CCC-Accelerated-Electrification-Vivid-Economics-Imperial-1.pdf
[94] G. Strbac, D. Pudjianto, and P. Djapic, "Power potential project: Evaluating synergies and conflicts of der services for distribution and transmission systems and market power assessment," Imperial College London, London, U.K., Tech. Rep., 2019. [Online]. Available: https://www.nationalgrideso.com/document/143386/download
[95] J. Kueck, B. Kirby, T. Rizy, F. Li, and N. Fall, "Reactive power from distributed energy," *Electr. J.*, vol. 19, no. 10, pp. 27–38, 2006.
[96] R. K. Varma, E. Siavashi, S. Mohan, and J. McMichael-Dennis, "Grid support benefits of solar PV systems as STATCOM (PV-STATCOM) through converter control: Grid integration challenges of solar PV power systems," *IEEE Electrific. Mag.*, vol. 9, no. 2, pp. 50–61, Jun. 2021.
[97] M. Braun, "Reactive power supplied by PV inverters–cost-benefitanalysis," in *Proc. 22nd Eur. Photovoltaic Solar Energy Conf. Exhib.*, vol. 3, 2007, p. 7.
[98] M. Braun, "Reactive power supplied by wind energy converters-cost-benefitanalysis," in *Proc. Eur. Wind Energy Conf. (EWEC)*, vol. 31, Brussels, Belgium, Jul. 2008, pp. 1–10.
[99] N. R. Ullah, K. Bhattacharya, and T. Thiringer, "Wind farms as reactive power ancillary service providers—Technical and economic issues," *IEEE Trans. Energy Convers.*, vol. 24, no. 3, pp. 661–672, Sep. 2009.
[100] O. Gandhi, C. D. Rodríguez-Gallegos, N. B. Y. Gorla, M. Bieri, T. Reindl, and D. Srinivasan, "Reactive power cost from PV inverters considering inverter lifetime assessment," *IEEE Trans. Sustain. Energy*, vol. 10, no. 2, pp. 738–747, 2018.
[101] R. Guideline, *Reliability Guideline: BPS-Connected Inverter-Based Resource Performance*. Atlanta, GA, USA: NERC, 2018.
[102] M. Liserre, R. Cárdenas, M. Molinas, and J. Rodriguez, "Overview of multi-MW wind turbines and wind parks," *IEEE Trans. Ind. Electron.*, vol. 58, no. 4, pp. 1081–1095, Apr. 2011.
[103] F. Li, J. D. Kueck, D. T. Rizy, and T. F. King, "A preliminary analysis of the economics of using distributed energy as a source of reactive power supply," Dept. Oak Ridge National Lab. (ORNL), Oak Ridge, TN, USA, Tech. Rep. 930730, 2006.
[104] A. Anurag, Y. Yang, and F. Blaabjerg, "Reliability analysis of single-phase PV inverters with reactive power injection at night considering mission profiles," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Sep. 2015, pp. 2132–2139.
[105] L. S. Gusman, H. Pereira, J. Callegari, and A. Cupertino, "Design for reliability of multifunctional PV inverters used in industrial power factor regulation," *Int. J. Electr. Power Energy Syst.*, vol. 119, Jul. 2020, Art. no. 105932.
[106] V. Terzija et al., "Wide-area monitoring, protection, and control of future electric power networks," *Proc. IEEE*, vol. 99, no. 1, pp. 80–93, Jan. 2011.
[107] Q. Hong et al., "Design and validation of a wide area monitoring and control system for fast frequency response," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 3394–3404, Jul. 2020.
[108] S. Chakrabarti, E. Kyriakides, T. Bi, D. Cai, and V. Terzija, "Measurements get together," *IEEE Power Energy Mag.*, vol. 7, no. 1, pp. 41–49, Jan. 2009.
[109] G. Jayasinghe and B. Bahrani, "Stability-enhancing measures for weak grids study," Monash Univ., Tech. Rep., 2021.
[110] A. A. Nazari, R. Keypour, M. H. Beiranvand, and N. Amjady, "A decoupled extended power flow analysis based on Newton-raphson method for islanded microgrids," *Int. J. Electr. Power Energy Syst.*, vol. 117, May 2020, Art. no. 105705.
[111] M. Cheng and Y. Zhu, "The state of the art of wind energy conversion systems and technologies: A review," *Energy Convers. Manage.*, vol. 88, pp. 332–347, Dec. 2014.
[112] V. Yaramasu, B. Wu, P. C. Sen, S. Kouro, and M. Narimani, "High-power wind energy conversion systems: State-of-the-art and emerging technologies," *Proc. IEEE*, vol. 103, no. 5, pp. 740–788, May 2015.
[113] A. T. Azar and N. A. Kamal, *Renewable Energy Systems: Modelling, Optimization and Control*. New York, NY, USA: Academic, 2021.
[114] B. Wu, Y. Lang, N. Zargari, and S. Kouro, *Power Conversion and Control of Wind Energy Systems*, vol. 76. Hoboken, NJ, USA: Wiley, 2011.
[115] R. Cardenas, R. Pena, S. Alepuz, and G. Asher, "Overview of control systems for the operation of DFIGs in wind energy applications," *IEEE Trans. Ind. Electron.*, vol. 60, no. 7, pp. 2776–2798, Jul. 2013.
[116] Z. Q. Zhu and J. Hu, "Electrical machines and power-electronic systems for high-power wind energy generation applications: Part II—Power electronics and control systems," *COMPEL Int. J. Comput. Math. Electr. Electron. Eng.*, vol. 32, no. 1, pp. 34–71, Dec. 2012.
[117] D. Yoosefian, H. Askarian-Abyaneh, M. Kheshti, Q. Li, and V. Terzija, "Fault ride through constrained protection scheme for distribution networks with DFIG-based wind parks," *Electr. Power Syst. Res.*, vol. 215, Feb. 2023, Art. no. 108973.
[118] T. Lund, P. Sørensen, and J. Eek, "Reactive power capability of a wind turbine with doubly fed induction generator," *Wind Energy*, vol. 10, no. 4, pp. 379–394, Jul. 2007.
[119] V. Vittal et al., *Impact of Increased DFIG Wind Penetration on Power Systems and Markets*, vol. 9. Madison, WI, USA: PSERC Publication, 2009.
[120] V. Terzija and V. Stanojevic, "Two-stage improved recursive Newton-type algorithm for power-quality indices estimation," *IEEE Trans. Power Del.*, vol. 22, no. 3, pp. 1351–1359, Jul. 2007.
[121] J. McDowell et al., "Reactive power interconnection requirements for PV and wind plants: Recommendations to NERC," Sandia Nat. Laboratories (SNL), Livermore, CA, USA, Tech. Rep. SAND2012-1098, 2012.
[122] R. Albarracín and M. Alonso, "Photovoltaic reactive power limits," in *Proc. 12th Int. Conf. Environ. Electr. Eng.*, Wroclaw, Poland, May 2013, pp. 13–18.
[123] *IEEE Guide for using IEEE Std 1547 for Interconnection of Energy Storage Distributed Energy Resources With Electric Power Systems*, Standard 1547.9-2022, E. S. BOG/SC21 Distributed Generation and I. S. Committee, 2022.
[124] H. Saboori, H. Mehrjerdi, and S. Jadid, "Mobile battery storage modeling and normal-emergency operation in coupled distribution-transportation networks," *IEEE Trans. Sustain. Energy*, vol. 13, no. 4, pp. 2226–2238, Oct. 2022.
[125] *Datasheet*, G.-G. S. S. Advanced Energy Storage for the Grid, NEC Energy Solutions, Inc., Westborough, MA, USA, 2024.
[126] *Datasheet*, I. Solis, Three Phase Inverter and.-K. Operation Manual For U.S., Ginlong Technol. Co., Doc, Ningbo,China, 2024.
[127] G. S. V. M. H. V. Vestas, *Datasheet*. Aarhus N, Aarhus, Denmark: Vestas Wind Syst. Co., Doc, 2010.
[128] T. Sathiyanarayanan and D. Vijay M., "Grid interconnection standards, grid code requirements and compliance for renewable integration," in *Power Systems Operation With 100% Renewable Energy Sources*. Amsterdam, The Netherlands: Elsevier, 2024, pp. 55–64.
[129] T. Ackermann et al., "Scaling up variable renewable power: The role of grid codes," Int. Renew. Energy Agency (IRENA), UAE, May 2016.
[130] *Commission Regulation (EU) 2016/631 of 14 April 2016 Establishing a Network Code on Requirements for Grid Connection of Generators*, European Union, Brussels, Belgium, 2016.
[131] *Reactive Power Management With Distributed Energy Resources*, document IEA-PVPS T14-16, 2024.
[132] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources With Associated Electric Power Systems Interfaces*, Standard 1547-2018, D. G. Photovoltaics and E. Storage, 2018.
[133] *IEEE Standard for Interconnection and Interoperability of Inverter-Based Resources (IBRS) Interconnecting With Associated Transmission Electric Power Systems*, Standard 2800-2022, 2022.
[134] *Reactive Power Requirements for Non-Synchronous Generation*, document Docket No. RM16-1-000, Order No, F. E. R. Commission, 2016, vol. 827.
[135] "Technische mindestanforderungen Für anschluss und parallelbetrieb von erzeugungsanlagen am niederspannungsnetz," V. V. D. E. E. Informationstechnik, VDE Assoc. Elect., Electron. Inf. Technol. (VDE), Frankfurt am Main, Germany, 2018.
[136] "EV VDE-ar-n 4110-technische regeln für den anschluss von kundenanlagen an das mittelspannungsnetz und deren betrieb (tar mittelspannung)," V. V. D. E. E. Informationstechnik, VDE Assoc. Elect., Electron. Inf. Technol. (VDE), Frankfurt am Main, Germany, Tech. Rep., 2018.
[137] "EV vde-ar-n 4120-technische regeln für den anschluss von kundenanlagen an das mittelspannungsnetz und deren betrieb (tar mittelspannung)," V. V. D. E. E. Informationstechnik, VDE Assoc. Elect., Electron. Inf. Technol. (VDE), Frankfurt am Main, Germany, Tech. Rep., 2018.
[138] "The grid code, issue 6, revision 25," Nat. Grid Electr. Syst. Operator, Warwick, U.K., Tech. Rep., 2024.
[139] *EirGrdi Grid Code Version 14.2*, EirGrdi, Dublin, Ireland, 2024.
[140] *Technical Regulation 3.2.2 for PV Power Plants Above 11 Kw*, Energinet, Erritsø, Denmark, 2016.
[141] *Technical Regulation 3.2.5 for Wind Power Plants Above 11 Kw*, Energinet, Erritsø, Denmark, 2016.
[142] J. Rocabert, A. Luna, F. Blaabjerg, and P. Rodríguez, "Control of power converters in AC microgrids," *IEEE Trans. Power Electron.*, vol. 27, no. 11, pp. 4734–4749, Nov. 2012.
[143] H. Zhang, W. Xiang, W. Lin, and J. Wen, "Grid forming converters in renewable energy sources dominated power grid: Control strategy, stability, application, and challenges," *J. Modern Power Syst. Clean Energy*, vol. 9, no. 6, pp. 1239–1256, Nov. 2021.
[144] S. Ansari, A. Chandel, and M. Tariq, "A comprehensive review on power converters control and control strategies of AC/DC microgrid," *IEEE Access*, vol. 9, pp. 17998–18015, 2021.
[145] R. Rosso, X. Wang, M. Liserre, X. Lu, and S. Engelken, "Grid-forming converters: Control approaches, grid-synchronization, and future trends—A review," *IEEE Open J. Ind. Appl.*, vol. 2, pp. 93–109, 2021.
[146] Y. Wu, H. Wu, F. Zhao, Z. Li, and X. Wang, "Influence of PLL on stability of interconnected grid-forming and grid-following converters," *IEEE Trans. Power Electron.*, vol. 39, no. 10, pp. 11980–11985, Oct. 2024.
[147] A. M. Bouzid, J. M. Guerrero, A. Cheriti, M. Bouhamida, P. Sicard, and M. Benghanem, "A survey on control of electric power distributed generation systems for microgrid applications," *Renew. Sustain. Energy Rev.*, vol. 44, pp. 751–766, Apr. 2015.
[148] R. Musca, A. Vasile, and G. Zizzo, "Grid-forming converters. A critical review of pilot projects and demonstrators," *Renew. Sustain. Energy Rev.*, vol. 165, Sep. 2022, Art. no. 112551.
[149] D. B. Rathnayake et al., "Grid forming inverter modeling, control, and applications," *IEEE Access*, vol. 9, pp. 114781–114807, 2021.
[150] Y. Gu and T. C. Green, "Power system stability with a high penetration of inverter-based resources," *Proc. IEEE*, vol. 111, no. 7, pp. 832–853, Jun. 2022.
[151] M. Ghazavi Dozein, B. C. Pal, and P. Mancarella, "Dynamics of inverter-based resources in weak distribution grids," *IEEE Trans. Power Syst.*, vol. 37, no. 5, pp. 3682–3692, Sep. 2022.
[152] A. Lasheen, M. E. Ammar, H. H. Zeineldin, A. Al-Durra, M. F. Shaaban, and E. El-Saadany, "Assessing the impact of reactive power droop on inverter based microgrid stability," *IEEE Trans. Energy Convers.*, vol. 36, no. 3, pp. 2380–2392, Sep. 2021.

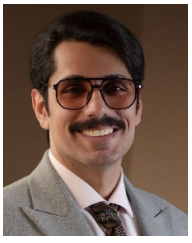
[153] H. Sun et al., "Review of challenges and research opportunities for voltage control in smart grids," *IEEE Trans. Power Syst.*, vol. 34, no. 4, pp. 2790–2801, Jul. 2019.
[154] K. Turitsyn, P. Sulc, S. Backhaus, and M. Chertkov, "Options for control of reactive power by distributed photovoltaic generators," *Proc. IEEE*, vol. 99, no. 6, pp. 1063–1073, Jun. 2011.
[155] J. Xu, S. Xie, and T. Tang, "Active damping-based control for grid-connected $LCL$-filtered inverter with injected grid current feedback only," *IEEE Trans. Ind. Electron.*, vol. 61, no. 9, pp. 4746–4758, Sep. 2014.
[156] A. A. A. Radwan and Y. A. I. Mohamed, "Power synchronization control for grid-connected current-source inverter-based photovoltaic systems," *IEEE Trans. Energy Convers.*, vol. 31, no. 3, pp. 1023–1036, Sep. 2016.
[157] A. Camacho, M. Castilla, J. Miret, J. C. Vasquez, and E. Alarcon-Gallo, "Flexible voltage support control for three-phase distributed generation inverters under grid fault," *IEEE Trans. Ind. Electron.*, vol. 60, no. 4, pp. 1429–1441, Apr. 2013.
[158] C. Bao, X. Ruan, X. Wang, W. Li, D. Pan, and K. Weng, "Step-by-step controller design for LCL-type grid-connected inverter with capacitor–current-feedback active-damping," *IEEE Trans. Power Electron.*, vol. 29, no. 3, pp. 1239–1253, Mar. 2014.
[159] B. K. Bose, "An adaptive hysteresis-band current control technique of a voltage-fed PWM inverter for machine drive system," *IEEE Trans. Ind. Electron.*, vol. 37, no. 5, pp. 402–408, Oct. 1990.
[160] X. Zhang, Y. Wang, C. Yu, L. Guo, and R. Cao, "Hysteresis model predictive control for high-power grid-connected inverters with output LCL filter," *IEEE Trans. Ind. Electron.*, vol. 63, no. 1, pp. 246–256, Jan. 2016.
[161] M. C. Chandorkar, D. M. Divan, and R. Adapa, "Control of parallel connected inverters in standalone AC supply systems," *IEEE Trans. Ind. Appl.*, vol. 29, no. 1, pp. 136–143, Aug. 1993.
[162] Y. Wang, C. Wang, L. Xu, J. Meng, and Y. Hei, "Adjustable inertial response from the converter with adaptive droop control in DC grids," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 3198–3209, May 2019.
[163] L. Meng, T. Dragicevic, J. Guerrero, J. Vasquez, M. Savaghebi, and F. Tang, "Agent-based distributed unbalance compensation for optimal power quality in islanded microgrids," in *Proc. IEEE 23rd Int. Symp. Ind. Electron. (ISIE)*, Jun. 2014, pp. 2535–2540.
[164] F. Mohammadi et al., "Robust control strategies for microgrids: A review," *IEEE Syst. J.*, vol. 16, no. 2, pp. 2401–2412, Jun. 2022.
[165] T. D. Do, V. Q. Leu, Y.-S. Choi, H. H. Choi, and J.-W. Jung, "An adaptive voltage control strategy of three-phase inverter for stand-alone distributed generation systems," *IEEE Trans. Ind. Electron.*, vol. 60, no. 12, pp. 5660–5672, Dec. 2013.
[166] S. Eren, M. Pahlevani, A. Bakhshai, and P. Jain, "An adaptive droop DC-bus voltage controller for a grid-connected voltage source inverter with LCL filter," *IEEE Trans. Power Electron.*, vol. 30, no. 2, pp. 547–560, Feb. 2015.
[167] M. Heidari, L. Ding, M. Kheshti, X. Zhao, and V. Terzija, "Adaptive inertial control for wind turbine generators in fast frequency response based on the power reduction period assessment," *IEEE Trans. Sustain. Energy*, vol. 16, no. 1, pp. 377–391, Jan. 2025.
[168] S. Ouchen, S. Abdeddaim, A. Betka, and A. Menadi, "Experimental validation of sliding mode-predictive direct power control of a grid connected photovoltaic system, feeding a nonlinear load," *Sol. Energy*, vol. 137, pp. 328–336, Nov. 2016.
[169] J. S. Lim, C. Park, J. Han, and Y. I. Lee, "Robust tracking control of a three-phase DC–AC inverter for UPS applications," *IEEE Trans. Ind. Electron.*, vol. 61, no. 8, pp. 4142–4151, Aug. 2014.
[170] S.-L. Jung and Y.-Y. Tzou, "Discrete sliding-mode control of a PWM inverter for sinusoidal output waveform synthesis with optimal sliding curve," *IEEE Trans. Power Electron.*, vol. 11, no. 4, pp. 567–577, Jul. 1996.
[171] N. Kumar, T. K. Saha, and J. Dey, "Sliding-mode control of PWM dual inverter-based grid-connected PV system: Modeling and performance analysis," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 4, no. 2, pp. 435–444, Jun. 2016.
[172] H. Alatrash, A. Mensah, E. Mark, G. Haddad, and J. Enslin, "Generator emulation controls for photovoltaic inverters," *IEEE Trans. Smart Grid*, vol. 3, no. 2, pp. 996–1011, Jun. 2012.
[173] Q.-C. Zhong, "Virtual synchronous machines: A unified interface for grid integration," *IEEE Power Electron. Mag.*, vol. 3, no. 4, pp. 18–27, Dec. 2016.
[174] H. Pishbahar, H. M. CheshmehBeigi, N. P. Yengijeh, and S. Bagheri, "Inertia emulation with incorporating the concept of virtual compounded DC machine and bidirectional DC–DC converter for DC microgrid in islanded mode," *IET Renew. Power Gener.*, vol. 15, no. 8, pp. 1812–1825, Jun. 2021.
[175] M. A. Awal and I. Husain, "Unified virtual oscillator control for grid-forming and grid-following converters," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 4, pp. 4573–4586, Aug. 2021.
[176] S. Azizi Aghdam and M. Agamy, "Virtual oscillator-based methods for grid-forming inverter control: A review," *IET Renew. Power Gener.*, vol. 16, no. 5, pp. 835–855, Apr. 2022.
[177] A. Trivedi and M. Singh, "Repetitive controller for VSIs in droop-based AC-microgrid," *IEEE Trans. Power Electron.*, vol. 32, no. 8, pp. 6595–6604, Aug. 2017.
[178] M. Ram, M. Child, A. Aghahosseini, D. Bogdanov, A. Poleva, and C. Breyer, *Comparing Electricity Production Costs of Renewables to Fossil and Nuclear Power Plants in G20 Countries*. Lappeenranta, Finland: Lappeenranta Univ. Technology (LUT), 2017, pp. 1–84.
[179] K. Ro and S. Rahman, "Two-loop controller for maximizing performance of a grid-connected photovoltaic-fuel cell hybrid power plant," *IEEE Trans. Energy Convers.*, vol. 13, no. 3, pp. 276–281, Sep. 1998.
[180] F.-J. Lin, K.-C. Lu, and T.-H. Ke, "Probabilistic wavelet fuzzy neural network based reactive power control for grid-connected three-phase PV system during grid faults," *Renew. Energy*, vol. 92, pp. 437–449, Jul. 2016.
[181] K. Passino, "Intelligent control: Anan overview of techniques," Perspect. Control Eng., Technol., Appl. New Directions, Dept. Electical Eng., Ohio State Univ., 2001, pp. 104–133.
[182] U. Yilmaz, A. Kircay, and S. Borekci, "PV system fuzzy logic MPPT method and PI control as a charge controller," *Renew. Sustain. Energy Rev.*, vol. 81, pp. 994–1001, Jan. 2018.

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